

NEURAL BELLMAN OPERATORS

QIAN QI  
Peking University

This paper develops Neural Bellman Operators for policy evaluation and feasible improvement in controlled economies. Centered continuation errors link economic transport to full-policy loss. A constructive neural backend retains the feasible actions attaining its labels and supports implementation under acquired states and endogenous capacity. We give an exact continuous-state compiler that preserves the original policy, including ties, while removing a redundant state-cover factor from its evaluation work. The construction combines classical distance transforms with witness-preserving off-grid evaluation and numerical certification. A priced observation technology determines precision from a certified economic account. Direct constrained-policy comparisons distinguish signed cost differences from the ability to recoup a controller replacement fee. Matched services include uniform and error-driven conventional methods, failed refinements, and three- and four-dimensional nonlinear stress cases. The results separate representation savings, complete work, and economic decisions without attributing a native-envelope capability exclusively to neural encoding. Broader applications retain their own comparison conditions.

KEYWORDS: Dynamic programming, Neural approximation, Policy certification, Recursive utility, Computational economics.

READING NOTICE FOR THE COMPLETE DEVELOPMENT EDITION

This volume retains the baseline development verbatim, including historical descriptions, followed by the current R52 argument. Its historical reference to a formerly authoritative revision is not a second current scientific claim. The active article, technical supplement and response govern the new evidence and numerical corrections. Every baseline result and its original assumptions remain available for examination.

Qian QI: [qiqian@pku.edu.cn](mailto:qiqian@pku.edu.cn)

## 1. INTRODUCTION

An economic decision uses a continuation value because its consequences extend beyond the current period. A learned continuation is useful only to the extent that the decisions made with it are accurate. Prediction loss, a Bellman residual at sampled states, and the value of the policy actually implemented are different objects. Their distinction is particularly important when a counterfactual changes future policies or when a recursive preference evaluates the continuation generated by those policies rather than the optimum.

This paper develops Neural Bellman Operators (NBOs) as a policy-evaluation and feasible-improvement method. An operator takes economic primitives, a current policy, and computational resources as inputs. It returns a continuation, a feasible policy, and an economic error account, or it records that the requested account was not obtained within its cap. The continuation is evaluated against its own fitted future. Current actions, policy-specific future values, numerical execution, and the cost of verification enter the same account. The construction is not a proposal to rename an arbitrary neural predictor as a Bellman operator.

There are four parts to the argument. First, only action-dependent continuation errors affect a current decision. Their oscillation, rather than their absolute level, controls loss. A transport bound identifies when an anchor continuation remains valid after the future economy changes. These results connect reuse and refresh decisions to economic tolerances. They also preserve the original formulation in controlled diffusions, recursive preferences, endogenous preferences, and games, subject to the distinct hypotheses of those applications.

Second, a signed residual certificate controls a complete returned policy. Its upper bound contains the widths of the residual enclosures, the suboptimality of the selected actions, and the terminal enclosure. It is invariant to additive date-specific shifts of the fitted continuation. This leads to a useful separation: a policy can be accurate even though a coarse verifier does not establish its accuracy. We prove that residual verification can be refined while retaining the original network, the original deployed actor, and its original allowance. The resulting bound measures

improved knowledge of the same policy, not a policy improvement disguised as additional verification.

Third, certification of each policy relative to the optimum does not rank two policies. We construct their cost difference directly. Under expected-cost preferences, an own-continuation Bellman telescoping identity produces an unbiased policy-cost score with a support width governed by the signed certificate. Coupling innovations across policies and enclosing every simulated path gives simultaneous finite-sample intervals for the difference in the original continuous-innovation economy. The statistical assertion is conditional on frozen policies and the stated independent simulation model; outward arithmetic and unresolved action-cell boundaries have their own explicit treatment.

Fourth, a deterministic neural construction connects primitive moduli to a complete policy tolerance. A minimum of Lipschitz cones, realized exactly by ReLU gates, fits each date’s own-future Bellman labels without inflating its Lipschitz bound when those labels are inexact. The state, action, and arithmetic budgets are specified before construction. This yields finite termination for a defined neural backend rather than a claim about arbitrary nonlinear training. A new full matched catalogue executes the same resolution ladder for this backend and a conventional spline, including every unsuccessful rung and three isolated timing repetitions.

The action that attains a Bellman label supplies additional information. We retain its index through the ReLU minimum circuit and deploy that action when its cone is selected. A one-sided policy sandwich then avoids charging the separate actor and critic accounts twice. Both the nearest-node neural ablation and the spline receive a strengthened certificate. Their complete matched frontiers isolate certificate sharpness from first-target work and actual policy cost.

A feasible policy must also survive the information and arithmetic used to implement it. We give a uniform acquisition-aware theorem that permits discontinuous witness selection, uncertain measured states and capacity-dependent repair. A coupled two-state investment economy realizes all covering, integration and repair operations explicitly, with a conventional fitted-value comparator. The original scalar witness policies receive direct expected-cost comparisons at every common first crossing, under a tolerance and simultaneous error account fixed before execution.

This establishes proximity within the declared decision band, not a signed neural ranking. A catalogue-choice theorem makes the result usable when implementation charges differ. The complete resource curves and an identical native-envelope control isolate what the construction, the certificate and the neural representation each contribute.

Three further results connect this construction to computation and economic implementation. First, a literal cone network repeats a scan over every state label whenever a Bellman query evaluates its future continuation. On a tensor cover, we compile this network once and answer an arbitrary continuous-state query from the cell's corner data. The compiler carries the original action witness and the original tie rule. Consequently the improvement in evaluation work does not come from changing the policy being evaluated. Classical separable distance transforms supply the nodal computation; the additional policy argument establishes the continuous off-grid identity and feasible witness realization. In the two-state construction this removes the redundant factor that makes dense cone evaluation scale as  $N^6$  rather than  $N^4$ . It does not remove the state cover or give a capability unavailable to the identical native envelope.

Second, we make observation accuracy a priced economic input. A finite-bit sensor reports a state cell. Robust repair selects an action feasible throughout that cell. The policy theorem then gives an augmented loss account consisting of the full-information construction bound, an acquisition term decreasing geometrically in the number of bits, and a linear information charge. Exact neighboring-bit comparisons identify the precision minimizing this guarantee. This is an information-design result for the stated technology, not an assertion that minimizing a bound also minimizes the unknown actual loss.

Third, we evaluate actual constrained-policy cost differences, independently of their all-state regret certificates. The two installed controllers face common continuous innovations and the same observation technology. A prospective replacement charge turns simultaneous expected-cost intervals into a decision about whether either replacement recoups its cost. That question differs both from the sign of the policy difference and from exact equality. A complete matched construction study, an error-driven conventional comparator, and higher-dimensional nonlinear stress

1 executions accompany the decision analysis. We charge compilation, adaptive pilots, 1  
 2 unsuccessful resolutions, repair and output, and report the joint policy-comparison 2  
 3 service separately. These distinctions permit a positive computational contribution 3  
 4 to the original NBO program without confusing a sharper certificate, a cheaper 4  
 5 representation, and a better economic policy. 5

6 The numerical implementation starts from economic primitives. In the scalar 6  
 7 investment economy, it fits a fresh ReLU continuation backward against targets 7  
 8 using its own previously fitted future network. In the two-capital economy, it trains 8  
 9 hidden locations in a convex squared-ReLU continuation, with continuous simplex 9  
 10 actions and continuous uniform innovations. The second experiment includes 10  
 11 a flexible nonnegative ridge dictionary as well as a quadratic projection. Each 11  
 12 comparator constructs its own future targets and uses the same final economic 12  
 13 criterion. Neither conventional values nor inherited neural weights are used as 13  
 14 labels for the neural services. 14

15 We report the complete 150-service catalogue, not only successful candidates. In 15  
 16 the original capped scalar services the neural procedure reaches the all-state target 16  
 17 0.06 in all twelve cases and 0.04 in six, whereas the spline reaches both targets in 17  
 18 every case. In the coupled services the neural procedure reaches 0.25 in six cases 18  
 19 and 0.10 in five; the flexible ridge reaches both in all six. These statements describe 19  
 20 the original service, including its original verification cap. 20

21 A subsequent diagnostic holds both the neural weights and the deployed policies 21  
 22 fixed. Refining only residual verification certifies all six previously unresolved scalar 22  
 23 policies at 0.04 and the previously unresolved coupled policy at 0.10. The original 23  
 24 failures are not relabeled. The diagnostic shows that they do not, by themselves, 24  
 25 identify a failure to construct an accurate neural policy. It does not identify the 25  
 26 population success probability of the optimizer, establish a new end-to-end runtime, 26  
 27 or change any policy’s cost. 27

28 The comparative findings are equally integral to the paper. The original scalar 28  
 29 signed policy comparisons frequently favor the spline. The new coupled direct 29  
 30 intervals all contain zero and therefore do not order the policies. The precision 30  
 31 controller genuinely escalates after rejected low-precision proposals, but its com- 31  
 32 plete recorded work does not improve on fixed binary64 in this experiment. These 32

findings distinguish the certified capability of the construction from a comparative advantage that would require additional evidence.

The paper retains its original economic subject and full theory. The exposition now follows the economic problem, the operator, the policy theorem, and the evidence. The complete previous article, proof supplement, and economic applications are materialized unchanged within the current source tree as preserved companions; the preservation map records their exact identities. No historical theorem or application is deleted to obtain a shorter active exposition. The present article and new technical supplement state and prove the additional results used by the new evidence.

### 1.1. *Relation to the literature*

Neural Bellman Operators sit within approximate dynamic programming, fitted-value methods, and policy iteration, not outside them. Regression-based continuation methods and simulation-based dynamic programming supply important conventional benchmarks (Judd, 1998, Powell, 2011, Longstaff and Schwartz, 2001, Ernst et al., 2005). Actor–critic and approximate policy-iteration methods separate evaluation and improvement (Konda and Tsitsiklis, 1999, Scherrer et al., 2015); safe-improvement arguments likewise distinguish a proposed action from a verified gain (Kakade and Langford, 2002, Pirodda et al., 2013). We use these distinctions to organize the economic object returned by the algorithm.

Neural methods for differential equations provide ways to approximate values and their derivatives (Sirignano and Spiliopoulos, 2018, Han et al., 2018, Hornik et al., 1990). Such approximation does not on its own verify a returned dynamic policy or its numerical execution. Our certificates can be applied to neural or conventional continuations. The neural-specific construction lies in fresh own-future training and, in the coupled experiment, learned hidden locations. Its comparative value must be isolated against flexible conventional methods, not inferred from the certificate’s validity.

The general economic formulation admits recursive utility (Epstein and Zin, 1989, Duffie and Epstein, 1992). The compact-domain policy theorem uses monotonicity and cash invariance, while the original Gaussian entropic construction has separate

moment and spectral conditions. These are different bridges, not interchangeable assumptions. The direct paired-cost identity below applies to conditional expectation; it is not asserted for a nonlinear certainty equivalent. The confidence step uses the empirical Bernstein inequality of [Maurer and Pontil \(2009\)](#). Our additional ingredients are the policy-specific telescoping score, its certificate-derived support, and the numerical enclosure of the original continuous-law comparison.

## 2. THE CONTROLLED ECONOMY AND THE OPERATOR

### 2.1. Policy-specific continuation

The original continuous-time economy has an admissible control  $a$  and state

$$dS_t = b(t, S_t, a_t) dt + \sigma(t, S_t, a_t) dW_t. \quad (1)$$

The feasible correspondence, invariant or stopped domain, and exit and terminal rewards are economic primitives. For recursive evaluation, the continuation of a fixed policy is represented, on its specified horizon, by

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v.$$

The value is  $V^a(t, s) = Y_t^a$  under the well-posedness and comparison hypotheses in the retained controlled-economy analysis. In particular, the recursive argument is the policy's own continuation. Replacing it by the optimal continuation would change the policy-evaluation problem.

The applications companion retains the full boundary specifications, infinite-horizon transversality conditions, utility-domain restrictions, temporal-self equations, and game conditions. A finite-domain residual calculation does not establish any of these by itself. In a game, certifying a specified deviation problem is not the same as establishing the equilibrium fixed point. In continuous time, the controlled generator contains state derivatives; freezing a critic during an actor update does not suppress legitimate derivatives of the represented state-dependent function.

For the constructive implementation and the main policy theorem, consider dates  $t = 0, \dots, T$ , compact state space  $X$ , and feasible correspondence  $A_t(x)$ . States evolve

as

$$X_{t+1} = F_t(X_t, a_t, Z_{t+1}). \quad (2)$$

Let  $c_t$  be cost,  $g$  terminal cost,  $\beta_t > 0$ ,  $B_0 = 1$ , and  $B_t = \prod_{j < t} \beta_j$ . We use a cost-minimization convention in the discrete implementation. It is equivalent to changing signs in an additive reward formulation, but does not replace the original recursive-utility assumptions by a sign convention.

Write

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z_{t+1}))), \quad (3)$$

$$\mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a), \quad V_T^* = g.$$

The conditional evaluations  $\rho_t$  are monotone and cash invariant on a domain containing every value and comparison function used below. Feasible measurable selectors exist; compactness, continuity of primitives, and continuity of the evaluated integrals are sufficient conditions in the experiments. The Bellman recursion is valid against all admissible adapted policies. For expectation,  $J^\pi$  is expected discounted cost. For a recursive evaluation it denotes the value generated by that evaluation's own recursion.

## 2.2. An executable NBO service

An implemented operator maps economic data and a current policy to

$$(\hat{f}, \hat{\pi}, \mathcal{C}, W) \quad \text{or} \quad (\text{unresolved}, W, \text{diagnostic}),$$

where  $\mathcal{C}$  is an economic certificate and  $W$  records the work used. Its stages are target construction, fitting, feasible action selection, independent certification, and durable storage. Critic fitting freezes its future inputs; action selection freezes the resulting critic. In the fresh backward construction, date  $t$  uses the already fitted  $\hat{f}_{t+1}$ , not an externally solved optimal continuation. The terminal target is the primitive  $g$ .

A finite optimizer schedule is an algorithm, not a convergence theorem. A service that has not met its target at its cap records failure, with the complete cost of its attempts. The retained factor-training analysis provides a finite constructive

theorem for a separate structured class. The nonlinear schedules below instead have an explicit, auditable capped-service estimand.

For a fixed catalogue  $\mathcal{K}$  and target  $\varepsilon$ , define

$$H_m(k, \varepsilon) = \mathbf{1}\{\text{method } m \text{ stores a valid certificate at } \varepsilon \text{ within its cap on object } k\}. \quad (4)$$

The empirical object is the full array of  $H_m$ , costs to the first qualifying checkpoint, and capped costs for failures. Its arithmetic average is a catalogue fraction. A probability over random initializations would require a specified law and a sampling design for that different estimand. A post-review refinement of a failed object's verifier is a new diagnostic and does not change (4) for the original service.

### 3. CONTINUATION ACCURACY, TRANSPORT, AND POLICY LOSS

#### 3.1. Centered errors and economic reuse

Consider a current objective  $r_\theta(s, a) + S_\theta(T_\theta(s, a))$  to be maximized. Let  $e = \hat{S}_0 - S_0$  be anchor approximation error and  $\Delta_\theta = S_\theta - S_0$  the change in the actual future continuation. Suppose the implemented action is  $\delta$ -optimal for  $r_\theta + \hat{S}_0 \circ T_\theta$ .

**PROPOSITION 1**—Centered decision and transport bounds: *The current decision loss satisfies*

$$L_\theta(s, \hat{a}) \leq \delta + \text{osc}_{A_\theta(s)}(e \circ T_\theta) + \text{osc}_{A_\theta(s)}(\Delta_\theta \circ T_\theta). \quad (5)$$

For a common finite action menu with positive weights  $p_j$ , set  $R_p = \sum_j p_j(e_j - \sum_i p_i e_i)^2$  and  $C_p = \max_{i \neq j}(p_i^{-1} + p_j^{-1})$ . If  $|\Delta_\theta| \leq b_\theta$  on the queried postdecision set, then  $L_\theta \leq \delta + \sqrt{C_p R_p} + 2b_\theta$ .

If common-innovation future transitions have Lipschitz constants  $L_k$  and perturbation bounds  $\epsilon_{\theta,k}$ , and future rewards have Lipschitz constants  $H_k$  and perturbations  $\eta_{\theta,k}$  on an invariant domain, put  $d_0 = 0$ ,  $d_{k+1} = L_k d_k + \epsilon_{\theta,k}$ . For nonnegative reward weights  $w_k$ ,

$$|S_\theta - S_0| \leq b_\theta := \sum_k w_k(\eta_{\theta,k} + H_k d_k). \quad (6)$$

The proof compares the true and fitted objective at the two selected actions and couples the future states. The weighted finite-menu bound is Cauchy–Schwarz

applied to  $e_i - e_j$ . Details are in the technical supplement. An additive continuation error common to all actions is irrelevant to the decision. A finite menu, however, does not certify a continuous action set merely because it includes both observed actions.

For recursive maps with sup-norm Lipschitz constants  $\beta_k$  on a common invariant utility domain, a one-step perturbation  $\nu_k$  propagates as  $b_k \leq \nu_k + \beta_k b_{k+1}$ . This is the recursive counterpart of (6). Its use requires the common domain; it cannot cross a singular utility boundary by extending a local constant. These transport bounds retain the original refresh logic: use an anchor while its economic allowance covers approximation and future change, and recertify or recompute when that condition is exhausted.

### 3.2. A direct all-state policy theorem

For continuous fitted functions  $f_t$  and a feasible measurable policy  $\pi$ , suppose the verifier supplies, uniformly in  $x$ ,

$$\ell_t \leq f_t(x) - \mathcal{T}_t f_{t+1}(x) \leq u_t, \quad t < T, \quad (7)$$

$$Q_t(f_{t+1}; x, \pi_t(x)) - \mathcal{T}_t f_{t+1}(x) \leq \eta_t,$$

$$\ell_T \leq f_T(x) - g(x) \leq u_T.$$

These are enclosures for the actual coefficients and actual returned actions. Neither an optimizer's success flag nor a residual on sampled states replaces them.

**THEOREM 2**—Centered full-policy certificate: *Under the conditions of Section 2 and (7), every initial state satisfies*

$$0 \leq J_0^\pi(x) - V_0^*(x) \leq G(f, \pi) := \sum_{t < T} B_t(u_t - \ell_t + \eta_t) + B_T(u_T - \ell_T). \quad (8)$$

More specifically,

$$V_0^*(x) \geq f_0(x) - \sum_{t < T} B_t u_t - B_T u_T, \quad (9)$$

$$J_0^\pi(x) \leq f_0(x) + \sum_{t < T} B_t(\eta_t - \ell_t) - B_T \ell_T.$$

Replacing  $f_t$  by  $f_t + d_t$ , with date-specific constants  $d_t$ , leaves the bound unchanged after the residual endpoints are shifted by  $d_t - \beta_t d_{t+1}$  and the terminal endpoints by  $d_T$ .

PROOF: Monotonicity and cash invariance imply  $\mathcal{T}_t(v + c) = \mathcal{T}_t v + \beta_t c$  and the analogous identity for the fixed-policy map. Starting from  $g \geq f_T - u_T$ , backward induction gives the first inequality in (9). Starting from  $g \leq f_T - \ell_T$  and using  $Q_t(f_{t+1}; x, \pi_t(x)) \leq f_t(x) - \ell_t + \eta_t$  gives the second. Subtraction proves the upper bound; optimality proves the lower bound. The shifted residuals have unchanged widths and unchanged action gaps. *Q.E.D.*

The theorem certifies a full adapted policy, not a finite action menu or a grid of initial states. With expected cost its proof is a Bellman telescoping argument; for a monotone cash-invariant recursive evaluation it is an order argument. The numerical experiment is not needed for its validity. Its numerical usefulness depends on producing small, valid enclosures.

COROLLARY 3—Retained-policy recertification: *Fix the functions  $f_t$ , the deployed policy  $\pi$ , valid actor allowances  $\eta_t^{\text{old}}$ , and a terminal enclosure. Let a second verifier return valid residual enclosures  $[\ell_t^{\text{new}}, u_t^{\text{new}}]$  for those same functions. Then*

$$J_0^\pi - V_0^* \leq \sum_{t < T} B_t(u_t^{\text{new}} - \ell_t^{\text{new}} + \eta_t^{\text{old}}) + B_T(u_T^{\text{old}} - \ell_T^{\text{old}}). \quad (10)$$

*No retraining or change in deployed actions is required. If desired, intersecting two valid residual enclosures makes the new bound no larger than the old one.*

The original actor allowance is valid on the entire state domain, so a finer critic verification mesh does not invalidate it. Conversely, replacing it by an allowance for a newly optimized fine-grid actor would certify a different policy. The implementation records the original actor hash and explicitly discards any fine-grid action proposals generated internally by the verifier. Those proposals' work and transient storage are still charged.

The corollary does not promise that arbitrary precision is attainable by mesh refinement. Nodal residual variation, the retained actor allowance, and the terminal enclosure can leave a positive floor. Nor is the corollary a claim that additional

verification improves welfare. It distinguishes the accuracy of a fixed economic decision rule from the sharpness of our knowledge of that accuracy.

### 3.3. Construction beyond post-training verification

The original trainable-factor construction remains a constructive part of NBO. For a quadratic own-policy continuation  $x'P_tx + c_t$ , a square-activation network  $\|W_tx\|^2/d + \hat{c}_t$  trains every entry of  $W_t$ . Its population derivative loss and gradient are

$$\mathcal{L}_t(W; P) = \frac{1}{4} \|W'W - dP\|_F^2, \quad \nabla_W \mathcal{L}_t = W(W'W - dP).$$

If  $\sigma_{\min}(W) \geq s > 0$ , then  $\|W'W/d - P\|_F \leq \|\nabla_W \mathcal{L}_t\|_F / (ds)$ . Together with the full-vector actor defect, this gives a policy-error account for a jointly trained hidden factor. The own-policy coefficient is re-evaluated for the returned policy, not silently borrowed from an earlier training policy.

The primitive-budget analysis in the retained article chooses a feasibility witness, coefficient envelopes, rank margins, and update budgets before training. In the Gaussian entropic extension it also verifies positive exponential-moment margins. Under its stated spectral hypotheses, a scaled orthogonal initialization and bounded gradient steps give the finite training construction proved there. The finite matrix and Gaussian formulas are not a nonlinear dimension-scaling experiment. They also do not imply that Adam or L-BFGS has a deterministic residual cap for a general squared-ReLU network. We preserve both results and keep the two logical statements separate.

## 4. FROM ECONOMIC PRIMITIVES TO NONLINEAR CERTIFICATES

### 4.1. Backward fitting and selected policies

For each method, initialize a fresh terminal fit from  $g$ . At date  $t$ , construct targets from  $\mathcal{T}_t f_{t+1}$  using that method's own already fitted future. Fit the current continuation under the frozen schedule. Then propose feasible actions, enclose the Bellman infimum and the selected-action gap, and validate residual coverage. Store a checkpoint before reporting its first target crossing. Continue through the declared ladder only as specified by the original study. The record includes all

attempted resolutions, unsuccessful target queries, target-generation work, and storage operations.

The scalar network has 63 ReLU features. Its training uses 513 states, a 257-point target action construction, 400 Adam steps at learning rate  $5 \times 10^{-5}$ , and a final output fit. The scalar verifier uses its own finer action and state ladders. The conventional spline is reconstructed from the same primitives with its own ladder and targets. A conventional target solver is not used to label a neural continuation.

The coupled neural continuation is

$$f_t(x) = a_t + b'_t x + \sum_{j=1}^{48} c_{tj} (w'_{tj} x + v_{tj})^2_+, \quad c_{tj} \geq 0. \quad (11)$$

Hidden locations and weights are trained. Nonnegative coefficient initialization and a finite L-BFGS schedule of 300 iterations act on fresh own-future targets. The ridge comparator instead fixes a 96-feature squared-hinge dictionary and fits nonnegative coefficients to its own targets; zero coefficients may be pruned. Its dictionary seed is fixed by date, so repeated training-seed labels do not create independent random ridge representations. A quadratic projection is retained as a restricted comparator, not treated as the strongest conventional alternative.

#### 4.2. All-state and continuous-action verification

At each state node, outward arithmetic encloses the actual network value and the action objective. In the coupled economy, convexity in actions and a positive strong-convexity constant yield a continuous-simplex primal–dual gap. A projected optimizer merely proposes the action; its stopping flag is not the certificate. State interpolation then extends the signed residual bounds off-grid. The nearest-node deployed actor receives a separate, uniform allowance for using its node action at the actual state.

Uniform innovations are integrated through positive-part moments, rather than replaced by a sampled innovation law. The elementary identity for a uniform variable  $U$  on  $[-h, h]$  is

$$\mathbb{E}(y + U)^2_+ = \frac{(y + h)^3_+ - (y - h)^3_+}{6h}, \quad h > 0.$$

Successive integration gives the corresponding formula for independent coordinates and affine neuron arguments; zero coefficients are handled by the lower-dimensional formula. Interval evaluation encloses cancellation and rounding. The supplement specifies the mesh and action bounds used in the inherited verifier.

The residual account distinguishes a nodal enclosure from state coverage, actor error, and terminal error. A nodal enclosure contains the effects of representation, fitting, Bellman optimization, and arithmetic; it is not an identified decomposition into a pure approximation error and a pure optimizer error. In particular, subtracting a training loss from the residual does not estimate either component. The diagnostic reports every separately available allowance without counting nested subcomponents twice.

#### 4.3. Work, dimension, and execution

For tensor state, action, and innovation covers of sizes  $K_x, K_a, m$ , a direct verifier pays work of order

$$\sum_{t < T} K_x K_a \{m(C_F + P_{t+1}) + C_\rho\} + \sum_{t \leq T} K_x P_t + W_{\text{storage}}, \quad (12)$$

where  $P_t$  counts network evaluation work. Actor storage is of order  $TK_x d_a$ . Analytic integration replaces  $m$  by the work of the moment formula, not by zero. With tensor covers,  $K_x$  scales as  $h_x^{-d_x}$ ; analogous factors arise for action and innovation discretizations. A dimension-explicit certificate is therefore not a dimension-free algorithm.

If a Lipschitz innovation approximation contributes at most  $\zeta_t$  to  $Q_t$ , widening residual endpoints by  $\zeta_t$  and the actor allowance by  $2\zeta_t$  adds  $4 \sum_t B_t \zeta_t$  to (8). This is a discretization account, not permission to replace a continuous law without charging its error. Independent policy queries may have their own scenario-tree or simulation cost and are recorded separately.

The complete service includes own-future targets, fitting, unsuccessful refinement, action extraction, verification, applicable policy queries, serialization, and durable checkpoint storage. Initialization and outer-process time are retained separately. New local diagnostic clocks are not added to old GitHub-hosted clocks to

manufacture an end-to-end estimate. All reported arithmetic certificates concern stored binary coefficients with outward enclosures. The inherited execution account treats rounded state acquisition and transition/cost evaluation separately from ideal policy loss.

## 5. DIRECT COMPARISONS OF ECONOMIC POLICY COSTS

A smaller upper bound on  $J^\pi - V^*$  need not mean a smaller  $J^\pi$ . We therefore compare the neural and ridge policies directly at common states, using common innovations and the expectation version of (3).

For method  $m$ , let  $X^m$  follow its own fixed policy and define

$$Z_m = f_0^m(x_0) + \sum_{t < T} B_t \{Q_t(f_{t+1}^m; X_t^m, \pi_t^m(X_t^m)) - f_t^m(X_t^m)\} + B_T \{g(X_T^m) - f_T^m(X_T^m)\}. \quad (13)$$

This is a control-variate score for actual policy cost, not the difference of two regret bounds.

**THEOREM 4**—Paired economic score and support: *For fixed policies and integrable continuations under conditional expectation,  $\mathbb{E}Z_m = J_0^{\pi^m}(x_0)$ . If method  $m$  satisfies (7), then every trajectory satisfies*

$$L_m \leq Z_m \leq U_m, \quad (14)$$

$$L_m = f_0^m(x_0) - \sum_{t < T} B_t u_t^m - B_T u_T^m,$$

$$U_m = f_0^m(x_0) + \sum_{t < T} B_t (\eta_t^m - \ell_t^m) - B_T \ell_T^m.$$

Consequently  $Z_N - Z_R$  has expectation  $J^{\pi^N} - J^{\pi^R}$  and support  $[L_N - U_R, U_N - L_R]$ , whose width is  $G_N + G_R$ . These statements hold under any coupling of the two methods that preserves each policy's innovation law.

**PROOF:** Conditional expectation gives  $\mathbb{E}Q_t(f_{t+1}^m; X_t^m, \pi_t^m) = \mathbb{E}c_t + \beta_t \mathbb{E}f_{t+1}^m(X_{t+1}^m)$ . Summing telescopes all continuation terms. The selected-action gap is between zero and  $\eta_t^m$ , so each centered summand in (13) is between

$-u_t^m$  and  $\eta_t^m - \ell_t^m$ . The terminal summand is between  $-u_T^m$  and  $-\ell_T^m$ . Summing  
proves the support statement and its width. Q.E.D.

To evaluate the original continuous law, simulate uniform innovation bins and  
enclose all within-bin innovations. Interval propagation includes every actor that  
can be selected when a state enclosure crosses a nearest-node boundary. Suppose the  
resulting endpoint pair satisfies  $A_i \leq Z_{N,i} - Z_{R,i} \leq D_i$  and lies in a known support  
interval of width  $B$ . For independent draws of bins, the lower and upper endpoints  
are bounded independent observations. Let  $\bar{A}, \bar{D}$  and  $s_A^2, s_D^2$  be their sample means  
and variances. For  $q$  predeclared comparisons, set

$$r(s^2) = \sqrt{\frac{2s^2 \log(4q/\alpha)}{n}} + \frac{7B \log(4q/\alpha)}{3(n-1)}. \quad (15)$$

Applying the one-sided empirical Bernstein inequality to the lower and upper  
endpoints separately, then taking a union bound, gives simultaneous coverage at  
least  $1 - \alpha$  for

$$[\bar{A} - r(s_A^2), \bar{D} + r(s_D^2)] \quad (16)$$

over all  $q$  comparisons. Mean, variance, logarithm, and square-root calculations  
are outward enclosed in the implementation; intersecting with the deterministic  
support preserves coverage. The proof and numerical details are in the supplement.

The coverage statement is conditional on frozen training outputs and an in-  
dependent simulation model. Fixed PCG64 streams make this implementation  
reproducible; a fixed stream is not itself a proof of probabilistic independence. No  
policy is selected using these simulations. Intervals crossing zero are reported as  
unresolved, not as equivalence or as evidence of a superior neural policy.

## 6. INVESTMENT ECONOMIES AND THE COMPLETE EVIDENCE

### 6.1. Design and provenance

The original science design has twelve scalar neural services, twelve scalar  
spline services, six coupled neural services, six coupled ridge services, six coupled  
quadratic services, and 36 services for each of three precision rules. The successful

source-bound execution contains all 150 records. The original first run, with four nonfinite-network exceptions, is retained. Its numerical inverse-softplus initialization was subsequently replaced by an algebraically equivalent stable expression; economic primitives, seeds, targets, budgets, and comparators were unchanged. We do not treat the failed first run as nonexistent or combine its clocks with the successful run.

The R44 protocol was committed before the additional diagnostics were executed. It is a post-review protocol, not an original preregistration. It selects the seven unresolved tight-target neural objects by their already observed outcomes and freezes their weights and policies. The paired comparison uses all six coupled pairs and all four originally declared states. A source manifest binds the original inputs, the diagnostic code, outcomes, and publication sources. Every original record and all five original science-source hashes were replayed.

## 6.2. *Scalar and coupled primitives*

The scalar experiment retains the original bounded investment state, price and risk cells, three declared training seeds, continuous feasible action interval, and the independent own-policy evaluator. Prices are one and four. Both methods construct their own continuations from the same primitive problem. The retained evidence and code specify every scalar primitive; the full economic derivation remains in the original investment section and applications.

In the coupled economy,  $x \in [0, 1]^2$ ,  $T = 4$ ,  $\beta = 15/16$ , and  $A = \{a \geq 0 : a_1 + a_2 \leq 1/8\}$ . With independent  $z_j \sim U[-1/32, 1/32]$ ,

$$F_j(x, a, z) = \frac{1}{8} + \frac{1}{2}x_j + \frac{1}{8}x_{3-j} + \frac{1}{16}x_j(1 - x_{3-j}) + a_j + z_j.$$

This transition maps the stated state-action-innovation domain into the state domain. Define

$$s(x) = \sum_{j=1}^2 (x_j - 11/16)^2 + 2(1/2 - x_1 - x_2)_+^2 + \frac{1}{4}(x_1 - x_2)^4.$$

The stage cost is  $s(x) + p \sum_j a_j^2 + \frac{1}{2}a_1a_2 + 4 \sum_j a_j^4$ ; terminal cost doubles the squared-target term in  $s$  and retains its remaining terms. Prices  $p \in \{1, 4\}$  penalize costly

investment. The shared simplex represents a common resource constraint, the transition couples the two capital stocks, and the state cost penalizes aggregate shortage and imbalance. This is a stylized controlled economy, not an empirically calibrated welfare exercise.

For both coupled methods the final all-state verification ladder is  $N = 16, 32, 64, 128$ . The action objective has strong-convexity lower bound  $2p - 1/2 > 0$ . Uniform positive-part moments integrate continuous uncertainty analytically. The finest original actor has  $4(129)^2 \cdot 2 = 133,128$  stored scalar actions. This actor cost is distinct from continuation-network storage.

### 6.3. Original attainment and comparative outcomes

TABLE I

ORIGINAL CAPPED-SERVICE TARGET ATTAINMENT

| Method            | All-state target | Certified |
|-------------------|------------------|-----------|
| Scalar neural     | 0.06             | 12/12     |
| Scalar neural     | 0.04             | 6/12      |
| Scalar spline     | 0.06             | 12/12     |
| Scalar spline     | 0.04             | 12/12     |
| Coupled neural    | 0.25             | 6/6       |
| Coupled neural    | 0.10             | 5/6       |
| Coupled ridge     | 0.25             | 6/6       |
| Coupled ridge     | 0.10             | 6/6       |
| Coupled quadratic | 0.25             | 0/6       |
| Coupled quadratic | 0.10             | 0/6       |

Table I reports the original capped-service outcomes. The original scalar neural bound is larger than its matched spline bound in every one of the twelve pairs. At the first common 0.06 crossing, 29 of 48 signed neural-minus-spline intervals have positive lower endpoints, none has a negative upper endpoint, and 19 contain zero. At 0.04, the six common pairs give 17 positive intervals, none negative, and seven overlaps. All lie within the prespecified descriptive margin  $\pm 0.005$ . This establishes

numerical closeness at those states but is not an equivalence design; where signs are identified, they favor the spline.

The coupled ridge reaches both targets in all six services. It has the smaller final gap in four pairs and the neural method in two. At 0.25 the ridge reaches the target faster in all six original observations. At 0.10 it is faster in four of the five pairs in which both pass, while the neural method is faster in one and fails in the remaining pair. These singleton nonlinear timing observations are descriptive. The ridge continuation has storage comparable to the neural continuation, so its strength cannot be dismissed as a comparison with an artificially restricted quadratic.

#### 6.4. Diagnosing tight-target failures without changing policies

TABLE II

ADDITIONAL VERIFICATION OF UNCHANGED NEURAL POLICIES

| Service | Original bound | Refined bound | Target | Attained |
|---------|----------------|---------------|--------|----------|
| 041     | 0.04092128     | 0.03068472    | 0.04   | Yes      |
| 089     | 0.04118465     | 0.03090917    | 0.04   | Yes      |
| 120     | 0.04112077     | 0.03085353    | 0.04   | Yes      |
| 124     | 0.04093160     | 0.03070697    | 0.04   | Yes      |
| 128     | 0.15223570     | 0.09775755    | 0.10   | Yes      |
| 145     | 0.04136567     | 0.03108538    | 0.04   | Yes      |
| 147     | 0.04101560     | 0.03077643    | 0.04   | Yes      |

For the six unresolved scalar policies, only the residual state grid is refined from  $N = 2048$  to  $N = 4096$ , retaining the original action verification resolution  $A = 1024$ . For the coupled policy, residual verification uses  $N = 256$  instead of 128. The original actor allowances and terminal enclosures remain in (10); all original action arrays and networks are hash-checked unchanged. Fine-grid actor proposals generated internally are discarded, not deployed.

Table II gives all seven results. The retained scalar policy bounds become approximately 0.03068–0.03109, below 0.04. The retained coupled policy bound becomes 0.09775756, below 0.10. There are zero training updates and zero newly stored

deployed-action scalars. The coupled diagnostic still pays for a fine verification mesh with 66,049 states, including transient fine-grid proposals; retaining the original actor does not make verification free.

The original coupled decomposition is informative. Its discounted state-cover contribution to nonterminal residual widths is about 0.07268224, its actor allowance is 0.03287249, its remaining nodal residual enclosure is 0.04574738, and its terminal contribution is 0.00093360. These sum to the original bound up to displayed rounding. The extra mesh quarters the principal second-order coverage term while retaining the actor and terminal terms. The resulting crossing does not establish that representation and optimization errors are separately small; it establishes the required full-policy bound for the unchanged policy.

The diagnostic answers a specific method question. The original capped service was less reliable at its tight target, but the saved neural objects were not necessarily inadequate policies. All seven selected policies admit the tighter certificates when verification is extended under a fixed rule. This finding does not alter the original attainment table or justify a comparison between a selectively refined neural certificate and an unrefined comparator as evidence of a work advantage.

### 6.5. *Direct coupled economic differences*

TABLE III

DIRECT NEURAL-MINUS-RIDGE COST COMPARISONS, SIMULTANEOUS 99 PERCENT INTERVALS

| Price | Comparisons | Positive | Negative | Unresolved |
|-------|-------------|----------|----------|------------|
| 1     | 12          | 0        | 0        | 12         |
| 4     | 12          | 0        | 0        | 12         |

For each of the six original neural/ridge pairs, select each method's first 0.25 crossing. Compare at  $(1/8, 1/8)$ ,  $(1/4, 1/2)$ ,  $(1/2, 1/4)$ , and  $(3/4, 3/4)$ . Each comparison uses 65,536 common innovation-bin trajectories, 40-bit uniform bins, and the own-policy score in (13). The total family error is 0.01 across 48 one-sided bounds. The complete 24 intervals, endpoint moments, seeds, ambiguity counts, and support bounds appear in the supplement and machine-readable records.

All 24 direct confidence intervals contain zero. Thus the coupled evidence now addresses the actual neural-minus-ridge policy-cost difference, but does not rank the methods. This is stronger information than subtracting regret bounds and different information from a smaller all-state certificate. The simultaneous interpretation pertains to the declared states and fixed policies under the simulation model, not to every state in the compact domain. The all-state policy guarantees remain the separate deterministic statements in Table I and Theorem 2.

### 6.6. Precision and complete recorded work

TABLE IV  
ORIGINAL REPEATED PRECISION CATALOGUE

| Method   | Certified | Accepted32 | Rejected32 | Accepted64 |
|----------|-----------|------------|------------|------------|
| Fixed32  | 6/36      | 147        | 30         | 0          |
| Fixed64  | 36/36     | 0          | 0          | 474        |
| Adaptive | 36/36     | 348        | 126        | 126        |

The precision experiment repeats twelve dimension/conditioning/target objects three times. Fixed binary64 and adaptive each certify all 36 services; fixed binary32 certifies six. Fixed64 and adaptive each make 474 accepted updates. Adaptive makes 348 accepted binary32 updates, 126 rejected binary32 proposals, and 126 accepted binary64 updates. Every rejection is charged.

Adaptive is slower in 26 of the 36 matched recorded services and faster in ten; aggregate recorded time is about 0.98% larger. The controller’s activation therefore demonstrates its numerical mechanism, not a complete-work saving in this implementation. This result does not invalidate the conditional numerical acceptance theorem. It prevents the theorem from being mistaken for an empirical efficiency conclusion.

### 6.7. What the evidence identifies

The evidence identifies a finite from-primitives nonlinear construction, all-state certificates for its returned policies, a useful retained-policy verification refinement,

and direct economic comparisons under the declared state and innovation design. It also identifies the continued strength of conventional comparators. It does not identify a general neural speed advantage, a nonlinear high-dimensional frontier, or a population probability of optimizer success.

An adaptive sparse-grid or adaptive-partition comparator has not been executed for the original neural optimizers. Section 15 adds a separate error-driven coordinate-FVI comparator and three- and four-dimensional stress catalogue for the constructive backend; neither retroactively changes the optimizer study. Neither has a nonlinear dimension frontier or repeated isolated timing of every original scalar and coupled service. Section 7.3 adds a separate predeclared horizon/price catalogue with repeated isolated timing for the deterministic constructive backend and spline; it does not replace the original experiments. The retained dimension-explicit analysis exposes the tensor factors those studies would have to confront, rather than treating a two-state experiment as their substitute. The general economic framework and conditional constructive results remain in force; the missing experiments concern comparative evidence for broader implementation claims.

## 7. A CONSTRUCTIVE NEURAL SERVICE WITH A PRIMITIVE ERROR BUDGET

The policy theorem is deliberately independent of the representation of the continuation. A construction theorem must do more: it must produce a representation, feasible actions, and verified residuals from economic primitives. This section supplies such a theorem for a deterministic ReLU backend. It does not assume convergence of the nonlinear optimizers used in the earlier experiments. Its distinguishing property is that the continuation has a prescribed Lipschitz constant even when its Bellman labels are inexact.

### 7.1. Primitive conditions and the neural architecture

Consider the compact Bellman economy of Section 2, with a common action set  $A \subset \mathbb{R}^a$  and state set  $X \subset \mathbb{R}^d$ . All distances in this section use the  $\ell^1$  norm. The innovation law and the monotone, cash-invariant functional  $\rho_t$  do not depend on the state or action. The transition remains in  $X$ . Suppose known nonnegative

constants satisfy, uniformly in the innovation,

$$|c_t(x, a) - c_t(y, b)| \leq C_{x,t} \|x - y\|_1 + C_{a,t} \|a - b\|_1, \quad (17)$$

$$\|F_t(x, a, z) - F_t(y, b, z)\|_1 \leq F_{x,t} \|x - y\|_1 + F_{a,t} \|a - b\|_1. \quad (18)$$

The terminal cost is  $L_g$ -Lipschitz. These assumptions are additional conditions on the constructive backend, not a replacement for the more general feasible correspondences in the original economic formulation. A state-dependent feasible correspondence requires its own feasible-cover and transport constants.

We require an effective numerical description of the primitives: certified finite state and action covers can be obtained at every positive radius; their nodes are feasible; and a numerical evaluator can enclose each Bellman query to any requested positive tolerance in finite work. The last condition includes evaluation of  $\rho_t$ , not just evaluation of the network. For expectation with a compact innovation domain, known cell probabilities and a certified innovation cover provide one such evaluator. If  $F_t$  is  $F_{z,t}$ -Lipschitz in the innovation, replacing each cell by a representative at distance at most  $h_z$  contributes at most  $\beta_t L_{t+1} F_{z,t} h_z$ . Arithmetic, probability-mass approximation, and any remaining integration error must be added. The experiment below instead integrates a piecewise-linear network analytically under the original continuous uniform law.

Set

$$L_T = L_g, \quad L_t = C_{x,t} + \beta_t F_{x,t} L_{t+1}, \quad L_t^a = C_{a,t} + \beta_t F_{a,t} L_{t+1}. \quad (19)$$

Choose state nodes  $x_{t,i}$  of covering radius  $h_t$  and action nodes  $a_{t,j}$  of covering radius  $k_t$ . Given the continuation already constructed at date  $t + 1$ , compute numbers  $\hat{q}_{t,ij}$  satisfying

$$|\hat{q}_{t,ij} - Q_t(f_{t+1}; x_{t,i}, a_{t,j})| \leq e_t. \quad (20)$$

Define  $y_{t,i} = \min_j \hat{q}_{t,ij}$ , choose a minimizing action index  $j(t, i)$  with a fixed tie rule, and construct

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \quad (21)$$

At date  $T$ , use labels within  $e_T$  of  $g$  and the same formula with constant  $L_g$ . At any other date, deploy the action  $a_{t,j(t,i(x))}$  attached to a nearest state node, again with a fixed tie rule. The action is feasible by construction. No future optimal value or conventional comparator is used as a label.

Formula (21) is an exact ReLU construction, not an appeal to a universal approximation theorem. Absolute values use  $|u| = \max\{u, 0\} + \max\{-u, 0\}$ ; minima use  $\min\{u, v\} = u - \max\{u - v, 0\}$ . A balanced tree of these gates implements the finite minimum. With  $K_t$  state nodes, the resulting sparse affine-ReLU circuit has  $O(dK_t)$  coefficients and  $O(1 + \log K_t)$  depth with unrestricted affine fan-in. Signed intermediate values can be carried by their positive and negative parts. The construction is related to the classical Lipschitz extension formula of McShane (1934); neither that formula nor its elementary ReLU realization is claimed as new. Here its role is to close an own-future dynamic construction, including the deployed actor and every numerical allowance.

**THEOREM 5—Constructive own-future neural policy:** *Under the preceding conditions, the network (21) is  $L_t$ -Lipschitz at every date, independently of the numerical label errors. Its residual and its deployed actor satisfy*

$$-e_t \leq f_t - T_t f_{t+1} \leq L_t^a k_t + e_t + 2L_t h_t, \quad (22)$$

$$Q_t(f_{t+1}; x, \pi_t(x)) - T_t f_{t+1}(x) \leq L_t^a k_t + 2e_t + 2L_t h_t. \quad (23)$$

*At the terminal date,  $-e_T \leq f_T - g \leq e_T + 2L_g h_T$ . Consequently the cost of the returned policy, relative to the optimum over all admissible adapted policies, is uniformly bounded by*

$$\mathcal{G}_{\text{con}} = \sum_{t < T} B_t \{2L_t^a k_t + 4e_t + 4L_t h_t\} + B_T \{2e_T + 2L_g h_T\}. \quad (24)$$

*For fixed nodes and  $L_t$ , changing every label by at most  $\delta$  changes the represented function by at most  $\delta$  in the uniform norm; it does not change the Lipschitz bound.*

**PROOF:** Monotonicity and cash invariance imply nonexpansiveness in the uniform norm. Thus  $Q_t(f_{t+1}; \cdot, a)$  and  $T_t f_{t+1}$  are  $L_t$ -Lipschitz, while the action modulus is  $L_t^a$ . At a node, minimization of the numerical queries gives  $-e_t \leq$

1  $y_{t,i} - T_t f_{t+1}(x_{t,i}) \leq L_t^a k_t + e_t$ . Each cone in (21) is  $L_t$ -Lipschitz, as is their finite min- 1  
 2 imum. The lower residual bound follows from the triangle inequality for every 2  
 3 cone; the upper bound uses a nearest node twice, once for the cone and once 3  
 4 for the Bellman value. At that node the selected action is within  $L_t^a k_t + 2e_t$  of the 4  
 5 continuous-action infimum. Transport to  $x$  costs at most  $2L_t h_t$ . The terminal argu- 5  
 6 ment is identical without action minimization. Substitution of these widths and 6  
 7 actor allowances into Theorem 2 gives (24). Finally, minima of two arrays whose 7  
 8 entries differ by at most  $\delta$  differ by at most  $\delta$ . The supplement gives the construction, 8  
 9 measurability, and work details. Q.E.D. 9

10 COROLLARY 6—Deterministic termination at an economic tolerance: *Let  $\varepsilon > 0$ ,* 10  
 11  *$H = \sum_{t=0}^T B_t$ , and  $s = \varepsilon/H$ . At nonterminal dates choose* 11

$$12 \quad e_t \leq s/8, \quad L_t^a k_t \leq s/8, \quad L_t h_t \leq s/16. \quad (25) \quad 13$$

14 *At the terminal date choose  $e_T \leq s/4$  and  $L_g h_T \leq s/4$ . Terms with zero coefficient impose* 14  
 15 *no restriction. Then the finite constructive service returns a feasible neural policy with loss* 15  
 16 *at most  $\varepsilon$ . No probability law over initializations and no optimizer stopping criterion is* 16  
 17 *needed.* 17

18 18  
 19 The result follows because each discounted bracket in (24) is at most  $s$ . Effec- 19  
 20 tive compactness and the numerical evaluator give finite work for these choices. 20  
 21 Arbitrarily small tolerances may require arbitrary precision; the corollary does 21  
 22 not promise termination in fixed binary64. A finite implementation ladder can 22  
 23 still return a recorded failure, even though the unbounded constructive algorithm 23  
 24 terminates under the stated hypotheses. 24

25 The earlier scalar constructive result in the retained paper implemented piecewise- 25  
 26 linear interpolation as a ReLU network. Its sufficient allocation controlled a label- 26  
 27 error-to-mesh ratio in the propagated slope. Formula (21) works in arbitrary finite 27  
 28 state dimension and enforces the slope bound directly. In particular, label and 28  
 29 action allowances can be allocated at order  $\varepsilon$  without a separate slope-preservation 29  
 30 requirement that those allowances be quadratic in the state spacing. This is an 30  
 31 improvement of that sufficient proof budget, not a lower bound for all spline 31  
 32 methods or evidence that conventional Lipschitz projection is unavailable. 32

## 7.2. What the construction costs

The construction does not eliminate state coverage. With  $K_t$  state nodes,  $J_t$  action nodes,  $M_t$  innovation representatives, and network evaluation cost  $P_{t+1}$ , direct target formation requires

$$\sum_{t < T} K_t J_t \{M_t (C_{F,t} + P_{t+1}) + C_{\rho,t} + C_{c,t}\} + O\left(\sum_{t=0}^T d K_t\right) \quad (26)$$

arithmetic operations under the indicated evaluation model.  $C_{\rho,t}$  includes the finite-risk functional and probability processing. Certified integration can replace the innovation sum but its own cost must be charged. Bit lengths, adaptive arithmetic, and nearest-node policy queries are additional explicit costs, not constants suppressed by a dimension claim. A rectangular state cover with radius  $h_t$  can require order  $h_t^{-d}$  nodes; a rectangular action cover can require order  $k_t^{-a}$  actions. The theorem is therefore a dimension-explicit construction, not an assertion that a nonlinear high-dimensional frontier has been executed or that the curse of dimensionality has been removed.

In one state dimension, the finite envelope can be compiled exactly into a piecewise-linear representation. Two passes compute the greatest  $L$ -Lipschitz minorant of the node labels. Between neighboring nodes the envelope is the minimum of the two adjacent affine cones. At most one additional intersection is needed per interval. Rational knots and values permit exact antiderivatives; an interval evaluator encloses their numerical evaluation. This compilation evaluates the defining network exactly and does not substitute labels from a separately solved Bellman problem. The spline comparator receives the same exact-integration acceleration, so it is not charged for an artificial dense neural evaluation loop.

## 7.3. A predeclared matched service catalogue

The new experiment tests complete services, not only previously failed neural candidates. It uses a further nonlinear investment laboratory with  $x \in [0, 1]$ ,  $a \in [0, 1/4]$ ,  $\beta = 15/16$ , and independent  $z$  uniform on  $[-1/32, 1/32]$ :

$$F(x, a, z) = \frac{1}{32} + \frac{11}{16}x + \frac{1}{16}x(1 - x) + a + z, \quad (27)$$

$$c_p(x, a) = (x - \frac{11}{16})^2 + pa^2 + 4a^4 + 2(\frac{3}{8} - x)_+^2, \quad (28)$$

$$g(x) = 2(x - \frac{11}{16})^2 + 2(\frac{3}{8} - x)_+^2. \quad (29)$$

The invariant domain follows from monotonicity in  $x, a, z$  and the two extreme transitions 0 and 1. Valid primitive state moduli are  $C_x = 23/8$ ,  $F_x = 3/4$ , and  $L_g = 17/4$ ; action moduli are  $C_a = p/2 + 1/4$  and  $F_a = 1$ . The downside term penalizes a low capital stock, and the quartic investment cost discourages rapid adjustment. This is a numerical economic laboratory, not an estimated or calibrated quantitative exercise.

The complete design was fixed before execution: horizons  $\{2, 4\}$ , prices  $\{1, 4\}$ , two methods, three isolated repetitions, and the full common state/action ladder  $N = A \in \{16, 32, 64, 128, 256, 512\}$ . The targets are 0.5, 0.25, and 0.125 in units of discounted cost. These are policy-loss tolerances, not equivalence margins for a neural-minus-spline contrast. Every rung restarts from terminal primitives and constructs its own future backward. The neural method uses (21); the spline uses linear interpolation of its own labels and propagates its actual exact slope bound. Both evaluate the same continuous innovation law analytically and round labels to dyadic multiples of  $2^{-40}$ . All numerical query errors are outwardly enclosed and enter the certificate.

All rungs are retained, including those preceding the first target crossing. The clock to a target includes every preceding rung, target formation, rational compilation, actor construction, verification, and durable checkpoint output. Fresh processes run sequentially on one selected CPU with one numerical-library thread and an identical non-economic warm-up. Imports and warm-up are excluded from the internal clock but included in a separate process clock. CPU frequency is not fixed. Independent policy-cost comparison is not performed in this catalogue and is not assigned a zero cost. Three timing repetitions describe local dispersion, not a population law for neural optimization.

The neural backend certifies all twelve services at 0.5 and 0.25, and nine of twelve at 0.125. The spline certifies all twelve at every target. The three neural failures are repetitions of the same horizon-four, price-four economy: the final gap is approximately 0.12503, just above 0.125. They remain failures rather than being rounded

TABLE V

FIRST CERTIFIED STATE / ACTION RESOLUTION IN THE COMPLETE CONSTRUCTIVE CATALOGUE

| Generator     | $T$ | $p$ | 0.5 | 0.25 | 0.125 | Final gap |
|---------------|-----|-----|-----|------|-------|-----------|
| ReLU envelope | 2   | 1   | 64  | 128  | 256   | 0.06132   |
| Spline        | 2   | 1   | 64  | 128  | 256   | 0.05619   |
| ReLU envelope | 2   | 4   | 128 | 256  | 512   | 0.06274   |
| Spline        | 2   | 4   | 64  | 128  | 256   | 0.05799   |
| ReLU envelope | 4   | 1   | 128 | 256  | 512   | 0.12236   |
| Spline        | 4   | 1   | 128 | 256  | 512   | 0.09711   |
| ReLU envelope | 4   | 4   | 256 | 512  | –     | 0.12503   |
| Spline        | 4   | 4   | 128 | 256  | 512   | 0.10221   |

Entries are first qualifying  $N = A$  on the fixed ladder. Each row represents three identical deterministic policy/certificate arrays, not three independent economic tasks. A dash is a retained failure at the declared cap  $N = 512$ . Final gaps use the cap for both methods.

TABLE VI

COMPLETE PREFIX WORK AT TARGET 0.125, OR AT THE FAILED CAP

| Generator     | $T, p$ | $N$ | Queries   | Median s | Range s          | Exit   |
|---------------|--------|-----|-----------|----------|------------------|--------|
| ReLU envelope | 2,1    | 256 | 176,586   | 0.3009   | [0.2982, 0.3094] | target |
| Spline        | 2,1    | 256 | 176,586   | 0.2144   | [0.2136, 0.2365] | target |
| ReLU envelope | 2,4    | 512 | 702,924   | 0.8450   | [0.8354, 0.8472] | target |
| Spline        | 2,4    | 256 | 176,586   | 0.2151   | [0.2119, 0.2153] | target |
| ReLU envelope | 4,1    | 512 | 1,405,848 | 1.5540   | [1.5497, 1.5626] | target |
| Spline        | 4,1    | 512 | 1,405,848 | 1.2654   | [1.2341, 1.2745] | target |
| ReLU envelope | 4,4    | 512 | 1,405,848 | 1.5495   | [1.4602, 1.5506] | cap    |
| Spline        | 4,4    | 512 | 1,405,848 | 1.2705   | [1.2625, 1.2810] | target |

Queries sum all Bellman evaluations on earlier failed rungs and the reported rung. Clocks include construction, verification and durable checkpoints. Ranges contain the three isolated observations. A cap row is not time to successful certification. Independent economic comparison is not included.

into attainment. Across the 33 common successful method/target/repetition comparisons, the spline reaches the target earlier in every recorded observation. Thus the new execution establishes reproducible complete construction and a transparent frontier, not neural cost superiority. The deterministic termination corollary concerns its specified sufficient budget; the predeclared finite ladder need not contain

that budget. The finite catalogue measures the deterministic backend just specified. It neither overwrites the original neural optimizer’s attainment array nor turns the seven retained-policy diagnostics into prospective observations. The theorem supplies a method-level termination statement for this backend; the recorded services establish only the achieved targets, costs, and storage on the declared economies. A smaller certificate is not a lower realized policy cost. The new numerical comparison and the earlier direct policy-cost comparisons therefore remain distinct throughout.

## 8. NEURAL CONTINUATION WITH AN ACTION WITNESS

A constructed continuation contains information about the action that produced each Bellman label. Discarding that information and subsequently controlling an unrelated actor can charge the same approximation twice. We now retain the label’s feasible action through the minimum circuit. This changes the returned policy, not merely the description of an old certificate. The construction continues to use the original economic primitives and its own fitted future.

### 8.1. A one-sided policy sandwich

Write  $\mathcal{T}_t^\pi f_{t+1}(x) = Q_t(f_{t+1}; x, \pi_t(x))$ . Suppose that, uniformly on the invariant domain,

$$f_t - \mathcal{T}_t f_{t+1} \leq u_t, \quad \mathcal{T}_t^\pi f_{t+1} - f_t \leq d_t, \quad \ell_T \leq f_T - g \leq u_T. \quad (30)$$

The quantities  $u_t$  and  $d_t$  are signed upper bounds. Their separate absolute values need not enter a policy-loss account.

**PROPOSITION 7—One-sided policy sandwich:** *Under the monotonicity, cash-invariance and Bellman comparison conditions of Section 2, inequalities (30) imply*

$$0 \leq J_0^\pi(x) - V_0^*(x) \leq \sum_{t < T} B_t(u_t + d_t) + B_T(u_T - \ell_T). \quad (31)$$

*The bound is invariant to additive date-specific shifts of the continuation. It holds for the stated recursive evaluations as well as expectation; no simulation identity is required.*

PROOF: Backward order comparison gives  $V_0^* \geq f_0 - \sum_{t < T} B_t u_t - B_T u_T$  and  $J_0^\pi \leq f_0 + \sum_{t < T} B_t d_t - B_T \ell_T$ . Subtraction proves the result. A shift  $f_t \mapsto f_t + b_t$  adds  $b_t - \beta_t b_{t+1}$  to  $u_t$  and subtracts it from  $d_t$ , leaving their sum unchanged. *Q.E.D.*

The centered theorem remains valid: its choice  $d_t = \eta_t - \ell_t$  recovers (8). The new proposition avoids that indirect bound when the selected-policy residual can be certified jointly with the continuation. It is an order comparison, not a new claim of neural superiority.

## 8.2. Retaining the minimizing cone's action

Use the primitive conditions, grids, query tolerances and labels of Section 7. Let  $a_{t,i}$  denote the action whose numerical query attains  $y_{t,i}$ . In place of the nearest-node actor, return

$$i_t(x) \in \arg \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad \pi_t^w(x) = a_{t,i_t(x)}. \quad (32)$$

A fixed tie convention makes this finite selector measurable. The common action set makes each selected action feasible, even when the attaining cone is not centered at a nearest node. The critic remains the exact affine-ReLU circuit. The actor additionally uses the circuit's comparison index; a discontinuous selector is not asserted to be a continuous ReLU output.

**THEOREM 8**—Witness-preserving neural construction: *Under the conditions of Theorem 5, the policy (32) satisfies*

$$Q_t(f_{t+1}; x, \pi_t^w(x)) - f_t(x) \leq e_t. \quad (33)$$

*Consequently its all-state loss relative to the optimum over all admissible adapted policies is at most*

$$\mathcal{G}_w = \sum_{t < T} B_t \{L_t^a k_t + 2e_t + 2L_t h_t\} + B_T \{2e_T + 2L_g h_T\}. \quad (34)$$

*The nonterminal allowance is half the separated sufficient allowance in (24); the terminal account is unchanged. This comparison of certified bounds does not assert that the two policies have ordered realized costs.*

PROOF: For the attaining index  $i = i_t(x)$ , the numerical label and its action obey  $Q_t(f_{t+1}; x_{t,i}, a_{t,i}) \leq y_{t,i} + e_t$ . Transporting this same action to  $x$  costs at most  $L_t \|x - x_{t,i}\|_1$ . Hence its objective is at most  $f_t(x) + e_t$ . The unchanged optimal-residual upper bound is  $u_t = L_t^a k_t + e_t + 2L_t h_t$ . Apply Proposition 7 with  $d_t = e_t$  and the unchanged terminal enclosure. *Q.E.D.*

The label, the cone and the selected action are one mathematical object in this argument. Retaining only a Lipschitz-regularized label value without an attaining original action index would not justify (33). The supplement proves that the scalar compiler preserves precisely this information, including dominated labels, rational intersections and ties.

COROLLARY 9—Resource and execution allowances: Let  $s = \varepsilon / \sum_{t=0}^T B_t$ . At non-terminal dates it suffices to choose

$$e_t \leq s/4, \quad L_t^a k_t \leq s/4, \quad L_t h_t \leq s/8,$$

and retain the terminal allocation of Corollary 6. Under the same effective-computability hypotheses the construction terminates with loss at most  $\varepsilon$ .

If the chosen cone exceeds the minimum by at most  $v_t$ , state acquisition error is at most  $r_t$  in the stated norm, and a feasible executed action is within  $\alpha_t$  of that cone's action, replace  $e_t$  in (33) by

$$e_t + v_t + 2L_t r_t + L_t^a \alpha_t.$$

Add the discounted execution terms to (34).

The relaxed allocation is a sufficient-budget improvement relative to the earlier proof, not a complexity lower bound for conventional methods. A tensor cover still incurs its dimension factors. Exact state acquisition is the reference experiment; unchecked lookup using rounded switchpoints does not inherit an exact-index guarantee.

### 8.3. A strengthened conventional comparison

For either the original nearest-node cone actor or the spline actor, let  $C_i$  be the node's closed nearest-neighbor cell and let  $f_t$  be that method's own continuation. The same selected-label argument gives the computable bound

$$d_t = e_t + D_t, \quad D_t = \max_i \sup_{x \in C_i} \{y_{t,i} + L_t \|x - x_{t,i}\|_1 - f_t(x)\}. \quad (35)$$

On the scalar domain, all terms are piecewise affine. Their maximum is attained among the continuation knots, state nodes and cell boundaries; both adjacent actors are included at a boundary. The exact rational maximum supplies a valid all-state allowance without an additional verification mesh. Both conventional interpolation and the nearest-node neural ablation receive this improved account. Thus the experiment does not contrast the new bound only against a deliberately loose baseline. The old separated bound remains recorded, and the smaller of two bounds for the same returned policy may be used.

### 8.4. Matched construction, policy and certificate frontiers

The following matched catalogue is the frozen R46 experiment. Its statements about direct policy cost refer to the evidence then available; Section 13 supplies the new actual-policy comparison, and Section 12 executes the state-dependent construction. The design was fixed in the repository before execution. It retains the nonlinear continuous-innovation economy (27), horizons 2, 4, prices 1, 4, and all six resolutions 16, 32, 64, 128, 256, 512. Three methods and three isolated timing repetitions produce 36 services and 216 rungs. Each rung restarts from primitives. The two cone methods have identical critics and node-action arrays; their deployed selectors differ. The spline constructs its own labels. All methods receive the same exact integration and outward numerical arithmetic.

The declared targets are 1/4, 1/8 and 1/16. Every unsuccessful rung and every capped failure is retained. Internal prefix clocks include target construction, witness or nearest-cell compilation, certificates and durable checkpoints; process clocks additionally include imports and the common non-economic warm-up. Runs use one selected CPU and one numerical-library thread, with method order rotated

across repetitions and economic cells. Frequency is not controlled. These are local measurements, not a hardware-independent work theorem. Primitive counts and rational storage are recorded separately from clocks, and are not called complete floating-point or bit-operation counts.

TABLE VII

FINAL ALL-STATE POLICY-LOSS BOUNDS AT  $N = 512$ 

| $T$ | $p$ | Witness neural | Nearest neural | Spline   |
|-----|-----|----------------|----------------|----------|
| 2   | 1   | 0.034308       | 0.050092       | 0.039641 |
| 2   | 4   | 0.035017       | 0.051637       | 0.040957 |
| 4   | 1   | 0.064387       | 0.093593       | 0.066369 |
| 4   | 4   | 0.065720       | 0.097039       | 0.069979 |

Bounds are rounded upward in the records; displayed decimals are summaries, not the target test. All three repetitions have identical checkpoints. Both nearest-node comparators use the strengthened one-sided certificate.

All three methods certify 12/12 services at 1/4 and at 1/8, and 6/12 at 1/16. The latter failures are all horizon-four services and are retained. The witness bound is strictly tighter than the nearest-node neural ablation in all 24 distinct economy/resolution comparisons, and tighter than the strengthened spline in 23 of 24. The exception is  $T = 4$ ,  $p = 1$ ,  $N = 16$ , where the witness and spline bounds are approximately 2.06039 and 2.03515. The declared dyadic targets have the same first-crossing resolution for all three methods in every economic cell. Thus the certificate improvement does not yield an earlier first crossing on this particular ladder. The spline is faster in all 30 common successful target/repetition clock comparisons; the witness construction is faster than the nearest-node neural ablation in all 30. These local clocks and tighter regret bounds establish neither policy-cost superiority nor a general neural speed advantage. The comparisons isolate the effect of carrying the action witness and of using a joint policy certificate. They do not reclassify earlier capped services or claim that the new selector was previously deployed. In particular, an improvement shared by the strengthened nearest-node ablation cannot be attributed solely to witness selection. A tighter regret certificate does not

identify the sign of a direct policy-cost contrast. The existing direct expected-cost intervals and adverse precision findings remain unchanged.

## 9. STATE-DEPENDENT CONSTRAINTS AND FEASIBLE WITNESSES

Investment opportunities often depend on current capital, collateral or remaining capacity. The common-action construction above does not by itself authorize transporting a node's action to a state where that action is inadmissible. We now incorporate a feasible repair into the same neural Bellman construction. The original controlled economy, rather than a relaxed control problem, remains the comparison object.

Use the invariant state domain, numerical-query conditions and common evaluation functional of Section 7. Replace its common action set by nonempty compact sets  $A_t(x)$  satisfying

$$d_H(A_t(x), A_t(z)) \leq K_t^A \|x - z\|_1, \quad (36)$$

where  $d_H$  is Hausdorff distance in the action norm. The primitive inequalities (17) are now required for pairs of feasible state-action points; neither costs nor dynamics need to be evaluated at infeasible actions. Assume effective finite  $k_t$ -nets of  $A_t(x_{t,i})$  and a feasible Borel repair  $\Gamma_{t,i}(x) \in A_t(x)$  of each stored action  $a_{t,i}$ , with

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \leq K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \quad (37)$$

The nonnegative  $\zeta_t$  is a certified displacement allowance, not permission to violate feasibility. Exact nearest-point selection gives  $\zeta_t = 0$  when available. The effective repair and feasible-net procedures are additional computational inputs, whose work must be charged.

Starting with  $L_T = L_g$ , define

$$A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}, \quad (38)$$

$$D_t^Q = C_{a,t} + \beta_t F_{a,t} L_{t+1}, \quad (39)$$

$$L_t = A_t^Q + D_t^Q K_t^A.$$

At each state node, minimize the certified queries over its own feasible action net. Retain the original minimizing action with its numerical label  $y_{t,i}$ . Form the same cone network  $f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}$ . Let  $i_t(x)$  attain this minimum and return  $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$ .

**THEOREM 10—Feasible-witness transport:** *Under these conditions and the Bellman comparison hypotheses of Section 2, the constructed continuation is  $L_t$ -Lipschitz, and the returned policy is feasible and measurable. Uniformly on the original state domain,*

$$-e_t \leq f_t - \mathcal{T}_t f_{t+1} \leq D_t^Q k_t + e_t + 2L_t h_t, \quad (40)$$

$$\mathcal{T}_t^\pi f_{t+1} - f_t \leq e_t + D_t^Q \zeta_t.$$

Consequently its loss relative to the optimum over all admissible adapted policies is at most

$$\sum_{t < T} B_t \{D_t^Q k_t + 2e_t + 2L_t h_t + D_t^Q \zeta_t\} + B_T (2e_T + 2L_T h_T). \quad (41)$$

No optimal continuation is supplied to the label generator. The bound concerns the policy after feasible repair, not the possibly infeasible array of unmodified node actions.

**PROOF:** The primitive inequalities imply a modulus  $A_t^Q$  in the state and  $D_t^Q$  in the action for feasible pairs. Given an action at  $x$ , the Hausdorff condition supplies an action at  $z$  within  $K_t^A \|x - z\|_1$ . Comparing the two objectives and reversing the roles of  $x, z$  shows that  $\mathcal{T}_t f_{t+1}$  is  $L_t$ -Lipschitz. The feasible node net therefore gives  $-e_t \leq y_{t,i} - \mathcal{T}_t f_{t+1}(x_{t,i}) \leq D_t^Q k_t + e_t$ . The cone argument proves the first enclosure. For the attaining index, transport between the two feasible pairs and (37) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \leq y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t^Q \zeta_t.$$

The cone equals  $f_t(x)$ . Apply Proposition 7 and the unchanged terminal enclosure.

*Q.E.D.*

This result closes a constructive route for a class of endogenous constraints within the original model. It does not assume that all feasible correspondences have a finite Hausdorff modulus. With  $s = \varepsilon / \sum_{t=0}^T B_t$ , sufficient nonterminal allocations are  $e_t \leq s/8$ ,  $D_t^Q k_t \leq s/4$ ,  $L_t h_t \leq s/8$  and  $D_t^Q \zeta_t \leq s/4$ . Together with the earlier

terminal allocation, they imply finite termination at tolerance  $\varepsilon$  when all listed operations are effective. Zero coefficients impose no restriction. When exact repair is available, the larger query allowance of Corollary 9 is again sufficient. Setting  $K_t^A = \zeta_t = 0$  recovers the common-action witness theorem.

9.0.0.1. *A capacity-constrained investment rule.* For two nonnegative state coordinates on  $[0, 1]^2$ , let  $A(x) = [0, (x_1 + x_2)/2]$ . This is a state-dependent capital-capacity constraint with  $K^A = 1/2$  in the  $\ell^1$  state norm. The repair  $\Gamma_i(x) = \min\{a_i, (x_1 + x_2)/2\}$  is feasible, satisfies (37) with  $\zeta = 0$ , and has an elementary affine-ReLU representation for each fixed witness. Thus changing capacity changes the actor and the constructive state modulus through an explicit term  $D^Q/2$ . The index selector remains a finite comparison procedure. The supplement provides exact rational checks for this example; these are checks of the theorem and admissibility contract, not an additional multidimensional performance study.

Neither feasible-net construction nor repair is free. With  $J_{t,i}$  node actions, target-query work is summed over  $\sum_i J_{t,i}$  rather than a common  $K_t J_t$ . Each deployed query also incurs the selected repair's work and storage. Under imperfect state acquisition, repair must be certified feasible at the true state, or throughout the acquisition set for a robust implementation. An error allowance for the objective cannot substitute for that feasibility condition.

## 10. FROM ACQUIRED STATES TO FEASIBLE POLICIES

A policy is implemented at an acquired state, not at an exact real number available without cost. This distinction matters for a witness selector: a small change in the state can change its index discontinuously. The appropriate object is therefore an objective allowance for the selected witness together with feasibility at the true state. A Lipschitz assumption on the actor would replace, rather than solve, this problem.

At date  $t$ , an observation  $\hat{x}$  specifies a nonempty acquisition set  $\mathcal{D}_t(\hat{x}) \subset X$  containing the true state, with  $\|x - \hat{x}\|_1 \leq r_t$  for every  $x \in \mathcal{D}_t(\hat{x})$ . Use the feasible-witness data of Section 9. Suppose the numerical selector returns an index  $i$  such that

$$y_{t,i} + L_t \|\hat{x} - x_{t,i}\|_1 \leq f_t(\hat{x}) + v_t. \quad (42)$$

The implemented action  $\hat{a}_t(\hat{x})$  must belong to  $A_t(x)$  for every  $x$  in the acquisition set. Assume, on that entire set,

$$\|\hat{a}_t(\hat{x}) - a_{t,i}\|_1 \leq K_t^A \|x - x_{t,i}\|_1 + \zeta_t^{\text{acq}}. \quad (43)$$

The observation and selection rules are measurable and causal. Enlarging the history by these observations leaves the stipulated conditional innovation evaluation unchanged; otherwise the full-information comparator must be defined on the enlarged information structure. The allowances are uniform, not fitted to a favorable sample of observation errors.

**THEOREM 11**—Acquisition-aware feasible realization: *Under the hypotheses of Theorem 10, the implemented action satisfies, uniformly over every permitted observation and true state,*

$$Q_t(f_{t+1}; x, \hat{a}_t(\hat{x})) - f_t(x) \leq e_t + v_t + 2L_t r_t + D_t^Q \zeta_t^{\text{acq}}. \quad (44)$$

Consequently the original full-information optimum bounds the implemented policy's value below, and its loss is bounded above by

$$\sum_{t < T} B_t \{D_t^Q k_t + 2e_t + 2L_t h_t + v_t + 2L_t r_t + D_t^Q \zeta_t^{\text{acq}}\} + B_T(2e_T + 2L_g h_T). \quad (45)$$

The result requires no continuity of the index or actor and no probabilistic independence of measurement error. It applies to the common monotone, cash-invariant recursive evaluation under its original comparison assumptions.

**PROOF:** Transport from the original feasible node action to the implemented feasible action gives

$$Q_t(f_{t+1}; x, \hat{a}_t) \leq y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t^Q \zeta_t^{\text{acq}}.$$

The cone at  $x$  is at most its value at  $\hat{x}$  plus  $L_t r_t$ . Apply (42), then use  $f_t(\hat{x}) \leq f_t(x) + L_t r_t$ . This proves (44). Backward order comparison, conditional also on the observation history when necessary, proves (45). Feasibility permits comparison with the optimum over all admissible adapted policies. The proof does not compare distances between the two selected actions at nearby states. *Q.E.D.*

COROLLARY 12—Capacity, measurement, and action quantization: Let  $A_t(x) = [0, \kappa_t(x)]$ , where  $\kappa_t$  is nonnegative and  $K_t^A$ -Lipschitz. Suppose a computed lower capacity satisfies

$$0 \leq \underline{\kappa}_t(\hat{x}) \leq \kappa_t(x), \quad \kappa_t(x) - \underline{\kappa}_t(\hat{x}) \leq 2K_t^A r_t + \chi_t \quad (x \in \mathcal{D}_t(\hat{x})).$$

For an action spacing  $\Delta_t > 0$ , implement

$$\hat{a}_t = \Delta_t \left\lfloor \frac{\min\{a_{t,i}, \underline{\kappa}_t(\hat{x})\}}{\Delta_t} \right\rfloor.$$

It is feasible on the whole acquisition set and satisfies (43) with  $\zeta_t^{\text{acq}} = 2K_t^A r_t + \chi_t + \Delta_t$ . Exact, unquantized robust repair uses  $\Delta_t = 0$  and the minimum itself.

PROOF: The downward quantization preserves nonnegativity and the lower capacity. Since  $a_{t,i} \leq \kappa_t(x_{t,i})$ , exact repair at  $x$  displaces the node action by at most  $K_t^A \|x - x_{t,i}\|_1$ . Replacing true capacity by its certified lower bound adds at most  $2K_t^A r_t + \chi_t$ ; downward quantization adds less than  $\Delta_t$ . Summing proves the displacement inequality. *Q.E.D.*

### 10.1. Ordinary numerical comparisons

The execution in Section 12 uses binary64 scores at dyadic acquired states. Stored labels and the coordinate distances are exactly representable at the declared input resolutions. Put  $u = 2^{-53}$ ,  $\gamma_2 = 2u/(1 - 2u)$ ,  $Y_t = \max_i |y_{t,i}|$ , and let  $\tilde{L}_t$  be the represented slope. In state dimension  $d$ , each computed cone score differs from its exact score by at most

$$\tau_t = d|\tilde{L}_t - L_t| + \gamma_2\{Y_t + d|\tilde{L}_t|\}. \quad (46)$$

The first term charges coefficient conversion; the second charges multiplication and addition. A minimum of the computed scores satisfies (42) with  $\nu_t = 2\tau_t$ , including ties. The supplement states the representability and arithmetic conditions and proves the error account. The general theorem also permits other certified score enclosures.

This implementation does not round a rational switchpoint and assume that its original cell is unchanged. For the scalar comparisons, all cells intersecting a state

enclosure are retained. For acquired constrained deployment, a numerical index has the objective allowance (46) and its action is repaired against the entire acquisition set. Both routes address discontinuity directly.

## 10.2. The identical non-neural envelope

The envelope construction must be separated from the language used to encode it. The classical Lipschitz extension (McShane, 1934), a native minimum computation, and an exact ReLU realization can represent the same function.

**PROPOSITION 13**—Representation control: *Fix the node locations, labels, slopes, original action witnesses, and selector tie rule. A conventional min-plus implementation and the affine-ReLU implementation of the same envelope return the same mathematical continuation and policy. They have the same exact policy certificate and direct policy cost. A balanced realization uses  $O(dK)$  arithmetic and comparison gates for  $K$  cones and depth  $O(\log K + \log d)$ . Merely changing the representation does not improve the covering or Bellman-query factors.*

**PROOF:** For every pair of real numbers,  $\min\{a, b\} = a - (a - b)_+$ . Applying this identity through a balanced reduction tree proves equality of the continuation. Carrying the same attaining index, with the prescribed tie rule, proves equality of the action witness and repaired action. The certificates and costs concern these mathematical objects. The gate count includes the coordinate absolute values and their sums. The supplement separately bounds floating-point gate errors; exact equality is not asserted for unchecked rounded evaluations. *Q.E.D.*

We execute this non-neural control on all 24 distinct scalar witness checkpoints. It shares the defining data rather than masquerading as an independently trained competitor. The native selector and its ReLU encoding are thus one construction with two representations. The learned hidden-location methods studied earlier are different candidate generators. Their assessment, and that of the deterministic backend, remains a comparison of actual policies and complete computational work.

## 11. WITNESS-PRESERVING COMPILATION OF THE NEURAL OPERATOR

The cost of a Bellman construction depends on how its continuation is evaluated. A literal evaluation of the cone network in Theorem 10 visits every state label at every action–innovation query. Repeating that scan is unnecessary on a tensor cover. The following result compiles the same continuation and original action witnesses; it neither changes the fitted function nor substitutes a newly optimized policy.

Let  $\mathcal{X} = \prod_{j=1}^d \{x_{j,0}, \dots, x_{j,n_j}\}$ , with strictly increasing coordinates and  $S = \prod_j (n_j + 1)$ . The rectangular domain is the product of the coordinate intervals. Index the nodes lexicographically. For arbitrary real labels  $y_i$  and  $L \geq 0$ , write

$$f(x) = \min_{i \leq S} \{y_i + L\|x - x_i\|_1\}, \quad i_*(x) = \min_{i \leq S} \argmin \{y_i + L\|x - x_i\|_1\}. \quad (47)$$

Each original index carries a feasible node action  $a_i$ . Labels need not be Lipschitz-consistent: the numerical errors allowed in the earlier construction are permitted here without alteration. Define the transformed label and its original owner at a grid node  $v$  by

$$z_v = \min_i \{y_i + L\|v - x_i\|_1\}, \quad o_v = \min_i \argmin \{y_i + L\|v - x_i\|_1\}. \quad (48)$$

The owner is an original label index, not the index of the node at which its transformed value is stored.

**THEOREM 14**—Continuous compilation with original witnesses: *For every closed tensor cell  $C$  and every  $x \in C$ , let  $\mathcal{V}(C)$  be its corner set. Then*

$$f(x) = \min_{v \in \mathcal{V}(C)} \{z_v + L\|x - v\|_1\}, \quad (49)$$

$$(f(x), i_*(x)) = \min_{\text{lex}} \{(y_{o_v} + L\|x - x_{o_v}\|_1, o_v) : v \in \mathcal{V}(C)\}. \quad (50)$$

*Forward and backward relaxations along each coordinate line compute all pairs  $(z_v, o_v)$  using  $O(dS)$  additions and comparisons and  $O(S)$  stored values and indices. After cell location, a value or exact witness query uses at most  $2^d$  corner pairs and  $O(d2^d)$  arithmetic operations. A binary search locates the cell in  $O(\sum_j \log(n_j + 1))$  comparisons; a uniform grid permits direct location.*

When node coordinates, labels and  $L$  are rational, exact rational preprocessing and comparisons implement these identities, including zero slopes, inconsistent labels and ties. Applying the same repair map to the original returned action gives exactly the same feasible policy and policy cost as the uncompiled construction. The residual and acquisition bounds of Theorems 10 and 11 are unchanged.

PROOF: The triangle inequality gives  $f(x) \leq z_v + L\|x - v\|_1$ . Choose an attaining original site  $x_i$ . In each coordinate choose the corner coordinate lying between  $x_j$  and  $x_{i,j}$ . The resulting  $v \in \mathcal{V}(C)$  satisfies  $\|x - x_i\|_1 = \|x - v\|_1 + \|v - x_i\|_1$ , proving (49). Choose now  $i = i_*(x)$ . At this corner an owner with a strictly smaller transformed value would improve the objective at  $x$ ; an owner with the same value and a smaller original index would contradict the prescribed tie rule. Hence  $o_v = i_*(x)$  for a suitable corner. Every candidate in (50) is an original cone, so its lexicographic minimum is the desired pair. Coordinatewise forward and backward minimizations evaluate the separable  $\ell^1$  transform; keeping original indices in each lexicographic comparison preserves ties. The supplement gives the line invariant, operation account and numerical enclosure details. Q.E.D.

The nodal distance transform is classical; the linear-time coordinate algorithm is described by Felzenszwalb and Huttenlocher (2012). We use it for a different computational obligation: continuous off-grid Bellman queries must retain the original feasible action and the specified tie rule. Equation (49) alone does not authorize using the action attached to  $v$ ; it is (50) that licenses the policy. The result belongs to the cone-based NBO construction, but an identical native min-plus implementation can use the same compiler. It does not establish a computational property exclusive to a neural encoding. Proposition 13 remains an equality statement, not a comparison between independently trained methods.

### 11.1. Complete construction work

Let  $J_{t,i}$  denote the size of the feasible action net at state node  $i$ , and let  $M_t$  be the number of integration points in a certified query. Put  $Q_t = M_t \sum_i J_{t,i}$ . For a tensor implementation, the representation part of a backward construction changes from

$O(\sum_t dS_{t+1}Q_t)$  to

$$O\left(\sum_t \left[ dS_{t+1} + Q_t \left\{ d2^d + \sum_j \log(n_{t+1,j} + 1) \right\} \right] \right). \quad (51)$$

Primitive evaluation, feasible-net construction, integration, error accounting, actor formation, repair and output are additional terms. Exact arithmetic has a bit cost; the count in (51) is not a claim that rational comparisons have unit physical cost. Our records therefore report coefficient bit lengths, storage and complete process and CPU times alongside the component counts.

In the two-state investment study,  $S = (N + 1)^2$ ,  $J = N + 1$  and  $M = N$ . At fixed dimension and horizon, compiling removes the extra state-cover factor from the dense representation workload:  $O(TN^6)$  becomes  $O(TN^4)$ , plus the explicitly charged setup and output. This does not remove the tensor cover, the action cover, or the innovation factor. In dimension  $d$ , both  $S$  and the  $2^d$  corner factor remain visible. The four-dimensional stress cases below are executions of this finite-dimensional construction, not evidence that the curse of dimensionality has disappeared.

**COROLLARY 15**—A resource allocation for implemented policy accuracy: *In the two-state economy of Section 12, let  $C_{T,p}$  be the explicit primitive constant in (54),  $S_T = \sum_{t < T} \beta^t$ , and choose a dyadic  $N$  with*

$$N \geq 4C_{T,p}/\varepsilon, \quad \xi, \xi_T \leq \frac{\varepsilon}{8(S_T + \beta^T)}.$$

*Let  $A$  be the acquisition coefficient in (59) and  $K = \sum_{t < T} \beta^t D_t^Q$ . With exact witness selection, a  $b$ -bit sensor and a common downward action spacing  $\Delta$  satisfying*

$$A2^{-b} \leq \varepsilon/4, \quad K\Delta \leq \varepsilon/4,$$

*the constructed implemented policy has loss at most  $\varepsilon$  against the full-information optimum. A zero coefficient imposes no corresponding precision restriction. An observation fee adds its explicit discounted charge to this loss account.*

*For fixed primitive parameters and horizon, choosing the smallest sufficient dyadic resolution gives  $N = O(\varepsilon^{-1})$ . The compiled algebraic construction uses  $O(TN^4)$  operations,*

versus  $O(TN^6)$  for the literal dense cone evaluation, and stores  $O(TN^2)$  labels, owners and actions. Executing the full dyadic ladder to this resolution has the same orders, with all preceding attempts charged. Numerical precision, coefficient bit lengths, uncertainty-box work and physical output latency remain separate costs; these orders are not a precision-free or machine-independent timing claim.

PROOF: The first two allocations make the right side of (54) at most  $\varepsilon/2$ . The acquisition and quantization terms add at most another  $\varepsilon/2$  by (59). Compilation preserves the selected policy, so it does not change this account. On the dyadic ladder, the sum of every positive power of the earlier resolutions is bounded by a constant times that power of its largest resolution. Summing the complete algebraic operation counts and retained arrays proves the resource assertions. Effective certified queries at the prescribed accuracy, rather than unchecked fixed-precision evaluations, are part of the hypotheses. *Q.E.D.*

The corollary turns the compiler into a sufficient work-to-economic-accuracy allocation. A finite experimental ladder can still end before the sufficient resolution; its failure must remain a failure. Nor does more precise observation eliminate construction error: the two portions of the budget are distinct.

### 11.2. Certified numerical queries and selector ambiguity

The executable compiler computes  $(z_v, o_v)$  in rational arithmetic. A value query encloses each corner expression outward and takes the minima of lower and upper endpoints. For a state box contained in one cell, the possible original actors are among its corner owners; for a box crossing cells, the union of their corner owners is used. An owner is discarded only when its lower objective enclosure exceeds another candidate's upper enclosure. Exact ambiguous point queries use rational comparisons and the original lexicographic rule. Uncertain-state queries retain the hull of all remaining actions and propagate that hull through repair and the dynamics.

Thus a floating-point tie is not silently assigned to a convenient policy. A numerical implementation that deliberately returns a nonattaining index instead has to pay

the  $\nu_t$  allowance in (42). The present frozen-object checks separate exact identities from outward-rounding diagnostics. Overlapping machine intervals are useful checks; the identity theorem, rather than that overlap, proves that the compiler represents the original policy.

## 12. AN ORACLE-EXPLICIT CONSTRAINED INVESTMENT ECONOMY

The following evidence is the frozen R47 study. Its results and limitations are retained; Section 15 gives the new execution and decision evidence. We now execute the feasible-witness construction in a coupled two-state economy. Capital affects both future production conditions and current investment capacity. This makes feasible repair part of the economic policy rather than an algebraic example appended to a common-action calculation.

Let  $x \in [0, 1]^2$ ,  $\beta = 15/16$ , and  $0 \leq a \leq \kappa(x) = 1/8 + (x_1 + x_2)/16$ . A common innovation  $z$  is continuously uniform on  $[-1/32, 1/32]$ , with opposite exposures in the two transitions:

$$F_1(x, a, z) = 1/16 + x_1/2 + x_2/8 + x_1(1 - x_2)/16 + a/2 + z, \quad (52)$$

$$F_2(x, a, z) = 1/16 + x_2/2 + x_1/8 + x_2(1 - x_1)/16 + a/4 - z.$$

Define the shortage and imbalance cost

$$b(x) = (x_1 - x_2)^2/4 + 2(1/2 - x_1 - x_2)_+^2.$$

Stage cost is  $\sum_{j=1}^2 (x_j - 5/8)^2 + b(x) + pa^2 + 4a^4$ ; terminal cost doubles the squared-target terms and retains  $b$ . The predeclared cells have  $T \in \{2, 3\}$  and  $p \in \{1, 4\}$ . These are normalized theoretical economies, not estimated parameters or calibrated welfare magnitudes.

**PROPOSITION 16**—Finite primitives and continuous-law queries: *The state domain is invariant under (52). In the  $\ell^1$  state norm, valid primitive constants are*

$$C_x = 13/4, \quad L_g = 9/2, \quad F_x = 11/16, \quad F_a = 3/4, \quad C_a = p/2 + 1/4, \quad K^A = 1/16. \quad (53)$$

The uniform tensor state grid with  $N$  subdivisions has covering radius  $h = 1/N$ . At each node,  $N + 1$  equally spaced feasible actions have covering radius at most  $k = 1/(8N)$ . The repair is  $\min\{a_i, \kappa(x)\}$  and has zero extra displacement allowance. If a continuation is  $L$ -Lipschitz,  $M$  uniform midpoint cells enclose its original continuous-law expectation after adding the deterministic remainder  $L/(32M)$  to the numerical integration error.

PROOF: The primitive formulas give  $F_1 \in [1/32, 29/32]$  and  $F_2 \in [1/32, 27/32]$  by interval bounds. The sums of absolute derivatives in each state column are at most  $11/16$ , and the action derivative has  $\ell^1$  norm  $3/4$ . Differentiating the costs on the two shortage regions gives (53). Capacity is  $1/16$ -Lipschitz and at most  $1/4$ , proving the grid and repair assertions. Within a uniform shock cell, the mean distance to its midpoint is  $1/(64M)$ . Since the transition has shock modulus two in  $\ell^1$ , the expectation error is at most  $L/(32M)$ . Thus quadrature does not change the innovation law to a discrete one. Q.E.D.

The two generators are the feasible cone witness and conventional fitted-value iteration in the tensor bilinear basis. Each works backward against its own fitted future, uses the same feasible action net, and evaluates the same continuous law with the preceding remainder. No optimal value or other method's continuation is supplied as a label. Rounded labels, their valid query errors, slopes, node actions and all policy bounds are stored.

The conventional actor selects a nearest state node and repairs its action at the current capacity. It receives the one-sided account rather than only the older separated certificate. For the bilinear interpolant, the Bellman-residual enclosure is

$$-e_t - L_t h \leq f_t - \mathcal{T}_t f_{t+1} \leq D_t^Q k + e_t + L_t h.$$

The interpolation weights are nonnegative and reproduce coordinates. Their weighted mean  $\ell^1$  distance to the surrounding vertices is at most  $1/N$ , which proves the stated extension from the nodal enclosure.

LEMMA 17—Complete cell-corner verification: *For the bilinear actor, the selected-policy allowance is  $e_t + D_t$ , where*

$$D_t = \max_i \sup_{x \in C_i} \{y_{t,i} + L_t \|x - x_i\|_1 - f_t(x)\}.$$

*Here  $C_i$  is the closed nearest-node cell and both adjacent choices are admitted on a boundary. It suffices to evaluate the at most nine vertices obtained by splitting each such cell along its node coordinates. Outward endpoint evaluation supplies a valid upper bound on  $D_t$  without an additional verification mesh.*

PROOF: On each resulting subrectangle the distance term is affine and the continuation is bilinear. An affine function minus a bilinear function attains its maximum at a vertex: first maximize over one coordinate, then over the other. The label's feasible action, repaired at the queried state, has objective at most  $y_{t,i} + e_t + L_t \|x - x_i\|_1$  by feasible transport. Taking the complete maximum proves the allowance. Q.E.D.

### 12.1. A cap obtained from economic accuracy

In this class the previously abstract covering, integration and repair operations are finite routines with explicit costs. Let  $S_T = \sum_{t < T} \beta^t$ ,  $a_p = C_x + K^A C_a$  and  $b = \beta(F_x + K^A F_a) = 705/1024$ . Define

$$\bar{L} = \max\{L_g, a_p / (1 - b)\}, \quad \bar{D} = C_a + \beta F_a \bar{L}.$$

The primitive recursion gives  $L_t \leq \bar{L}$  and  $D_t^Q \leq \bar{D}$  at every horizon. Suppose that, apart from the quadrature remainder, each nonterminal numerical query has error at most  $\xi$  and terminal labels have error at most  $\xi_T$ . With  $M = N$ , Theorem 10 implies

$$\mathcal{G}_N \leq C_{T,p}/N + 2S_T \xi + 2\beta^T \xi_T, \tag{54}$$

$$C_{T,p} = S_T \{\bar{D}/8 + 2\bar{L} + \beta \bar{L}/16\} + 2\beta^T L_g.$$

Thus a dyadic  $N \geq 2C_{T,p}/\varepsilon$  and  $\xi, \xi_T \leq \varepsilon/[4(S_T + \beta^T)]$  suffice for loss at most  $\varepsilon$ . Acquisition and action quantization add the explicit terms of (45); these must

receive part of the same tolerance. Fixed binary64, a fixed label lattice and a finite three-rung study are not asserted to achieve every positive  $\varepsilon$ .

For this two-state, one-action, one-shock implementation, Bellman queries number  $T(N + 1)^3$ . Each has  $N$  midpoint evaluations. Direct cone evaluation examines  $(N + 1)^2$  cones, giving  $O(TN^6)$  primitive arithmetic work. Bilinear future evaluation uses a local patch, giving  $O(TN^4)$  work for that implementation. Both store  $O(TN^2)$  critic labels and actions. The present batched program additionally holds  $O(N^3)$  state-action arrays and  $O(BN^2)$  temporary cone entries for fixed batch size  $B$ . Feasible acquisition and repair add their recorded deployment work, rather than being included at zero cost.

With  $p_b$ -bit arithmetic and multiplication cost  $M(p_b)$ , the corresponding arithmetic upper bounds acquire a factor  $M(p_b + T + \log N)$ , with coefficient conversion, comparison and serialization charged separately. The precision must also meet the query enclosure requirement. These are algorithmic upper bounds, not lower bounds for all conventional methods. Proposition 13 shows that the identical native envelope has the same leading factors as its ReLU realization. No representation label removes the tensor dependence.

## 12.2. Execution and acquired deployment

The fixed ladder is  $N = M = 4, 8, 16$ , with three isolated processes per method and economic cell. The 24 services store all 72 rungs, including failed attempts. Targets are 4, 2, 1, 1/2 in the normalized cost units. They test the construction-to-certificate service; a mathematically valid coarse bound is not automatically an economically adequate tolerance. The full attained bounds and work therefore accompany the target counts.

Both methods attain the target 4 in all twelve services, at the last rung, and neither attains 2, 1 or 1/2 within this cap. The witness's final bound is smaller in each economic cell, but bilinear fitted-value iteration is faster in all twelve common successful comparisons. These coarse all-state certificates establish an executed feasible construction, not an economically precise solution at the unachieved tolerances. The

TABLE VIII

COUPLED CONSTRAINED ECONOMY: FULL-PREFIX WORK AT  $N = 16$ 

| $T$ | $p$ | Method       | Bound  | Queries | Seconds | Time range      |
|-----|-----|--------------|--------|---------|---------|-----------------|
| 2   | 1   | Witness      | 2.3148 | 11,534  | 7.045   | [7.003,7.070]   |
| 2   | 1   | Bilinear FVI | 2.7960 | 11,534  | 0.326   | [0.325,0.352]   |
| 2   | 4   | Witness      | 2.3691 | 11,534  | 6.992   | [6.975,7.188]   |
| 2   | 4   | Bilinear FVI | 2.8288 | 11,534  | 0.324   | [0.323,0.325]   |
| 3   | 1   | Witness      | 3.3212 | 17,301  | 10.519  | [10.480,10.552] |
| 3   | 1   | Bilinear FVI | 3.8915 | 17,301  | 0.490   | [0.487,0.500]   |
| 3   | 4   | Witness      | 3.4107 | 17,301  | 10.482  | [10.425,10.538] |
| 3   | 4   | Bilinear FVI | 3.9321 | 17,301  | 0.494   | [0.489,0.520]   |

Query counts and clocks include all three rungs, including failed attempts. Seconds are medians over three isolated process observations; the range reports their minimum and maximum. CPU affinity and library thread counts are fixed, but CPU frequency is not controlled.

sufficient-cap formula states what further resources the mathematical guarantee requires without relabeling these failures.

For each of the twelve distinct saved witness policies we also execute the selector on a 33-by-33 measurement grid, with coordinate acquisition radii 0,  $2^{-12}$  and  $2^{-8}$ . Exact robust repair and downward action spacing  $2^{-12}$  give 72 deployment cases. Capacity is bounded on the entire acquisition box; it is not checked only at its midpoint. The binary64 score allowance is uniform by (46); the grid records exercise implementation and repair activity rather than replacing that proof.

Repair is active rather than a vacuous operation, and downward action quantization is charged explicitly. At exact acquired input the small nonzero numerical selection allowance remains in the exact records even when it rounds to zero in this table. At a nonzero radius the acquisition term is positive; the total bound is not reported as the exact-state certificate.

### 13. POLICY COSTS AND ECONOMIC METHOD CHOICE

The following evidence is the frozen R47 study. Its results and limitations are retained; Section 14 gives the new execution and decision evidence. We next evaluate the actual witness policy, not its distance bound relative to an unknown optimum.

TABLE IX

ACQUIRED-STATE AND QUANTIZED-ACTION DEPLOYMENT

| Coordinate radius | Action spacing | Active repairs | Quantized | Largest added bound |
|-------------------|----------------|----------------|-----------|---------------------|
| 0                 | 0              | 4,817          | 0         | 0.000000            |
| 0                 | 1/4096         | 4,817          | 0         | 0.004619            |
| 1/4096            | 0              | 6,609          | 0         | 0.022513            |
| 1/4096            | 1/4096         | 6,609          | 6,609     | 0.027132            |
| 1/256             | 0              | 6,609          | 0         | 0.360204            |
| 1/256             | 1/4096         | 6,609          | 0         | 0.364823            |

Each row covers all twelve distinct constrained witness policies and 32,670 measured state–date queries. Feasibility holds on the entire acquisition box. The additional loss is a uniform theoretical allowance for the executed arithmetic, acquisition and action rule, not an estimated average policy cost.

The policies are the frozen objects at every common first crossing in the original witness catalogue. There are ten such economic-cell and target combinations. Three timing repetitions of an identical checkpoint are not three independent policies.

The primary initial-state law is uniform on  $[0,1]$ . The additional fixed states  $1/8$ ,  $1/2$  and  $7/8$  test particular economic starting positions. At each of these forty groups, the three pairwise contrasts concern witness, nearest-cone and spline policies. A fixed design specifies 131,072 common paths, forty-bit innovation bins and family error 0.01 for all 120 contrasts. The economic comparison uses the same continuous innovations as the original economy.

Let  $[\underline{z}_m, \bar{z}_m]$  enclose  $g - f_T^m$  and let the initial continuation difference have range  $[a_0, b_0]$  over the chosen initial-state law. With the one-sided residuals of (30), the score difference of two policies has deterministic support

$$\begin{aligned}
 a &= a_0 + \sum_{t < T} B_t(-u_t^m - d_t^n) + B_T(\underline{z}_m - \bar{z}_n), \\
 b &= b_0 + \sum_{t < T} B_t(d_t^m + u_t^n) + B_T(\bar{z}_m - \underline{z}_n).
 \end{aligned} \tag{55}$$

For a common random initial state, one bounds the difference  $f_0^m - f_0^n$ , not the two separate ranges. Its extrema are obtained exactly from the union of the rational

1 piecewise-affine knots. Conditional expectation and then integration over the initial 1  
2 law extend Theorem 4 to this estimand. 2

3 Every simulated bin generates lower and upper score endpoints. When a state 3  
4 enclosure meets a selector boundary, the calculation encloses all possible actors and 4  
5 subsequent states. It neither drops the path nor substitutes an unverified midpoint 5  
6 actor. The mean and variance computations have rational roundoff accounts. For 6  
7 the declared family,  $\log(4 \cdot 120/0.01) < 11$ ; a finite positive Taylor sum for  $\exp(11)$  7  
8 certifies this inequality. Using 11 in (15) is conservative and avoids an unchecked 8  
9 logarithm evaluation. The supplement gives the complete endpoint construction 9  
10 and arithmetic audit. 10

11  
12 THEOREM 18—Simultaneous policy comparison and adoption: *Conditional on the 12  
13 frozen policies and the independent uniform-bin model, the endpoint intervals constructed 13  
14 with (55) and (15) cover all declared expected policy-cost differences simultaneously with 14  
15 probability at least 0.99. If an interval  $[L_{mn}, U_{mn}]$  has  $U_{mn} < 0$ , policy  $m$  has smaller 15  
16 expected cost than  $n$  on this event. For a previously specified tolerance  $\eta$ ,  $U_{mn} \leq \eta$  certifies 16  
17 noninferiority within that tolerance, whereas  $[L_{mn}, U_{mn}] \subset [-\eta, \eta]$  certifies two-sided 17  
18 proximity. 18*

19 More generally, fix a reference policy  $b$  and known implementation charges  $k_m$ . Suppose 19  
20 simultaneous intervals  $[L_m, U_m]$  enclose  $J_m - J_b$  for a finite candidate catalogue, including 20  
21  $[0, 0]$  for  $b$ . Select  $\hat{m}$  to minimize  $U_m + k_m$ . On the coverage event, 21

$$J_{\hat{m}} + k_{\hat{m}} \leq \min_m \{J_m + k_m + (U_m - L_m)\}. \quad (56)$$

22  
23  
24 Thus an economic choice made after seeing all intervals retains its declared uncertainty 24  
25 account. The catalogue comparison is within one common economic task and initial-state 25  
26 law, not across different prices, horizons or evaluations. 26  
27

28 PROOF: The score identity identifies the expected difference. Its interval end- 28  
29 points are bounded independent observations within each group. Apply the one- 29  
30 sided empirical Bernstein inequality of Maurer and Pontil (2009) to each lower 30  
31 and upper endpoint and take a union bound over 240 tails. The numerical enclo- 31  
32 sures only enlarge the valid intervals. Independence between different contrasts is 32

unnecessary. On the resulting event,  $J_{\hat{m}} - J_b \leq U_{\hat{m}}$  and  $U_m \leq J_m - J_b + (U_m - L_m)$ .  
Minimize the former upper bound plus the known charge to obtain (56). *Q.E.D.*

The study fixes  $\eta = 1/1024$  in its normalized cost units before execution. It is a stated decision tolerance, not an estimated welfare threshold. Zero remains the superiority threshold. An interval merely intersecting zero meets neither criterion automatically. The fixed pseudorandom implementation supplies repeatable inputs; it does not prove the independent sampling model. The theorem states that model explicitly. No claim about initialization risk or the distribution of newly trained policies follows from these frozen-policy comparisons.

TABLE X

DIRECT EXPECTED-COST COMPARISONS UNDER A PREDECLARED DECISION TOLERANCE

| Contrast               | Groups | Signed | Within 1/1024 | Largest endpoint |
|------------------------|--------|--------|---------------|------------------|
| Witness – Cone nearest | 40     | 0      | 40            | 0.0001161        |
| Witness – Spline       | 40     | 0      | 40            | 0.0001975        |
| Cone nearest – Spline  | 40     | 0      | 40            | 0.0002078        |

Signed counts exclude intervals containing zero. “Within” requires the entire interval inside  $[-1/1024, 1/1024]$ . Largest endpoint is the largest absolute endpoint over the 40 groups. All 120 intervals use one 0.01 family error account, conditional on the frozen policies and independent-bin model. Each group uses 131,072 common paths.

Every direct interval contains zero, so the experiment identifies no signed policy ranking. Every interval is also contained in the predeclared decision band. It therefore certifies two-sided expected-cost proximity for the specified initial laws at that tolerance, rather than inferring equivalence from a failure to reject equality. The uniform initial-state law is a distributional conclusion, not an all-state assertion. The individual initial states are separate declared estimands.

### 13.1. *Work as a function of the certified bound*

A coarse target list can hide a genuine difference between certificate frontiers. For the original saved rungs  $j$ , let  $G_{m,j}$  be the valid bound and let  $C_{m,j}$  include all work

through rung  $j$ . The complete first-crossing curve is

$$W_m(\varepsilon) = C_{m,j_m(\varepsilon)}, \quad j_m(\varepsilon) = \min\{j : G_{m,j} \leq \varepsilon\}, \quad (57)$$

with  $W_m = +\infty$  when the cap is exhausted. Its breakpoints are the recorded bounds; both methods' breakpoints partition the entire positive tolerance axis. We reconstruct every interval of this partition, not only intervals favorable to the witness. This is an exact descriptive account of frozen evidence, not a retrospective change to the original targets.

TABLE XI  
COMPLETE TOLERANCE-AXIS PARTITION OF THE ORIGINAL SCALAR FRONTIERS

| Relation on a partition interval        | Number |
|-----------------------------------------|--------|
| Both attain; witness uses fewer queries | 19     |
| Both attain; spline uses fewer queries  | 1      |
| Both attain; same prefix queries        | 20     |
| Only witness attains within the cap     | 4      |
| Only spline attains within the cap      | 0      |
| Both exhaust the cap                    | 4      |

The complete curves contain 19 nonempty tolerance intervals on which both methods attain and the witness uses fewer Bellman queries, one favoring the spline, and twenty with identical counts. Four further intervals are attainable only by the witness; four below both final bounds are unattained. Interval counts are not probabilities or measures of economic importance. Above the largest breakpoint in each cell both first-rung counts coincide. The original targets remain unchanged and still have identical crossing resolutions. These descriptive intervals identify where a sharper certificate changes a resource decision without manufacturing new prospective targets.

### 13.2. *Economic interpretation and the NBO program*

The new evidence connects a feasible implementation to two distinct economic statements. First, the acquisition-aware theorem bounds the loss of a policy that can

actually be executed under uncertain state measurement and capacity constraints. Second, the direct experiment certifies proximity of the original scalar policy costs within an explicitly chosen decision tolerance, so that a user of that catalogue may consider implementation charges without treating separate regret bounds as values. These statements are stronger than a certificate comparison alone.

Neither statement implies that a neural representation is indispensable. The identical native envelope is a mathematical control, and the conventional fitted-value and spline implementations remain strong alternatives. The constrained experiment demonstrates two-state continuous-uncertainty construction and nontrivial repair, not a high-dimensional scaling result or an estimated economic application. Its tight targets remain unachieved. The direct cost experiment does not cover the new constrained policies or nonlinear recursive utility; its expectation-specific conclusion is not transferred to those settings. The controlled-diffusion, recursive-preference, endogenous-preference, temporal-self and game developments below retain their own conditions and original role in the NBO program. Their preservation is not counted as a new comparative execution.

#### 14. PRICED OBSERVATION AND DIRECT CONSTRAINED-POLICY DECISIONS

A state-error radius is an economic input when acquiring more precise information is costly. We specify a finite-bit observation technology and a controller replacement charge. These are primitives of a theoretical decision problem in resource units, not estimates from a calibrated economy. They permit a decision interpretation without treating an interval containing zero as equality or treating an all-state regret bound as a ranking of two policies.

##### 14.1. *Choosing the observation precision*

On  $[0, 1]^d$ , a  $b$ -bit sensor reports, in each coordinate, the containing dyadic interval and its midpoint. The right endpoint 1 belongs to the last interval. The resulting acquisition set has  $\ell^1$  radius

$$r(b) = d2^{-(b+1)}. \quad (58)$$

For  $A_t(x) = [0, \kappa_t(x)]$ , use the lowest capacity on the reported box, then round the selected action downward to spacing  $\Delta_t$ . Feasibility holds throughout that box. In the affine capacity economy this lower capacity is computed exactly from the lower box coordinates. The acquisition repair therefore has  $\chi_t = 0$  in Corollary 12.

Suppose the same  $b$  is chosen before date zero and each coordinate-bit costs  $\lambda > 0$  per date. The observation fee is paid in the same additive resource units as the objective. With  $B_t = \prod_{s < t} \beta_s$ , it is  $Cb$ , where  $C = \lambda d \sum_{t < T} B_t$ . Write  $G$  for the full-information witness loss certificate with exact selector and repair. The acquisition theorem implies an augmented loss bound, relative to the full-information optimum without the observation fee,

$$J^{\pi, b}(x) + Cb - V^*(x) \leq G + K_\Delta + A2^{-b} + Cb, \quad (59)$$

$$A = d \sum_{t < T} B_t (L_t + D_t^Q K_t^A), \quad K_\Delta = \sum_{t < T} B_t D_t^Q \Delta_t.$$

A known selector allowance is added to  $G$  when necessary. The fee is a deterministic date cost, so the same argument also applies to a common cash-invariant recursive evaluation for which the earlier comparison theorem holds. Direct sampling below is, separately, expectation-specific.

**PROPOSITION 19**—Optimal precision for the certified economic account: *On a declared integer range  $b_{\min} \leq b \leq b_{\max}$ , the smallest minimizer of the right side of (59) is characterized by the neighboring inequalities*

$$C \leq A2^{-b} \quad \text{if } b > b_{\min}, \quad C \geq A2^{-(b+1)} \quad \text{if } b < b_{\max}, \quad (60)$$

*with the smallest minimizer selected when equality gives two. The criterion can be decided by rational comparisons, without an unverified logarithm. It minimizes a proved upper account; it need not minimize the unknown actual policy cost as a function of precision.*

**PROOF:** The difference between the accounts at  $b + 1$  and  $b$  is  $C - A2^{-(b+1)}$ , which is nondecreasing in  $b$ . The two neighboring comparisons are therefore necessary and sufficient, with the indicated endpoint convention. Equation (59) follows by substituting (58) and  $\zeta_t^{\text{acq}} = 2K_t^A r(b) + \Delta_t$  into (45). Q.E.D.

This calculation provides an explicit information-price comparative static: an increase in  $\lambda$  weakly decreases the smallest minimizing precision, while an increase in the sensitivity account  $A$  weakly increases it. The result concerns the guarantee, not a fitted distribution of measurement errors. The executed curves use every  $b = 4, \dots, 16$  and every  $\lambda \in \{1/16384, 1/4096, 1/1024\}$ . All curves and endpoint choices are retained, rather than reporting only an attractive radius.

#### 14.2. *The actual constrained-policy cost*

Fix the two-state witness and bilinear-FVI policies stored at the finest original R47 resolution,  $N = 16$ . In each economic cell, the two policies are evaluated with the same observation contract: exact state, six bits per coordinate, or ten bits per coordinate. The finite-bit policies use robust capacity repair and action spacing  $1/4096$ ; the exact-state policies are unquantized. Thus there are three distinct pairs of implemented policies, not three observations of an assumed continuous actor.

For an initial law  $\mu$ , the estimand is

$$D_{\mu,b} = \mathbb{E}_{\mu} \left[ \sum_{t < T} \beta^t \{c(X_t^W, A_t^W) - c(X_t^F, A_t^F)\} + \beta^T \{g(X_T^W) - g(X_T^F)\} \right]. \quad (61)$$

The observation fees cancel within a pair because its two controllers use the same sensor. The paired policies share innovations but follow their own endogenous state paths. This is their actual cost difference, not a difference of regret certificates. A raw path score is used here; the earlier own-continuation score remains a valid alternative when its signed support is sharper.

For the two-state primitives,  $0 \leq c \leq (99 + 4p)/64$  and  $0 \leq g \leq 37/16$ . Consequently the path difference belongs to  $[-H, H]$ , with

$$H = \frac{99 + 4p}{64} \sum_{t < T} \beta^t + \frac{37}{16} \beta^T. \quad (62)$$

This uniform support is conservative. It is independent of the observed contrast and is fixed before simulation.

**THEOREM 20**—Interval simulation and replacement decisions: *For each of  $K$  fixed policy-pair tasks, partition the underlying initial and innovation uniforms into equal bins*

and draw independent uniform bin indices. Suppose a numerical path routine returns bounded random endpoints  $L_j, U_j$  enclosing the exact path difference for every continuous realization inside the selected bins, including every possible actor, observation and repair branch. Let  $[a, b]$  be a deterministic support enclosure and clip endpoints to it. For  $n \geq 2$ , let  $\underline{m}_L, \overline{m}_U$  enclose their sample means and  $s_L^2, s_U^2$  upper-bound their unbiased sample variances. Put

$$R(s^2) = \sqrt{\frac{2s^2\ell}{n}} + \frac{7(b-a)\ell}{3(n-1)}, \quad \ell \geq \log(4K/\alpha). \quad (63)$$

Then, under the independent-bin sampling model, simultaneously for all  $K$  tasks with probability at least  $1 - \alpha$ ,

$$D_{\mu,b} \in [\max\{a, \underline{m}_L - R(s_L^2)\}, \min\{b, \overline{m}_U + R(s_U^2)\}]. \quad (64)$$

If replacing either installed controller costs  $k > 0$  once at date zero, an interval strictly inside  $(-k, k)$  proves that neither replacement recoups that fee in expected resource cost for that task. An interval with upper endpoint below zero ranks the witness policy below FVI; a lower endpoint above zero ranks it above FVI. Containing zero does not establish equality.

PROOF: Conditional on a bin sequence, integrating the continuous uniform variables gives a conditional path expectation between its enclosing endpoints. Averaging over uniform bin indices reproduces the original continuous law, so  $\mathbb{E}L_j \leq D_{\mu,b} \leq \mathbb{E}U_j$ . Apply the empirical-Bernstein inequality of [Maurer and Pontil \(2009\)](#) to the lower and upper endpoint samples, with a union bound across  $2K$  one-sided assertions; certified moment and arithmetic allowances enlarge, rather than shrink, the intervals. Finally, switching from FVI to witness changes cost by  $D_{\mu,b} + k$ , while switching in the opposite direction changes it by  $k - D_{\mu,b}$ . Both are positive when  $|D_{\mu,b}| < k$ . Q.E.D.

The experiment fixes  $K = 48$ ,  $n = 262144$  and  $\alpha = 0.01$  before execution. An exact rational exponential bound verifies that  $\ell = 10$  is sufficient. Forty-bit bins cover the continuous initial and innovation laws. The initial laws are independent uniform coordinates and the three fixed vectors  $(1/8, 1/8)$ ,  $(1/2, 1/2)$  and  $(7/8, 7/8)$ , crossed with all four horizon-price cells and three sensors. A fixed pseudorandom stream

1 makes the records reproducible; it is not a proof that those indices are independent 1  
 2 random variables. 2

3 The replacement fee is  $k = 1/64$ , one quarter of the maximal quadratic investment 3  
 4 expenditure at  $p = 1$ . It is an explicit theoretical transaction charge. We do not 4  
 5 calibrate it to welfare data, choose it after inspecting the intervals, or infer an all- 5  
 6 state ranking from the four initial laws. Reporting the zero threshold separately 6  
 7 keeps the decision statement distinct from a claim of neural superiority. 7

## 8 15. COMPLETE COMPILED-POLICY SERVICES AND ECONOMIC COMPARISONS 9

10 The new study separates three questions: the cost of representing a fixed NBO 10  
 11 policy, the complete cost of constructing and certifying a policy, and the actual cost 11  
 12 of using one controller rather than another. A representation benchmark cannot by 12  
 13 itself answer either of the latter questions. Conversely, a conventional comparator 13  
 14 receives the same innovation law, feasibility conditions and all-state comparison 14  
 15 principle, not a deliberately inefficient implementation chosen to favor the neural 15  
 16 operator. 16

### 17 15.1. *Design, stopping rules and conventional baselines* 18

19 The design was fixed before the complete execution. The principal catalogue uses 19  
 20 the original constrained two-state economy of Section 12, horizons  $T = 2, 3$ , prices 20  
 21  $p = 1, 4$ , and  $N = M \in \{4, 8, 16, 32, 64\}$ . The three methods are the compiled cone 21  
 22 witness, uniform tensor FVI and error-driven nonuniform tensor FVI. There are 22  
 23 three isolated process repetitions for each method and economic cell: 36 services and 23  
 24 180 rungs. Every rung starts from primitives and uses its own future continuation. 24  
 25 All rungs are executed, including those after a target crossing; first-target work 25  
 26 charges the complete prefix, including every prior failure. 26  
 27 27

28 The conventional adaptive method forms an own-future pilot on a  $5 \times 5$  grid, 28  
 29 uses its coordinatewise second differences to allocate nonuniform dyadic knots, and 29  
 30 certifies with the actual largest coordinate gaps. Its pilot, allocation and repeated 30  
 31 target evaluations are charged. The section on nonuniform fitted-value comparison 31  
 32 in the supplement specifies the fixed heuristic and proves the strengthened nearest- 32

node certificate on these unequal grids. It is a conventional adaptive-coordinate FVI comparator, not a sparse-grid method or an unrestricted optimal adaptive partition.

All methods share outward arithmetic, feasible action fractions and midpoint integration with its deterministic remainder under the continuous uniform law. Each service includes a finite acquired-state deployment checksum and durable checkpoint output. Complete prefix wall and CPU times, method-specific evaluation counts, coefficient bit lengths, stored actor scalars, checkpoint bytes and peak resident memory are recorded. The internal clock excludes startup and the common non-economic warm-up; a separate process clock includes them. No independent cost comparison is hidden inside that boundary: its joint service is reported separately below.

TABLE XII

COMPLETE TWO-STATE CONSTRUCTION SERVICES AT THE DECLARED CAP

| $T$ | $p$ | Method | Loss bound | Median (s) | Range (s)        | Peak MiB |
|-----|-----|--------|------------|------------|------------------|----------|
| 2   | 1   | C      | 0.579      | 48.001     | [47.981, 49.481] | 92.0     |
| 2   | 1   | U      | 0.719      | 36.308     | [36.280, 36.426] | 74.5     |
| 2   | 1   | A      | 0.719      | 36.689     | [36.685, 36.696] | 74.6     |
| 2   | 4   | C      | 0.592      | 48.077     | [47.790, 48.600] | 91.3     |
| 2   | 4   | U      | 0.727      | 36.391     | [36.389, 37.257] | 73.5     |
| 2   | 4   | A      | 0.727      | 37.025     | [36.669, 37.110] | 73.7     |
| 3   | 1   | C      | 0.830      | 71.933     | [71.549, 72.464] | 93.1     |
| 3   | 1   | U      | 1.003      | 54.649     | [54.146, 54.650] | 74.8     |
| 3   | 1   | A      | 1.003      | 55.677     | [55.256, 57.886] | 75.5     |
| 3   | 4   | C      | 0.853      | 71.963     | [71.798, 72.082] | 93.5     |
| 3   | 4   | U      | 1.013      | 54.151     | [53.822, 54.812] | 74.9     |
| 3   | 4   | A      | 1.013      | 55.026     | [54.930, 55.110] | 75.3     |

C denotes compiled witness, U uniform tensor FVI, and A error-driven coordinate FVI. Each entry ends at  $N = M = 64$  and includes the complete five-rung prefix through durable record output. The range is over three isolated repetitions with identical checkpoints, not three economic draws. Peak resident memory includes the interpreter and warm-up. Bounds are displayed to three decimals; exact rational tests determine attainment.

Against uniform FVI, compiled witness is earlier in 3 of 30 common successful target/repetition comparisons, later in 27, and tied in 0. Against error-driven FVI,

TABLE XIII  
PROSPECTIVE TARGET ATTAINMENT IN THE TWO-STATE CATALOGUE

| Method | 4     | 2     | 1     | 1/2  | 1/4  |
|--------|-------|-------|-------|------|------|
| C      | 12/12 | 12/12 | 12/12 | 0/12 | 0/12 |
| U      | 12/12 | 12/12 | 6/12  | 0/12 | 0/12 |
| A      | 12/12 | 12/12 | 6/12  | 0/12 | 0/12 |

All failures and every prior rung are retained. Counts concern twelve deterministic services per method, crossing four economic cells and three timing repetitions. A repeated failure is not an independent estimate of optimizer reliability. No historical failure is reclassified.

compiled witness is earlier in 15 of 30 common successful target/repetition comparisons, later in 15, and tied in 0. These are observed complete-prefix wall times, not a hardware-general speed law. The comparison uses the prospective targets rather than retrospectively selecting a favorable breakpoint. The full positive-tolerance partition, including one-sided and joint nonattainment, is reported in the supplement and machine-readable frontier. The error-driven rule returns uniform grids in all 210 recorded date models; all 60 adaptive checkpoints have exactly the same continuations and actors as their uniform-FVI counterparts. Thus its additional observed cost is pilot and allocation overhead, not evidence of a realized nonuniform-grid improvement. The nonuniform verification theorem is proved and tested, but an effective spatial-adaptation advantage is not established by this catalogue. One prospective tolerance illustrates the distinction between common-resolution speed and target work. At horizon three, price four and target two, witness crosses at  $N=32$  while uniform FVI first crosses at  $N=64$ . Median complete-prefix times are 5.498 and 54.146 seconds, respectively. This is one economic cell with three timing repetitions. At target one, witness attains all twelve services and each FVI implementation attains six; none attains one half or one quarter at its cap.

## 15.2. Representation identity and higher-dimensional execution

The fixed-object diagnostic uses all twelve distinct original constrained witness checkpoints, crossing  $T = 2, 3$ ,  $p = 1, 4$  and  $N = 4, 8, 16$ . Both the original dense evaluator and the compiler evaluate the same fixed point catalogue. Compiler setup

is charged. Exact rational off-grid checks verify values and original winning indices, including the deterministic tie convention; outward machine intervals are also checked. These clocks do not include new target generation and are not added to historical training times.

TABLE XIV

FIXED-POLICY REPRESENTATION COSTS, INCLUDING COMPILER SETUP

| $N$ | Observations | Median ratio | Minimum | Maximum | Compiled (s) |
|-----|--------------|--------------|---------|---------|--------------|
| 4   | 12           | 1.93         | 1.87    | 2.37    | 0.0074       |
| 8   | 12           | 2.92         | 2.80    | 4.10    | 0.0137       |
| 16  | 12           | 4.48         | 4.42    | 4.52    | 0.0385       |

The ratio is dense load-plus-query time divided by compiled setup-plus-query time, over the same original continuation and points. Each resolution crosses four economic cells and three repetitions. This is a same-policy representation experiment, not a construction-to-certificate speedup or an independently trained native comparator. Exact witness-index checks accompany the numerical interval checks.

To expose, rather than suppress, dimension factors, a cycle-coupled extension executes  $d = 3, 4$  with continuous common uncertainty, state-dependent capacity and a nontrivial scalar action. Its normalized stage and terminal costs recover the two-state economy exactly at  $d = 2$ . The full primitive formulas and valid moduli appear in the supplement. The stress catalogue fixes  $p = 1$ ,  $T = 2, 3$ , and the compiled witness and uniform multilinear FVI methods. Dimension three uses  $N = M = 4, 8, 16$ ; dimension four uses  $N = M = 2, 4, 8$ . This adds eight services and 24 rungs. The smaller four-dimensional cap is explicit. These are memory-limited stress designs with one observation per cell, not repeated fixed-accuracy scaling estimates. They test an executed nonlinear implementation in dimensions above two without turning the finite-dimensional theorem into a dimension-free performance claim. The retained tensor state count, action factors, innovation work and  $2^d$  corner terms determine what this implementation can afford.

TABLE XV

EXECUTED NONLINEAR DIMENSION STRESS CASES

| $d$ | $T$ | Method | $N$ | States | Loss bound | Seconds | MiB  |
|-----|-----|--------|-----|--------|------------|---------|------|
| 3   | 2   | C      | 16  | 4913   | 2.330      | 9.505   | 64.9 |
| 3   | 2   | U      | 16  | 4913   | 2.814      | 6.677   | 52.3 |
| 3   | 3   | C      | 16  | 4913   | 3.349      | 13.906  | 66.5 |
| 3   | 3   | U      | 16  | 4913   | 3.915      | 9.905   | 52.6 |
| 4   | 2   | C      | 8   | 6561   | 4.630      | 7.733   | 69.7 |
| 4   | 2   | U      | 8   | 6561   | 5.456      | 7.368   | 50.3 |
| 4   | 3   | C      | 8   | 6561   | 6.642      | 11.550  | 71.2 |
| 4   | 3   | U      | 8   | 6561   | 7.589      | 10.982  | 51.2 |

Price is one. Each service includes all three declared rungs and has one repetition. The maximum resolution differs by dimension and is not a matched-accuracy scaling design. The common innovation remains continuous; its quadrature remainder, state-dependent capacity and all-state loss account are retained.

### 15.3. *Direct constrained-policy decisions and information prices*

The direct comparison freezes the original  $N = 16$  witness and FVI policies. It does not select the new R48 policies after examining their costs. Forty-eight policy-pair tasks cross the four economic cells, four initial laws and three observation contracts of Section 14. The full study contains  $48 \times 262144 = 12582912$  paired paths. Numerical path widths, every unresolved actor or sensor branch, endpoint moments, simultaneous intervals and decisions are deposited. Table XVI summarizes interval hulls across the four declared initial laws; its rows are not all-state confidence bands. The supplement reports every interval separately. The zero threshold and the replacement-fee threshold  $1/64$  are different decision questions. A no-recoupment result is conditional on that predeclared theoretical fee. It neither establishes exact policy equality nor measures empirical welfare equivalence.

The sensor-price experiment evaluates all thirteen precisions  $b = 4, \dots, 16$  at each of the three declared information prices, for every original witness checkpoint. Table XVII gives the finest original resolution; the evidence includes all coarser checkpoints and every candidate precision, not only the minimizing choice. The optimum minimizes the uniform augmented upper account. Since the full-information

TABLE XVI

DIRECT WITNESS-MINUS-FVI CONSTRAINED-POLICY COSTS

| $T$ | $p$ | Sensor | Interval hull over four laws | Signed | No fee recovery |
|-----|-----|--------|------------------------------|--------|-----------------|
| 2   | 1   | exact  | $[-0.001247, 0.002717]$      | 1/4    | 4/4             |
| 2   | 1   | 6      | $[-0.001113, 0.002716]$      | 1/4    | 4/4             |
| 2   | 1   | 10     | $[-0.001220, 0.002717]$      | 1/4    | 4/4             |
| 2   | 4   | exact  | $[-0.001142, 0.001840]$      | 0/4    | 4/4             |
| 2   | 4   | 6      | $[-0.001014, 0.001602]$      | 0/4    | 4/4             |
| 2   | 4   | 10     | $[-0.001151, 0.001832]$      | 0/4    | 4/4             |
| 3   | 1   | exact  | $[-0.001501, 0.002627]$      | 1/4    | 4/4             |
| 3   | 1   | 6      | $[-0.001364, 0.002528]$      | 1/4    | 4/4             |
| 3   | 1   | 10     | $[-0.001455, 0.002639]$      | 1/4    | 4/4             |
| 3   | 4   | exact  | $[-0.001356, 0.001806]$      | 0/4    | 4/4             |
| 3   | 4   | 6      | $[-0.001342, 0.001783]$      | 0/4    | 4/4             |
| 3   | 4   | 10     | $[-0.001365, 0.001808]$      | 0/4    | 4/4             |

Each hull contains the four separate simultaneous intervals for the declared initial laws; it is not an all-state band. There are 48 contrasts, each with 262,144 common interval paths, and family error at most 0.01 under the independent-bin model. Signed counts identify intervals excluding zero. The final column counts intervals strictly inside  $(-1/64, 1/64)$ , ruling out recouping either replacement fee. Display endpoints are rounded outward. The observation fee cancels when the two controllers use the same sensor.

certificate  $G$  is unchanged across observation precisions, the integer choice responds to the objective sensitivity and the price of information, not to an arbitrary selection of an attractive state-error radius. A coarse state cover can still make the total account large; paying for more bits cannot repair that separate construction error.

#### 15.4. *What changes in the evidence*

The compiler theorem supplies a constructive work reduction for a defined NBO backend without changing its policy. The complete catalogue measures that backend against uniform and error-driven conventional FVI with all failed resolutions included. Direct constrained costs and priced observation support explicit economic decisions for fixed policies under declared information contracts. These results extend the original program, rather than replace it with a different subject.

TABLE XVII

PRICED PRECISION FOR THE ORIGINAL FINEST WITNESS POLICIES

| $T$ | $p$ | Bit price         | $b_*$ | Acquisition | Fee    | Total account |
|-----|-----|-------------------|-------|-------------|--------|---------------|
| 2   | 1   | $\frac{1}{16384}$ | 16    | 0.0027      | 0.0038 | 2.3212        |
| 2   | 1   | $\frac{1}{4096}$  | 14    | 0.0040      | 0.0133 | 2.3320        |
| 2   | 1   | $\frac{1}{1024}$  | 12    | 0.0092      | 0.0455 | 2.3694        |
| 2   | 4   | $\frac{1}{16384}$ | 16    | 0.0034      | 0.0038 | 2.3763        |
| 2   | 4   | $\frac{1}{4096}$  | 14    | 0.0047      | 0.0133 | 2.3871        |
| 2   | 4   | $\frac{1}{1024}$  | 12    | 0.0101      | 0.0455 | 2.4247        |
| 3   | 1   | $\frac{1}{16384}$ | 16    | 0.0043      | 0.0056 | 3.3310        |
| 3   | 1   | $\frac{1}{4096}$  | 14    | 0.0063      | 0.0193 | 3.3468        |
| 3   | 1   | $\frac{1}{1024}$  | 12    | 0.0145      | 0.0661 | 3.4017        |
| 3   | 4   | $\frac{1}{16384}$ | 16    | 0.0054      | 0.0056 | 3.4215        |
| 3   | 4   | $\frac{1}{4096}$  | 15    | 0.0061      | 0.0207 | 3.4374        |
| 3   | 4   | $\frac{1}{1024}$  | 13    | 0.0103      | 0.0716 | 3.4925        |

The original  $N = 16$  checkpoint is fixed. Prices are per coordinate-bit per date;  $b$  ranges from four to sixteen and action spacing is  $1/4096$ . The selected bit count minimizes the exact augmented uniform upper account, not measured policy cost. All thirteen candidate precisions and the coarser original checkpoints remain in the evidence. Displayed nonnegative accounts are rounded upward.

The historical adverse findings remain: the original nonlinear optimizer catalogue, selected residual-only recertifications, unresolved neural-ridge signs, and mixed-precision timing records are not reclassified. Identical native and ReLU envelopes remain one mathematical construction. The new deterministic repetitions do not estimate a population success probability for nonconvex optimization; the direct confidence statements concern their fixed policies under the sampling model; the stress clocks do not identify hardware-general scaling. The comparative claims in this article are restricted to the objects actually proved or executed, while the controlled-economy and application-specific theoretical scope remains intact.

## 16. ECONOMIC APPLICATIONS AND CONCLUSION

The original applications concern policy-dependent recursive utility, endogenous preference specifications, temporal selves, controlled diffusion problems, and dynamic games. Their complete economic statements and proofs are retained without

alteration. The centered and transport bounds explain how an NBO can support economic counterfactuals when changes in future primitives are controlled. The policy theorem explains what must be verified for the actual policy. Neither statement supplies an equilibrium or transversality condition absent from the application.

The nonlinear investment evidence adds an executable link from primitives to learned continuations and directly certified policies. Its retained-policy diagnostic demonstrates that a failed tight certificate can reflect the resolution of verification rather than the quality of the deployed policy. Its direct coupled comparison answers a different question—the relative cost of two policies—without confusing separate regret bounds with a method contrast.

The acquired-state result makes that returned policy implementable under a stated information and arithmetic contract. The constrained economy supplies finite covering, integration and repair routines; the direct scalar experiment supplies simultaneous tolerance-based cost comparisons and a catalogue-adoption rule. Their distinct assumptions and work accounts remain visible.

The compiler adds a policy-preserving computational step to this program. It connects the explicit neural continuation to an executable representation whose per-query work no longer scans the full state cover. The economic implementation adds two further choices: how precisely to acquire the state, and whether replacing an installed controller pays for itself. The first is governed by a priced, certified information account; the second by direct expected costs with an explicit replacement charge. They are not inferred by ranking regret bounds.

The empirical content is correspondingly separated. Frozen-object experiments identify representation costs for the same policy; complete services compare fresh own-future constructions against conventional FVI; direct intervals concern fixed policies and declared initial laws. The higher-dimensional executions expose the remaining tensor and innovation costs. They do not constitute a dimension-free approximation theorem, a calibration to welfare data, or a reliability law for unrestricted neural training. Within these boundaries, the operator supplies a constructive route from economic primitives through a feasible implemented policy to a verified resource decision.

Neural Bellman Operators thus organize construction, certification, economic reuse, and resource accounting around the policy actually returned. Their scientific assessment requires all of these objects. Flexible conventional methods remain important competitors; numerical activation is not a work saving, and a fixed catalogue is not a general reliability theorem. Theorem 5 and Corollary 6 now supply a separate deterministic reliability statement for an explicitly constructed neural backend. They do not retroactively prove convergence of the earlier optimizers. Preserving those distinctions makes the original economic program testable while retaining its full theory and applications.

#### APPENDIX A: READING AND PRESERVATION MAP

The main article is the authoritative R48 exposition of the existing Neural Bellman Operators paper. It includes the controlled-economy formulation, centered transport and policy comparison, constructive continuation and original witnesses, acquired-state feasibility, exact compilation, and the economic decision analysis. The technical supplement supplies the proofs and numerical accounts needed for those results, as well as every new direct interval and the complete first-target time partitions. All R47 main and supplement labels, including their inherited results, remain in the corresponding active documents. The current response addresses the R47 report item by item.

The experimental sections distinguish frozen historical evidence from new executions. Earlier optimizer, precision, scalar and coupled policy records are not regenerated or relabeled. The full historical theory and application companions are retained as ordinary files in the revision tree; preservation is not asserted to be a new validation of every application. The main finite-horizon certificate needs the monotonicity, cash-invariance, feasibility and comparison assumptions stated here. The direct sampling result additionally uses conditional expectation and its sampling model. Controlled diffusions, infinite horizons, recursive utility domains and equilibrium applications retain their separate analytic conditions.

The source tree contains the ordinary manuscript, proofs, bibliography, table generator, scientific sources, complete frozen records and build instructions. A clean

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|----|--------------------------------------------------------------------------------------|----|
| 1  | rebuild audits the records and regenerates the documents without a new scientific    | 1  |
| 2  | run or an expiring artifact download. The publication audit binds the review anchor, | 2  |
| 3  | scientific source, evidence, tests and compiled documents. These checks establish    | 3  |
| 4  | a reproducible submitted object; they are not a substitute for mathematical peer     | 4  |
| 5  | review.                                                                              | 5  |
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## APPENDIX: CURRENT R52 DEVELOPMENT

## APPENDIX B: INTRODUCTION

A continuation value is useful to an economist because it changes a decision. Its prediction error is not the loss of the policy actually implemented. Nor is a smaller upper bound on that loss evidence that one policy has lower cost than another. When observation, action selection and verification are costly, the numerical procedure and the economic decision have to be evaluated together.

We develop Neural Bellman Operators as a policy-evaluation and feasible-improvement method. The economic object remains a controlled economy with policy-specific continuations. The operator constructs each date's continuation using its own fitted future, returns a feasible policy, and accounts for its loss against the Bellman optimum. A general signed policy sandwich permits arbitrary fitted candidates. A constructive affine-ReLU backend additionally produces the candidate from primitive moduli, without assuming that a nonconvex training algorithm eventually succeeds. Exact compilation retains the original action witness through off-grid evaluation and state-dependent feasibility repair.

The principal question in this revision is how that construction produces a certified economic answer with finite resources. State coverage, action search and integration have different error coefficients. Tying them to one grid size conceals the allocation problem. We derive a separated primitive budget and an optimal continuous allocation for its leading work proxy, including lower resource bounds and dyadic implementation. The empirical comparison then records the complete positive-tolerance frontier of a fixed common resource ladder, not only an isolated favorable target. All failed earlier attempts belong to the work required to return a policy.

The next question is what the returned policy does. We evaluate the actual policies at the declared first crossings, even when two generators cross at different resolutions. Independent interval simulation preserves the original continuous initial and innovation laws and encloses discontinuous actors and robust capacity repair. Conditioning on construction information permits simultaneous cost inference after this selection. The result does not infer a policy ranking from two certificates and

does not extrapolate three deterministic timing repetitions into an optimizer-success probability.

Two economic decisions make this distinction operational. First, a controller replacement is worthwhile only if its actual gain exceeds the replacement fee. We identify the directional gain intervals and the full break-even fee account, retaining but no longer relying on a single generous charge. Second, observation precision is chosen on the same finite bit grid on which the implemented policies are evaluated. A simultaneous upper-net-cost selector has a regret bound relative to the best precision in that catalogue. The bound holds for every nonnegative information price on one coverage event. This is a result about actual net policy cost, different from minimizing a conservative all-state loss account.

These advances do not depend on calling a classical envelope neural. A native min-plus envelope and its exactly equivalent ReLU circuit are one method with two encodings. The original-owner compiler is available to both. We compare that method with uniform multilinear fitted-value iteration (FVI) and two separately recorded conventional refinement rules. The original curvature-bisection block remained uniform; the later graded block was nonuniform. A new residual-driven comparator below records its realized geometry and all pilot work. Each generates its own future labels and receives the same feasible action menu, continuous-law integration and one-sided economic criterion. The evidence must determine the cost of the returned policy as well as certificate sharpness. No uniform neural dominance is built into the design.

**B.0.0.1. *Related methods.*** Lipschitz extension supplies the cone construction ([McShane, 1934](#)), and separable distance transforms supply its linear-work nodal engine ([Felzenszwalb and Huttenlocher, 2012](#)). Approximation in min-plus spaces has an established dynamic-programming literature: [Lakshminarayanan and Bhatnagar \(2014\)](#) study min-plus projected Bellman equations and supremum-norm errors. Bellman-inequality methods likewise construct lower value bounds and policies; [Wang et al. \(2015\)](#) use iterated inequalities with linear or semidefinite optimization. Our argument does not claim priority for these principles. Its subject is their connection to an own-future feasible policy, original-owner continuous compilation,

acquired-state implementation, complete resource allocation and direct economic comparison of the returned objects. The empirical-Bernstein component uses the bounded-sample inequality of [Maurer and Pontil \(2009\)](#); conditional selection and numerical path enclosures specify the estimand to which it is applied.

The broader NBO program includes controlled diffusions, recursive utility, endogenous preferences, temporal selves and games. Those results and their assumptions are retained in the complete development edition and its full proof supplement, together with the original training, matrix-factor and precision experiments. The present article supplies a self-contained constructive theorem–evidence chain. Its direct-cost statistical theorems concern expected discounted costs, not an undeclared extension to every nonlinear recursive application. This organization preserves the paper’s economic subject while making the current contribution and the actual comparison burden explicit.

The additional improvement result addresses the economically important case where the certified construction returns a more costly policy than a conventional competitor. We preserve that incumbent and permit only actions whose actual reference advantage is certified nonpositive after the numerical and observation allowances. At the final date of the original investment model, exact continuous-law integration makes this gate executable without assuming a successful policy-value fit. Both generators receive the same repair. The resulting experiment isolates removable final-action loss from the policy gap that remains; it does not rename the compiler as a uniquely neural advantage.

#### APPENDIX C: THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates  $t = 0, \dots, T$  and an invariant compact state domain  $X \subset \mathbb{R}^d$ . At date  $t$ , the action belongs to a nonempty compact set  $A_t(x)$ . The transition is  $F_t(x, a, Z_{t+1})$ , the current cost is  $c_t(x, a)$ , and the terminal cost is  $g(x)$ . Write  $B_0 = 1$  and  $B_t = \prod_{j < t} \beta_j$ , with  $\beta_j > 0$ . An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation  $v$ , define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \quad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \quad (65)$$

Throughout the deterministic comparison, the evaluations  $\rho_t$  are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the experiments  $\rho_t$  is conditional expectation, costs are bounded and innovations have their original continuous uniform law. The direct-cost inference in Section I is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (65). It forms targets using its *own* fitted  $f_{t+1}$ , returns a neural continuation and a feasible policy, and records an economic error allowance. It does not receive the optimal continuation as a training label. Arbitrary trained networks can be evaluated by the first result below. The subsequent constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

### C.1. Action-dependent errors and complete-policy loss

Let  $H(a)$  be a true current cost and  $\hat{H}(a)$  its continuation approximation. If  $\hat{a}$  is  $\delta$ -optimal for  $\hat{H}$  on the same feasible action set, then

$$H(\hat{a}) - \inf_a H(a) \leq \delta + \text{osc}_a \{\hat{H}(a) - H(a)\}. \quad (66)$$

Indeed, compare  $\hat{a}$  with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 21—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \leq u_t, \quad \mathcal{T}_t^\pi f_{t+1} - f_t \leq v_t, \quad \ell_T \leq f_T - g \leq u_T.$$

*Then the actual feasible policy satisfies*

$$0 \leq J_0^\pi - V_0^* \leq \sum_{t < T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \quad (67)$$

*The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.*

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where  $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$ . The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (67). A shift  $f_t \mapsto f_t + c_t^0$  adds  $c_t^0 - \beta_t c_{t+1}^0$  to  $u_t$  and subtracts it from  $v_t$ . The terminal width is unchanged. *Q.E.D.*

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (67) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

#### APPENDIX D: CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus  $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$  and action modulus  $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$ . Let

$$d_H(A_t(x), A_t(z)) \leq K_t^A \|x - z\|_1, \quad L_t = A_t^Q + D_t K_t^A, \quad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that  $\mathcal{T}_t f_{t+1}$  is  $L_t$ -Lipschitz whenever  $f_{t+1}$  is  $L_{t+1}$ -Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes  $x_{t,i}$  with covering radius  $h_t$ , form effective feasible  $k_t$ -nets. Evaluate their own-future objectives with certified error at most  $e_t$ . Minimize the computed labels and retain the corresponding *original* action  $a_{t,i}$ . A Borel repair  $\Gamma_{t,i}(x)$  is feasible at  $x$  and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \leq K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \quad (68)$$

The allowance  $\zeta_t$  concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \quad (69)$$

Return  $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$ . Terminal labels approximate  $g$  with error  $e_T$  and use the same cone construction with modulus  $L_g$ .

**THEOREM 22**—Own-future construction with feasible witnesses: *The functions in (69) have exact affine-ReLU realizations and Lipschitz moduli  $L_t$  independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \leq f_t - \mathcal{T}_t f_{t+1} \leq D_t k_t + e_t + 2L_t h_t, \quad (70)$$

$$\mathcal{T}_t^\pi f_{t+1} - f_t \leq e_t + D_t \zeta_t,$$

and hence

$$J_0^\pi - V_0^* \leq \sum_{t < T} B_t (D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T (2e_T + 2L_g h_T). \quad (71)$$

*With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.*

**PROOF:** The selected node label lies between the node Bellman infimum minus  $e_t$  and that infimum plus  $D_t k_t + e_t$ . The Lipschitz property of the infimum implies that every cone lies above it minus  $e_t$ . At a nearest node its cone exceeds the infimum by at most  $D_t k_t + e_t + 2L_t h_t$ . At the attaining original index, the two feasible pairs and

(68) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \leq y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (70). The terminal residual belongs to  $[-e_T, e_T + 2L_g h_T]$ ; Proposition 21 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities  $|z| = \text{ReLU}(z) + \text{ReLU}(-z)$  and  $\min(u, v) = u - \text{ReLU}(u - v)$  give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. *Q.E.D.*

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are *one method with two encodings*. They return the same function and policy. Neither Theorem 22 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

#### D.1. Original-owner compilation on a tensor cover

On a tensor cover  $\mathcal{X} = \prod_{j=1}^d \{x_{j,0}, \dots, x_{j,n_j}\}$ , put  $S = \prod_j (n_j + 1)$ . For arbitrary labels  $y_i$ , including inconsistent inexact labels, compute at each grid node  $v$

$$(z_v, o_v) = \min_{\text{lex}, i} (y_i + L \|v - x_i\|_1, i).$$

The index  $o_v$  is the original label owner, not the transformed node  $v$ .

**THEOREM 23**—Continuous original-owner identity: *For a closed tensor cell  $C$  with corners  $\mathcal{V}(C)$ , and  $x \in C$ ,*

$$f(x) = \min_{v \in \mathcal{V}(C)} (z_v + L \|x - v\|_1), \tag{72}$$

$$(f(x), i_*(x)) = \min_{\text{lex}, v \in \mathcal{V}(C)} (y_{o_v} + L \|x - x_{o_v}\|_1, o_v).$$

*Coordinatewise forward and backward relaxations compute  $(z, o)$  in  $O(dS)$  additions and comparisons and  $O(S)$  storage. After locating the cell, a query uses  $O(d2^d)$  arithmetic*

operations. Exact rational comparisons preserve all ties, including  $L = 0$ . Using the original owner's repaired action preserves the uncompiled policy, its cost and its certificate exactly.

PROOF: The triangle inequality bounds  $f(x)$  above by each corner expression. Choose the lexicographically first attaining site  $i_*(x)$ . In every coordinate choose the cell endpoint between  $x_j$  and  $x_{i_*,j}$ . At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at  $x$ ; a tied smaller owner would contradict the original tie rule. Thus this corner has owner  $i_*(x)$ , proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

The nodal algorithm is the classical distance transform ([Felzenszwalb and Huttenlocher, 2012](#)); preserving the original action through continuous off-grid queries is the policy obligation. If there are  $J_{t,i}$  node actions and  $M_t$  integration representatives, write  $Q_t = M_t \sum_i J_{t,i}$ . Literal dense cone evaluation costs  $O(\sum_t dS_{t+1}Q_t)$  in representation work. Compilation replaces this by  $O(\sum_t [dS_{t+1} + Q_t d2^d])$ , plus cell location, primitive evaluation, integration, feasible repair, certificate and output. Rational bit lengths and state boxes crossing many cells incur additional costs. There is no dimension-free complexity assertion.

## D.2. Acquired states and conventional actors

A  $b$ -bit coordinate sensor reports a dyadic cell and its midpoint  $\hat{x}$ . On the unit cube,  $\|x - \hat{x}\|_1 \leq r(b) = d2^{-b-1}$ . For the investment capacity  $\kappa(x)$  below, clip the original action to the smallest capacity throughout that cell and round down to an action quantum  $\Delta_t$ . This is feasible at every state in the cell. Relative to an exact true-state repair, its displacement is at most  $2K_t^A r(b) + \Delta_t$ . Transporting the witness cone from  $\hat{x}$  to  $x$  adds at most  $2L_t r(b)$ , plus any certified selector objective error  $v_t$ . Thus the implemented-policy account adds

$$\sum_{t < T} B_t \{ (2L_t + 2D_t K_t^A) r(b) + D_t \Delta_t + v_t \}. \quad (73)$$

No continuity of the discrete owner selector is assumed. At uncertain numerical comparisons the simulator retains all possible actors, rather than choosing a favorable one.

For conventional FVI the continuation is multilinear on its realized tensor grid. Its actual modulus is bounded by the largest coordinate edge slope. A nearest-node actor uses a feasible node minimizer and true-state repair. Let  $h_t$  be the largest half-cell  $\ell^1$  radius,  $L_t^{\text{gr}}$  the Bellman graph modulus, and  $e_t$  the node query allowance. Its optimal residual upper bound is  $D_t k_t + e_t + L_t^{\text{gr}} h_t$ . Its selected-policy upper bound is  $e_t + E_t$ , where

$$E_t = \sup_x \{y_{i(x)} + L_t^{\text{gr}} \|x - x_{i(x)}\|_1 - f_t(x)\}.$$

Partitioning cells by nearest-node boundaries makes the expression separately affine in each coordinate. Its maximum is therefore at a partition vertex; the implementation encloses all such vertices. Both uniform and curvature-refined FVI receive this sharper one-sided account, not an artificially doubled baseline. Their acquired-state allowance is derived separately in (103).

## APPENDIX E: COST-DIRECTED FEASIBLE IMPROVEMENT

A policy certificate answers a safety question relative to the optimum. It need not answer the comparison with an installed controller. We now make actual cost reduction a constraint on the improvement step. The continuation representation, including a neural realization, supplies proposals; a centered, policy-specific evaluation determines whether to implement them. The incumbent is always an admissible fallback.

Fix a feasible implemented policy  $\pi$  and bounded functions  $h_0, \dots, h_T$  in the common evaluation domain of Section C. Its observation and rounding rules are part of  $\pi$ . Suppose verified constants satisfy

$$\ell_t \leq \mathcal{T}_t^\pi h_{t+1} - h_t \leq u_t \quad (t < T), \quad \ell_T \leq g - h_T \leq u_T. \quad (74)$$

Define  $a_T = \ell_T$ ,  $b_T = u_T$  and

$$a_t = \ell_t + \beta_t a_{t+1}, \quad b_t = u_t + \beta_t b_{t+1}, \quad w_t = b_t - a_t.$$

All inequalities in this section are uniform over the true state domain. Bounds at sampled points alone do not meet this requirement.

PROPOSITION 24—Centered reference advantage: *Under (74),  $a_t \leq J_t^\pi - h_t \leq b_t$ . For any feasible action  $v$  at state  $x$ , put*

$$d_t(x, v) = Q_t(h_{t+1}; x, v) - Q_t(h_{t+1}; x, \pi_t(x)).$$

*If  $\bar{d}_t(x, v)$  is a valid upper enclosure of this difference, then*

$$Q_t(J_{t+1}^\pi; x, v) - J_t^\pi(x) \leq \bar{d}_t(x, v) + \beta_t w_{t+1}. \quad (75)$$

*The width allowance and the action contrast are invariant to date-specific constant shifts of the fitted continuations.*

PROOF: Monotonicity and cash invariance propagate the two error endpoints backward. Applying them once more to each of the two feasible actions bounds each continuation-evaluation error between  $\beta_t a_{t+1}$  and  $\beta_t b_{t+1}$ . Their difference is at most  $\beta_t w_{t+1}$ . Date-specific shifts translate both endpoints by the same amount and cancel in the action contrast. *Q.E.D.*

The proposition is deliberately a reference-policy result. Substituting a bound on  $h - V^*$  without controlling  $h - J^\pi$  changes the statement. Likewise, accepting an action merely because it lowers  $Q_t(h_{t+1})$  ignores the reference-evaluation allowance.

THEOREM 25—Incumbent-preserving Bellman improvement: *Suppose a measurable feasible proposal  $v_t$  and a nonnegative measurable margin  $\kappa_t$  are given. At a state where a verified enclosure satisfies*

$$\bar{d}_t(x, v_t(x)) + \beta_t w_{t+1} \leq -\kappa_t(x), \quad (76)$$

*implement  $v_t(x)$ ; otherwise implement the original  $\pi_t(x)$ . Call the resulting policy  $\mu$ . For a cell-based observation rule, require feasibility and (76) for every true state in the reported cell, using the same proposed action on that cell. Then  $J_t^\mu(x) \leq J_t^\pi(x)$  at every date and state.*

Under conditional expectation, let  $D_t$  be the accepted-state set and let the state trajectory follow  $\mu$ . Then the stronger quantitative assertion is

$$J_0^\mu(x) - J_0^\pi(x) \leq -\mathbb{E}_x^\mu \sum_{t < T} B_t \kappa_t(X_t) \mathbf{1}_{D_t}(X_t). \quad (77)$$

The qualitative assertion holds for the stipulated monotone cash-invariant conditional recursion. The occupancy formula is asserted only for expectation.

PROOF: At accepted states Proposition 24 gives  $\mathcal{T}_t^\mu J_{t+1}^\pi \leq J_t^\pi - \kappa_t$ . Elsewhere the two actions coincide, so the advantage is exactly zero, not  $\beta_t w_{t+1}$ . Backward monotone comparison, starting from a common terminal cost, gives the first assertion. Under expectation subtract  $J_t^\pi(X_t)$  along the  $\mu$  trajectory and telescope the conditional expectations. The resulting identity is the expected discounted sum of true reference advantages, each bounded above by the accepted negative margin or by zero. *Q.E.D.*

The improvement principle is related to approximate policy iteration with safeguarded updates, including Metelli et al. (2021). We do not claim that monotone policy improvement itself is new. Here the relevant connection is between a signed own-policy evaluation account, action-centered numerical errors, original-owner construction, and the actual information-constrained action. The test is available to neural and conventional proposals alike. An exact native encoding of a ReLU continuation does not become a second independent method.

### E.1. A fully executed final-date specialization

A useful test of this result leaves every earlier decision fixed and improves only the final action. Set  $h_T = g$ . The continuation width is then zero, so the gate concerns the actual conditional final cost, without a fitted policy-evaluation error. For the investment economy in Section K, write

$$y_j = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + \gamma_j a, \quad s_j = (-1)^{j+1}, \quad \sigma_z^2 = \frac{1}{3072},$$

where the cyclic convention and  $\gamma_j$  are as in that section. Put  $b = 1/2 - 2\sum_j y_j/d$ ,  
 $r = |\sum_j s_j|/(16d)$ , and

$$\Psi_2(b, r) = \begin{cases} b_+^2, & r = 0, \\ ((b+r)_+^3 - (b-r)_+^3)/(6r), & r > 0. \end{cases} \quad (78)$$

Exact integration under the original continuous innovation law gives

$$\begin{aligned} \mathbb{E}g(y + sZ) &= \frac{4}{d} \sum_j (y_j - 5/8)^2 + \frac{1}{4d} \sum_j (y_j - y_{j+1})^2 + 2\Psi_2(b, r) \\ &\quad + 4\sigma_z^2 + \frac{\sigma_z^2}{4d} \sum_j (s_j - s_{j+1})^2. \end{aligned} \quad (79)$$

Thus no terminal innovation mesh is needed for this improvement step. Its computation is nevertheless charged. The formula includes the odd-dimensional shortage uncertainty; setting it to zero in dimension three would change the model.

**COROLLARY 26**—Acquired-cell final improvement: *In the declared investment economies, final conditional cost  $H(x, a) = c(x, a) + \beta \mathbb{E}g(F(x, a, Z))$  is strictly convex in  $a$ , with second derivative at least  $2p$ . Fix an observation cell  $C$  and the implemented incumbent action  $a_C$ . Choose a lattice minimizer  $a_C^*$  at the reported midpoint on the cell-wide feasible set. Accept it only if an outward enclosure of  $H(x, a_C^*) - H(x, a_C)$  is nonpositive for all  $x \in C$ . Keep the incumbent on every other cell and at all earlier dates. The resulting acquired policy weakly improves actual expected discounted cost for every initial state. This guarantee does not compare it to a different full-information incumbent.*

The lattice is  $\{k/4096 : k \in \mathbb{N}_0\}$ , truncated by the minimum capacity on the eight-bit cell. Convex derivative bisection identifies the two neighboring lattice actions; their actual cost contrast selects the midpoint proposal. The gate compares its cost with the incumbent on the entire cell, not just at the midpoint. Exact rational point comparison resolves a numerically ambiguous proposal tie. An uncertain true-state box may intersect several cells; the simulator retains their action hulls rather than selecting a favorable cell.

For each generator, old and repaired policies share their earlier trajectory. Their discounted cost difference therefore equals the conditional final contrast. Centering cancels state-only costs and tightens its interval enclosure. When two generators are compared, their trajectories differ and are simulated separately using common innovations. Both comparisons concern actual policies; neither is obtained by subtracting two policy-loss certificates.

#### APPENDIX F: FROM SAFE IMPROVEMENT TO FINITE-SWEEP ACCURACY

The preceding improvement result compares an implemented policy with its own incumbent. We now connect that comparison to the original Bellman optimum. This distinction matters economically: a policy can improve strictly while remaining more costly than a conventional alternative. The result below preserves the original objective and gives explicit sufficient conditions for closing its optimality gap; it does not infer those conditions from a small empirical residual.

Consider the finite-horizon expectation model of the paper, with bounded costs, feasible measurable Markov policies, and discount factors  $\beta_t \geq 0$ ,  $0 \leq t < T$ . Write

$$Q_t(h; x, a) = c_t(x, a) + \beta_t \mathbb{E}[h(X_{t+1}) \mid x, a], \quad (T_t h)(x) = \inf_{a \in A_t(x)} Q_t(h; x, a),$$

and  $V_T^* = g$ ,  $V_t^* = T_t V_{t+1}^*$ . The benchmark is the original fully observed optimum. An acquired-state policy remains feasible for this benchmark; a restriction to cell-wise actions may, however, impose a positive approximation floor. No vanishing information error is assumed.

Starting from a feasible  $\pi^0$ , at pass  $k$  let  $h^k$  have certified signed policy-evaluation bands

$$a_t^k \leq J_t^{\pi^k}(x) - h_t^k(x) \leq b_t^k, \quad w_t^k = b_t^k - a_t^k. \quad (80)$$

These can be obtained from the signed-residual recursion already proved; the approximator need not have a small absolute level error. Let a feasible proposal  $v_t^k$  satisfy, uniformly in  $x$ ,

$$Q_t(h_{t+1}^k; x, v_t^k(x)) \leq (T_t h_{t+1}^k)(x) + \alpha_t^k. \quad (81)$$

For its centered comparison with the exact incumbent, define  $d_t^k(x) = Q_t(h_{t+1}^k; x, v_t^k(x)) - Q_t(h_{t+1}^k; x, \pi_t^k(x))$ . A quantitative upper enclosure satisfies

$$d_t^k(x) \leq \bar{d}_t^k(x) \leq d_t^k(x) + \zeta_t^k. \quad (82)$$

Here  $\zeta_t^k$  includes acquisition-cell variation and integration or arithmetic enclosure width, not just the machine unit roundoff. The new policy uses the proposal exactly when  $\bar{d}_t^k(x) + \beta_t w_{t+1}^k \leq 0$  and otherwise uses  $\pi_t^k(x)$  unchanged. For an acquired cell, feasibility and this comparison must hold throughout that cell, with a single implementable action and acceptance decision. Bounds (81)–(82) are required on the same cells, including their boundaries.

**THEOREM 27**—Finite-sweep accuracy with incumbent preservation: *Suppose (80)–(82) hold at each pass, with nonnegative finite  $\alpha_t^k$  and  $\zeta_t^k$ . Put*

$$\epsilon_t^k = \alpha_t^k + \zeta_t^k + 2\beta_t w_{t+1}^k, \quad E_t^k = \|J_t^{\pi^k} - V_t^*\|_\infty.$$

*Then  $J_t^{\pi^{k+1}} \leq J_t^{\pi^k}$  at every date and state, and*

$$E_t^{k+1} \leq \beta_t E_{t+1}^k + \epsilon_t^k, \quad E_T^k = 0. \quad (83)$$

*Set  $B_{t,0} = 1$  and  $B_{t,j} = \prod_{r=t}^{t+j-1} \beta_r$ . For  $1 \leq m \leq T - t$ ,*

$$E_t^m \leq B_{t,m} E_{t+m}^0 + \sum_{j=0}^{m-1} B_{t,j} \epsilon_{t+j}^{m-1-j}. \quad (84)$$

*In particular, if  $\epsilon_t^k \leq \bar{\epsilon}_t$  uniformly in  $k$ , then after  $T$  passes*

$$E_t^T \leq \sum_{j=0}^{T-t-1} B_{t,j} \bar{\epsilon}_{t+j}. \quad (85)$$

*With exact policy evaluation up to a datewise constant, exact greedy proposals, and exact centered comparisons, the policy is optimal after at most  $T$  passes, and no intermediate pass raises its cost.*

The hypotheses separate three computational tasks. A trained continuation must be evaluated relative to the policy actually retained; the action search must ap-

proximate the original feasible infimum; and the comparison must be valid over the available information set. A guarantee for only one of these tasks does not imply the theorem. The same conclusion applies to native and neural representations when they implement the same objects. In particular, it does not turn their common compilation saving into a neural-specific advantage. Safe policy improvement is an established principle; the result here supplies the signed-width and acquisition-aware error account for the incumbent-preserving NBO construction.

**PROPOSITION 28**—An implementable cellwise proposal bound: *Let an acquired cell  $C$  have representative  $x_C$  and radius  $r_C$ . Suppose  $Q_t(h; \cdot, a)$  is  $L_t$ -Lipschitz on the cell for every menu action, and  $Q_t(h; x, \cdot)$  is  $D_t$ -Lipschitz on feasible actions. Let  $M_C \subseteq \bigcap_{x \in C} A_t(x)$  be a nonempty finite menu such that every  $a \in A_t(x)$ , for every  $x \in C$ , lies within  $\delta_{A,C}$  of some member of  $M_C$ . If  $v_C \in M_C$  is an  $\eta_C$ -minimizer at  $x_C$ , then it satisfies (81) on  $C$  with*

$$\alpha_t^k \leq \eta_C + 2L_t r_C + D_t \delta_{A,C}.$$

The directed-cover assumption is substantive when constraints move with the state. It is not implied by a dense grid at the representative alone. A failed cover or a cell that cannot certify a comparison retains its incumbent; it must not be counted as a successful accuracy refinement.

**PROPOSITION 29**—Sharpness of the evaluation-width coefficient: *Under the theorem’s assumptions, the coefficient two on  $\beta_t w_{t+1}^k$  in (83) cannot be uniformly replaced by a smaller constant, even for a one-period deterministic problem with two actions, an exact greedy proposal, and exact contrast arithmetic.*

The theorem is constructive but conditional on verified error budgets. It does not assert that a continuous neural fit uniformly approximates a discontinuous acquired-policy value, or that refining a fixed acquisition contract eliminates its information floor. The executed investment study below certifies a final-date improvement. It is not an execution of  $T$  full improvement passes, nor numerical evidence of the finite-sweep near-optimality conclusion. Exact finite-state regression tests exercise

the new identities, rejection cases, and sharpness example separately from the economic benchmark.

## APPENDIX G: POLICY-RELEVANT CONTINUATION ERRORS

A continuation error matters for an investment decision through the actions it changes. The signed-width account above removes a datewise level error. It can nevertheless charge for a nonconstant error that has the same conditional expectation under every feasible action. We now obtain a sharper account without changing the Bellman optimum, the acquisition contract, or the incumbent-preserving rule.

Write  $P_t^a f(x) = \mathbb{E}[f(X_{t+1}) \mid x, a]$ , and at pass  $k$  put  $e_t^k = J_t^{\pi^k} - h_t^k$ . An *action-contrast certificate* is a measurable nonnegative bound  $R_t^k(x; a, b)$  satisfying

$$|(P_t^a - P_t^b)e_{t+1}^k(x)| \leq R_t^k(x; a, b) \leq \chi_t^k \quad (a, b \in A_t(x)). \quad (86)$$

Both inequalities are required uniformly over true states and feasible actions. The scalar certificate  $R_t^k = w_{t+1}^k$  always remains available. A bound computed for the selected proposal alone is sufficient for safety, but not for the accuracy conclusion, which compares with the entire feasible infimum.

**THEOREM 30**—Contrast-certified finite-sweep improvement: *In the expectation model of Theorem 27, retain the feasible proposal and comparison-enclosure assumptions (81)–(82). Suppose (86) holds. Accept  $v_t^k(x)$  precisely when*

$$\bar{d}_t^k(x) + \beta_t R_t^k(x; v_t^k(x), \pi_t^k(x)) \leq 0; \quad (87)$$

*otherwise retain the exact incumbent. On an acquired cell, the proposal, acceptance decision, feasibility, and all enclosures must be valid on the whole cell. Then every pass is nonworsening at every state and date. Moreover,*

$$E_t^{k+1} \leq \beta_t E_{t+1}^k + \tilde{\epsilon}_t^k, \quad \tilde{\epsilon}_t^k = \alpha_t^k + \zeta_t^k + 2\beta_t \chi_t^k. \quad (88)$$

*The finite-sweep unrolling and T-pass bound of Theorem 27 hold with  $\epsilon$  replaced by  $\tilde{\epsilon}$ . In particular, exact greedy search and comparisons with  $\chi_t^k = 0$  give an optimal policy after at most  $T$  passes, even when the global evaluation widths are positive. The coefficient two remains sharp over this class of certificates.*

The theorem concerns expected discounted cost. It does not silently extend the linear action-kernel difference to a nonlinear recursive preference. The earlier monotone-recursion safety results remain in force under their own hypotheses. Nor does a small sampled action contrast establish (86). The next result constructs an upper bound using residuals and couplings with verified marginals; it never calls the unknown policy value.

**PROPOSITION 31**—Coupled residual certificate: *Fix a policy  $\pi$  and critic  $h$ . Let  $r_T = g - h_T$  and  $r_t = \mathcal{T}_t^\pi h_{t+1} - h_t$ , and let  $w_t$  be valid signed-residual widths. For every pair of states let  $\Lambda_t^\pi(x, y)$  be a measurable coupling of  $P_t^{\pi_t(x)}(x, \cdot)$  and  $P_t^{\pi_t(y)}(y, \cdot)$ . Suppose verified nonnegative functions  $H_t$  enclose  $|r_t(x) - r_t(y)|$ . Define*

$$\begin{aligned} D_T(x, y) &= \min\{w_T, H_T(x, y)\}, \\ D_t(x, y) &= \min\left\{w_t, H_t(x, y) + \beta_t \int D_{t+1}(u, v) d\Lambda_t^\pi(x, y)(u, v)\right\}. \end{aligned} \quad (89)$$

*Then  $|e_t(x) - e_t(y)| \leq D_t(x, y)$ . For any measurable coupling  $\Lambda_t(x; a, b)$  of the two feasible action kernels,*

$$R_t(x; a, b) = \min\left\{w_{t+1}, \int D_{t+1}(u, v) d\Lambda_t(x; a, b)(u, v)\right\} \quad (90)$$

*is a valid action-contrast certificate. Replacing either integral by a verified nonnegative upper enclosure preserves validity. No optimal-transport minimizer or smooth incumbent is required.*

The construction includes the work needed to enclose residual differences, transport them, and take the action supremum. It is an alternative certificate, not a free evaluation oracle. In a finite model, a common-uniform coupling gives an explicit terminating implementation. In a continuous model, an available common innovation gives a coupling directly, but integration and cellwise certification still have to be supplied. With coarse enclosures the minimum in (90) returns the original scalar guarantee.

### G.1. Action-null components and neural realizations

Define the action-null space at date  $t$  by

$$\mathcal{N}_t = \{n : (P_t^a - P_t^b)n(x) = 0 \text{ for every } x \text{ and } a, b \in A_t(x)\}.$$

This is a linear space of bounded measurable functions, not a statement about the accuracy of a training procedure. Constants belong to it; exogenous payoff components can make it larger. Values differing by an element of  $\mathcal{N}_t$  have identical action contrasts at date  $t$ , although their levels and state dependence need not agree.

**COROLLARY 32**—Certified removal of action-null error: *Suppose  $e_{t+1}^k = n_{t+1}^k + u_{t+1}^k$ , where membership  $n_{t+1}^k \in \mathcal{N}_t$  is established and  $u_{t+1}^k$  has a verified range of width  $v_{t+1}^k$ . Then  $R_t^k \leq v_{t+1}^k$  is available, independently of the amplitude of  $n_{t+1}^k$ . A constructive way to obtain this width is to evaluate the corrected critic  $h^k + n^k$  by the original signed-residual recursion. The action comparisons using  $h_{t+1}^k$  and  $h_{t+1}^k + n_{t+1}^k$  coincide.*

The qualification “established” is important. Arbitrarily declaring part of a fit to be null invalidates the certificate. If an exact null component is known, symbolic cancellation before interval arithmetic avoids subtracting two large levels. If it is only approximately null, its verified nonzero action contrast must be added to the allowance.

**PROPOSITION 33**—A nonconstant ReLU component in the original economy: *In the investment dynamics (106), for each declared dimension  $d \geq 2$ , every bounded measurable function of  $y_1 - 2y_2$  is action-null under the common innovation. In particular,*

$$n_M(y) = M(y_1 - 2y_2 + 1)_+, \quad M \geq 0,$$

*is a one-unit ReLU realization with oscillation  $2M$  on the state cube. At the final date, the critic  $h_T = g - n_M$  has global evaluation width  $2M$  but zero action-contrast error. The contrast-aware terminal policy is therefore exactly the unperturbed policy of Corollary 26; its statewise cost guarantee is unchanged for every  $M$ .*

This result supplies a representation-level robustness property of NBO, not an identified advantage over a native encoding of the same continuation. It also iden-

1 tifies an avoidable source of conservatism in a global critic-error account. It does 1  
 2 not remove an economically relevant continuation error, a capacity constraint, or an 2  
 3 information floor. Full proofs and an exact finite-state reference implementation are 3  
 4 in the supplement. 4

## 6 APPENDIX H: ALLOCATING RESOURCES TO AN ECONOMIC TOLERANCE 6

7 A single resolution parameter imposes an economic restriction that the Bellman 7  
 8 problem does not impose. State coverage, action search and integration have differ- 8  
 9 ent error coefficients and different effects on work. Refining all three at the same rate 9  
 10 can therefore spend most of the service budget on the wrong source of error. We sep- 10  
 11 arate these choices while retaining the continuation, feasible-witness interpretation 11  
 12 and all-state comparison developed above. 12

### 14 H.1. A separated primitive budget 14

15 Consider the investment family specified in Section K, with  $d \in \{2, 3, 4\}$  state 15  
 16 coordinates, discount  $\beta = 15/16$ , and uniform coordinate grids with  $N$  intervals. 16  
 17 At a state node use  $K + 1$  feasible action fractions. Integrate the common scalar 17  
 18 innovation using  $M$  equal-probability bins and their midpoints. The three inte- 18  
 19 gers need not agree. Let  $L_t$  be the primitive graph modulus of the constructed 19  
 20 witness continuation, and let  $D_t$  be its action modulus. A downward numerical 20  
 21 displacement of the action grid is denoted by  $\delta_a$ . 21

22 The state and action covering radii satisfy  $h_t = d/(2N)$  and  $k_t \leq 1/(8K) + \delta_a$ . The 22  
 23 innovation lies in an interval of length  $1/16$ . Within a bin its expected absolute 23  
 24 distance from the midpoint is  $1/(64M)$ . Since changing the scalar innovation by 24  
 25  $z - z'$  changes the next state by at most  $d|z - z'|$  in the  $\ell^1$  norm, a certified numerical 25  
 26 query has an error allowance 26

$$28 \quad e_t \leq \frac{\beta d L_{t+1}}{64M} + \xi_t. \quad (91) \quad 28$$

29 Here  $\xi_t$  covers arithmetic, interval evaluation and label rounding. It does not sup- 29  
 30 press the integration error. The terminal label allowance is  $e_T$ . 30  
 31 31  
 32 32

PROPOSITION 34—Separated-resource policy account: *Under the feasible-witness hypotheses, the returned policy satisfies*

$$0 \leq J^\pi - V^* \leq \frac{a}{N} + \frac{b}{K} + \frac{q}{M} + \eta, \quad (92)$$

where

$$\begin{aligned} a &= d \sum_{t < T} \beta^t L_t + d \beta^T L_g, & b &= \frac{1}{8} \sum_{t < T} \beta^t D_t, \\ q &= \frac{d}{32} \sum_{t < T} \beta^{t+1} L_{t+1}, & \eta &= \sum_{t < T} \beta^t (2\tilde{\zeta}_t + D_t \delta_a) + 2\beta^T e_T. \end{aligned} \quad (93)$$

The primitive coefficients are independent of the computed labels. The leading number of innovation evaluations in target formation is

$$T(N+1)^d(K+1)M. \quad (94)$$

The compiled continuation additionally incurs its corner-evaluation and preprocessing costs. Pilot selection, policy extraction, verification, arithmetic bit lengths and output are separate charged components.

PROOF: Insert the state and action radii and (91) into the feasible-witness loss bound. The selected-policy upper account contributes the second copy of  $e_t$ , not a second action-search allowance. Collecting the coefficients of  $N^{-1}, K^{-1}, M^{-1}$  gives (93). There are  $(N+1)^d$  state nodes,  $K+1$  actions per node and  $M$  innovation representatives per action at each nonterminal date. No independence of fitting errors is used. *Q.E.D.*

The account explains why a common further state rung need not be a diagonal refinement. In the  $T=3, p=4, d=2$  cell, the exact primitive coefficients are approximately  $a=51.1569, b=2.3650$  and  $q=1.0486$ . The choice  $(N, K, M) = (128, 16, 8)$  has account about  $0.6786 + \eta$ , below target one when the numerical remainder fits the remaining budget. The diagonal choice  $(128, 128, 128)$  is more accurate, but its target-generation innovation count at that same state grid is  $129 \cdot 128 / (17 \cdot 8)$  times larger. This is an arithmetic comparison of two declared allocations, not an

observation of the unexecuted diagonal service or a runtime ratio. Both witness and conventional FVI receive the separated allocation in the experiment.

## H.2. The optimal continuous allocation and its boundary cases

A useful planning problem minimizes the leading product  $N^d KM$  subject to (92). This proxy isolates the primitive evaluation term; it is not the entire operation count or the measured wall clock. Write  $E = \varepsilon - \eta > 0$  and suppose first that  $a, b, q$  are positive and all optimal resources exceed their required lower bounds.

PROPOSITION 35—Error shares and clipped allocation: *The unique interior minimizer of the continuous proxy has*

$$\frac{a}{N} = \frac{d}{d+2}E, \quad \frac{b}{K} = \frac{1}{d+2}E, \quad \frac{q}{M} = \frac{1}{d+2}E. \quad (95)$$

For lower resource bounds  $r_i \geq \underline{r}_i > 0$ , put  $c = (a, b, q)$ ,  $p = (d, 1, 1)$  and  $\bar{x}_i = c_i / \underline{r}_i$ . If  $E < \sum_i \bar{x}_i$ , there is a unique positive  $\lambda$  such that

$$x_i = \min\{\bar{x}_i, p_i \lambda\}, \quad \sum_i x_i = E; \quad r_i = c_i / x_i. \quad (96)$$

These resources minimize the constrained continuous proxy. If  $E \geq \sum_i \bar{x}_i$ , all lower resource bounds suffice. Coordinates with zero error coefficient remain at their resource lower bound. When dyadic lower bounds are used, rounding each optimal resource upward to a power of two preserves the error guarantee and multiplies this proxy by less than  $2^{d+2}$ .

PROOF: Set  $x_i = c_i / r_i$ . Minimizing the resource product is equivalent to maximizing  $\sum_i p_i \log x_i$  over  $0 < x_i \leq \bar{x}_i$  and  $\sum_i x_i \leq E$ . Strict concavity and the multiplier conditions give  $p_i / x_i$  equal on unconstrained coordinates and larger on capped coordinates. This is precisely (96). The sum of its right sides is strictly increasing until all caps bind. The unconstrained case yields (95). Upward dyadic rounding never enlarges an error term and multiplies  $N^d KM$  by less than the stated factor. Neither argument bounds rational-arithmetic bit cost or computer time. *Q.E.D.*

The optimization is a standard separable resource-allocation calculation. Its role here is to expose the error coefficients of the actual own-future policy construction,

including integration and implementation, rather than to claim novelty for the multiplier argument. A finite experimental menu need not contain the continuous optimum. The prototype uses a common, frozen dyadic menu for all generators and records all of it. It does not describe this menu as a globally optimal tuning policy for either FVI or the witness method.

### H.3. *A complete tolerance frontier*

For method  $m$ , let a prospectively fixed ladder return certificates  $g_{m,r}$  and cumulative costs  $w_{m,r}$ , including failed earlier rungs. Define

$$r_m(\varepsilon) = \min\{r : g_{m,r} \leq \varepsilon\}, \quad W_m(\varepsilon) = w_{m,r_m(\varepsilon)}. \quad (97)$$

An empty set is a capped failure, not an interpolated success. Sort the finite union of all certificate values and zero. On each left-closed, right-open interval between successive values, every method's first crossing and complete work are constant. Evaluating the first-crossing rule at the left endpoint therefore reconstructs the entire positive-tolerance frontier exactly, even when successive certificates are not monotone. An unbounded final interval is retained. Repetitions have separate costs, while their mathematical checkpoints must be checked for identity.

This finite partition supplies the appropriate interpretation of a threshold advantage. At a fixed resolution the witness method may cost more. At a tolerance where it crosses one rung earlier, the smaller certificate may more than offset that cost. Whether this happens, and over which tolerance interval, is determined by (97); it is not determined by a favorable single threshold. Under an additional homogeneous model  $g_m(n) = A_m/n$  and  $w_m(n) = C_m n^\kappa$ , the continuous frontier ratio is  $(C_W/C_F)(A_W/A_F)^\kappa$ . This last expression is a conditional explanation, not a fitted scaling law for the experiments. The release supplies the exact finite partition, sensitivity around one, complete prefixes and separate fixed-resolution observations.

## APPENDIX I: FROM RETURNED POLICIES TO ECONOMIC RESOURCE DECISIONS

An all-state upper bound and a comparison of actual policy costs answer different economic questions. We now connect the policies returned by a construction frontier

to direct cost inference and then to a choice of implementation resources. The construction data select the policies; independently generated evaluation paths assess them. This separation permits a valid comparison of first-crossing policies with different resolutions.

### I.1. *Simultaneous inference after construction*

Let  $\mathcal{H}$  denote all construction information: primitive inputs, numerical labels, fitted continuations, returned policies and complete certificate ladders. Conditional on  $\mathcal{H}$ , fix a finite catalogue of implemented-policy contrasts. Each implementation includes its observation rule, original witness or nearest-node selector, true-state feasible repair and action quantization. For a contrast  $i$ , let  $D_i$  be the difference in expected discounted costs under a specified initial law. A common-innovation path score evaluates the two policies on their own endogenous trajectories. It is not the difference between two regret bounds.

Suppose the initial and innovation uniforms are partitioned into equal bins. Independent uniform bin indices are drawn, and interval simulation encloses the score for every continuous realization inside the drawn bins. All unresolved state, selector and repair branches remain in the enclosure. If  $[A_i, B_i]$  is a deterministic support interval, clip the path endpoints to that support. Write  $\underline{m}_i, \overline{m}_i$  for outward sample-mean bounds on the lower and upper endpoint samples and  $v_{i,L}, v_{i,U}$  for upper bounds on their unbiased sample variances. For  $n \geq 2$  define

$$R_i(v) = \sqrt{2v\ell/n} + \frac{7(B_i - A_i)\ell}{3(n-1)}, \quad \ell \geq \log(4K/\alpha), \quad (98)$$

where  $K$  bounds the number of sampled contrasts.

**THEOREM 36**—Construction-conditional economic intervals: *Under the stated independent-bin model, conditional on any valid construction history  $\mathcal{H}$ , with probability at least  $1 - \alpha$  all catalogue contrasts satisfy*

$$D_i \in [l_i, u_i] := [\max\{A_i, \underline{m}_i - R_i(v_{i,L})\}, \min\{B_i, \overline{m}_i + R_i(v_{i,U})\}]. \quad (99)$$

*The same coverage holds unconditionally. Subsequent selection among these covered contrasts, or evaluation at any deterministic implementation-price vector, needs no additional*

union bound. This conclusion applies only to policies and initial laws included in the catalogue; it does not create direct-cost observations for every unstudied tolerance or initial state.

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than  $D_i$ , and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of [Maurer and Pontil \(2009\)](#) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every  $\mathcal{H}$  implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

In particular, selecting the first policy to meet a specified certificate is permitted when that selection uses construction records rather than the subsequent evaluation paths. Repeated construction clocks are not independent draws from a neural-optimizer population. The statistical randomness here belongs to the evaluation design. A fixed pseudorandom stream implements a reproducible experiment; it is not itself a proof of the independent-bin assumption.

## I.2. Replacement-fee identification

Write  $D = J^W - J^F$ . If witness is installed, replacing it by FVI changes total cost by  $k - D$ , where  $k \geq 0$  is the replacement fee. If FVI is installed, the change is  $k + D$ . Thus the two directional gains before fees are  $D$  and  $-D$ .

COROLLARY 37—The break-even fee frontier: *On an event where  $D \in [l, u]$ , the two directional gains belong to  $[l, u]$  and  $[-u, -l]$ . The unknown absolute cost difference satisfies*

$$|D| \in [\text{dist}(0, [l, u]), \max\{|l|, |u|\}]. \quad (100)$$

For every  $k \geq \max\{0, u, -l\}$  neither replacement strictly lowers expected cost. By contrast,  $k \geq \max\{0, l, -u\}$  only removes a certified strictly profitable replacement. The two thresh-

olds generally differ. Strict inequalities in the first criterion certify that both replacements strictly increase cost.

PROOF: Subtract the gain intervals from  $k$  and compare the resulting interval endpoints with zero. Taking the image of  $[l, u]$  under the absolute-value map gives (100). The assertions hold simultaneously for every nonnegative fee on the original coverage event. Q.E.D.

The fee frontier is more informative than a conclusion at one generous charge. The earlier 1/64 replacement decision is retained with its original interpretation and confidence account. The new analysis reports the two directional gain intervals and both thresholds instead of selecting a favorable fee after seeing the data. There is no welfare calibration hidden in the units: a decision maker must supply the economically relevant charge, or use the full frontier to see what remains unidentified.

### I.3. Observation costs on the same bit grid

Let  $\mathcal{B} = \{4, \dots, 16\}$  be the available bit precisions. For one fixed returned controller, write  $J_b$  for the expected cost of its actual  $b$ -bit, robustly repaired and quantized implementation. Use  $b_0 = 16$  as a common reference and obtain simultaneous intervals for  $D_b = J_b - J_{b_0}$ . The identity interval at  $b = b_0$  is exactly  $[0, 0]$ . Let  $s_b = db \sum_{t < T} B_t$  be the discounted number of coordinate-bits purchased; the net cost is  $J_b + \lambda s_b$ .

THEOREM 38—Certified choice of actual net implementation cost: Suppose all  $D_b \in [l_b, u_b]$  simultaneously. For any  $\lambda \geq 0$  define

$$L(\lambda) = \min_b \{l_b + \lambda s_b\}, \quad U(\lambda) = \min_b \{u_b + \lambda s_b\}, \quad (101)$$

$$\hat{b}(\lambda) = \min \arg \min_b \{u_b + \lambda s_b\}.$$

Then, simultaneously for all  $\lambda \geq 0$ ,

$$0 \leq J_{\hat{b}} + \lambda s_{\hat{b}} - \min_{b \in \mathcal{B}} (J_b + \lambda s_b) \leq U(\lambda) - L(\lambda). \quad (102)$$

The lower and upper functions are finite piecewise-affine envelopes. All changes in their active lines occur at nonnegative intersections of catalogue lines, with exact comparisons deciding ties. The smallest selected precision  $\hat{b}(\lambda)$  is weakly decreasing in  $\lambda$ .

PROOF: Subtract the common  $J_{b_0}$ . The selected implementation has net difference at most  $U(\lambda)$ , and the best catalogue implementation has net difference at least  $L(\lambda)$ . This proves the regret bound pointwise on the one simultaneous event and hence for every  $\lambda$ . A minimum of finitely many affine functions changes its active affine expression only at an intersection. For  $\lambda_2 > \lambda_1$ , sum the two minimizing inequalities to obtain  $(\lambda_2 - \lambda_1)(s_{\hat{b}(\lambda_2)} - s_{\hat{b}(\lambda_1)}) \leq 0$ . Since  $s_b$  increases strictly with  $b$ , the precision comparative static follows. *Q.E.D.*

This result makes a statement about *actual* net cost within the declared bit catalogue. It differs from minimizing a uniform all-state loss bound. Both criteria are useful: the latter gives distribution-free protection relative to the full-information optimum, while (102) uses a specified initial law to compare feasible implementations of one returned policy. Neither criterion gives an optimum over arbitrary sensor technologies or all policies.

For a fair comparison the conventional actor receives its own acquisition allowance. Suppose its nearest-node policy has a certified selected-policy account at a reported state, obtained from the graph modulus  $L_t^{\text{gr}}$ , and its actual interpolated critic has modulus  $L_t^f$ . Transporting the same selected node action to the true state, then using cell-wide capacity repair, adds at most

$$(L_t^{\text{gr}} + L_t^f + 2D_t K_t^A)r(b) + D_t \Delta_t. \quad (103)$$

Indeed, transport the feasible node pair directly to the true state, bound its node distance by the reported-state distance plus  $r(b)$ , and bound  $f_t(\hat{x}) - f_t(x)$  by  $L_t^f r(b)$ . Robust repair adds  $D_t(2K_t^A r(b) + \Delta_t)$ . This proves (103) without assuming actor continuity. For a witness critic  $L_t^f = L_t^{\text{gr}}$ , it reduces to the preceding witness allowance. The empirical comparison therefore evaluates the all-state-account minimizer, the interval upper-net-cost selector, and a descriptive midpoint selector on the *same* integer bit and price grid. The midpoint selector is not called the true optimum.

## APPENDIX J: JOINT POLICY-COST AND RESOURCE DECISIONS

Let a finite catalogue  $\mathcal{P}$  contain implemented policies, including incumbents, repaired policies and conventional alternatives. Construction, verification and direct evaluation are distinct charged activities. Denote their measured deployment-relevant resource vector by  $r_j$ ; the published timing convention must specify whether an evaluation shared by several candidates is a common fixed charge or an attributable marginal charge. In particular, a joint paired evaluation cannot be counted as zero or silently charged only to the preferred generator.

Conditional on construction information, suppose a simultaneous event satisfies  $J_j \in [L_j, U_j]$  for every  $j \in \mathcal{P}$ . For nonnegative prices  $\theta$ , let  $k_j(\theta)$  be a known implementation charge, for example  $\theta \cdot r_j$  plus an information fee. Define

$$\hat{j}(\theta) \in \arg \min_{j \in \mathcal{P}} \{U_j + k_j(\theta)\}, \quad R(\theta) = U_{\hat{j}} + k_{\hat{j}}(\theta) - \min_{j \in \mathcal{P}} \{L_j + k_j(\theta)\}. \quad (104)$$

A fixed lexicographic tie rule makes the selection measurable.

**PROPOSITION 39—Actual net-cost frontier:** *On the simultaneous event, for every nonnegative price vector, including one selected after observing the intervals,*

$$0 \leq J_{\hat{j}} + k_{\hat{j}}(\theta) - \min_j \{J_j + k_j(\theta)\} \leq R(\theta).$$

*A proposed replacement of incumbent  $i$  by  $j$  is certified nonworsening in net cost whenever a simultaneous upper bound on  $J_j - J_i$ , plus  $k_j(\theta) - k_i(\theta)$  and the replacement fee, is nonpositive. Otherwise a rule retaining  $i$  remains safe. Policy-specific certificate widths can be displayed alongside this frontier, but do not determine its ordinate.*

**PROOF:** Bound the selected true cost above by its upper line and the best catalogue cost below by the minimum lower line. All inequalities hold on the same finite event for every price, so a new price query does not consume another probability allocation. The replacement assertion applies the paired upper bound directly. *Q.E.D.*

Absolute intervals can be considerably wider than centered paired differences. For a decision between an incumbent and its final repair, the centered interval is therefore the appropriate replacement input. A statistically unresolved interval

is not evidence of equivalence. In contrast, Corollary 26 supplies a deterministic nonincrease, although it does not necessarily certify a strictly positive improvement large enough to cover an implementation fee.

### J.1. *Information technology and the price of conservatism*

An explicit theoretical observation technology gives the bit price units. Suppose a device reports a dyadic cell in each coordinate at date  $t$ . With  $b$  bits per coordinate it transmits  $db$  bits and requires  $\tau db$  resource units, where  $\tau$  is a technological conversion factor. At unit resource price  $w$ , and with a setup cost  $K_b$ , its present cost is

$$C_b(w, \tau) = K_b + w\tau db \sum_{t < T} B_t. \quad (105)$$

Each reported cell entails its own feasible action contract. In the retained sensor experiment,  $K_b = 0$  and the normalized price  $\lambda = w\tau$  takes the declared values. This supplies a structural interpretation and a sensitivity variable, not an estimate of actual device costs. The dataset contains no empirical measurements from which  $w$  or  $\tau$  could be calibrated.

Theorem 38 and Proposition 39 apply to the finite bit catalogue with charges (105). Their upper-minus-lower envelope is a valid bound on the cost of conservative choice relative to the unknown true catalogue minimizer. Replacing that minimizer by the lowest observed sample mean and calling it the ex post true optimum is not justified. The retained exact price envelopes and their regret bounds answer the identified decision question without making this substitution. A richer technology with fixed charges, nonlinear bandwidth costs or precision-dependent delays can be assessed by changing the known charge functions, but new unobserved policies require new cost bounds.

A similarly explicit interpretation applies to terminal repair. A controller that incurs cost  $K$  to install the additional gate, and  $q$  per deployed final action, is worth adopting only when its expected discounted policy gain covers those charges. The published centered gain interval defines the entire break-even range. Its mathe-

maternal safety does not by itself imply economic adoption at positive computation prices.

## APPENDIX K: CERTIFIED COMPUTATION AND ECONOMIC DECISIONS

### K.1. *The controlled investment family*

There are  $d \in \{2, 3, 4\}$  capital coordinates on  $[0, 1]^d$ , one investment action, and a common continuous shock. The feasible set is

$$0 \leq a \leq \kappa(x) = \frac{1}{8} + \frac{1}{8d} \sum_{j=1}^d x_j.$$

With cyclic coordinate indices,  $\gamma_j = 1/2$  for odd  $j$  and  $1/4$  for even  $j$ , and  $z \sim \text{Unif}[-1/32, 1/32]$ , the dynamics are

$$F_j(x, a, z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + \gamma_j a + (-1)^{j-1} z. \quad (106)$$

Current costs are

$$c(x, a) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2 \left( \frac{1}{2} - \frac{2}{d} \sum_j x_j \right)_+^2 + p a^2 + 4 a^4. \quad (107)$$

Terminal cost doubles the first coefficient and retains the imbalance and shortage terms, with no action cost. Capital enters both adjustment incentives and feasible capacity. The nonlinear transition and shortage penalty couple the coordinates; the model is not separable linear-quadratic control. Prices  $p \in \{1, 4\}$  and horizons  $T \in \{2, 3\}$  are theoretical primitives, not calibrated estimates.

For the declared dimensions, valid primitive moduli in  $\ell^1$  are

$$C_x = \frac{13}{2d}, \quad L_g = \frac{9}{d}, \quad F_x = \frac{11}{16}, \quad F_a = \sum_j \gamma_j, \quad C_a = \frac{p}{2} + \frac{1}{4}, \quad K^A = \frac{1}{8d}.$$

The transition is invariant by direct interval bounds. The state column sums of its Jacobian are at most  $11/16$ . The costs have piecewise-affine gradients; the cube vertices and the shortage-boundary vertices give the stated bounds. The supplemental

proof restricts this explicit formula to the declared dimensions rather than presuming it for an arbitrary number of coordinates. The general constructive theorem uses whichever valid primitive moduli the economy supplies.

## K.2. *Matched services and realized adaptation*

The execution-before protocol fixes three generators: compiled witness, uniform multilinear FVI, and curvature-refined coordinate FVI. Every generator forms its own future labels and receives the same state-node count, feasible action fractions and integration bins. For each two-state price-horizon cell the common resource ladder is

$$(N, K, M) = (4, 2, 1), (8, 2, 1), (16, 4, 2), (32, 4, 2), (64, 8, 4), (128, 16, 8).$$

The entire ladder is run in three isolated processes per cell and method, rotating method order. This yields 36 two-state services and 216 rungs. State dimension three uses four declared rungs through  $(32, 8, 4)$ ; dimension four uses three through  $(16, 8, 4)$ , with the witness and uniform generators, both horizons, price one and three repetitions. Those additional 24 services contain 84 rungs. The comparison across dimensions uses the same target five, not a dimension-specific certificate threshold. This is a finite common-accuracy experiment at a coarse tolerance, not evidence of an asymptotic or dimension-free rate.

The curvature comparator obtains an own-future pilot on a nine-node coordinate grid. For each axis it takes maxima of adjacent exact second differences across the other coordinates. It then bisects the interval maximizing squared length times the largest pilot curvature over that interval, with deterministic leftmost ties. A negligible positive floor permits a defined rule when the pilot is affine; the dominant unit baseline of the earlier heuristic is absent. Pilots, allocation and the realized axes are stored and charged. Realized nonuniformity, rather than the method's name, determines whether the experiment supplies an adaptive comparison. This remains a coordinate tensor algorithm; it is not presented as a sparse-grid or general adaptive-cell method.

Each rung starts from primitives. The internal complete-prefix clock includes own-future integration, all earlier failed rungs, pilots, exact compilation, actor construction, all-state certification, deployment checks and durable checkpoints. Separate process clocks include startup and the identical non-economic warm-up. A single CPU is selected, numerical-library threads are fixed to one, and CPU frequency is not controlled. Peak resident memory includes the process. Primitive query counts are component counts, not complete FLOPs or bit-operation counts. Independent policy evaluation has a separate joint service and is never added to an unrelated historical clock.

TABLE K.I: Block A separated-resource attainment

| Method | 4     | 2     | 1     | 1/2  |
|--------|-------|-------|-------|------|
| W      | 12/12 | 12/12 | 12/12 | 6/12 |
| F      | 12/12 | 12/12 | 12/12 | 3/12 |
| A      | 12/12 | 12/12 | 12/12 | 3/12 |

W: compiled witness; F: uniform FVI; A: pilot-curvature bisection FVI. Twelve services per method are four economies, each repeated three times. They are not twelve independent economic tasks or optimizer draws.

TABLE K.II: Complete positive-tolerance frontiers in two states

| $(T, p)$ | Tolerance interval  | W: $N$ ; sec. | F: $N$ ; sec. | A: $N$ ; sec. |
|----------|---------------------|---------------|---------------|---------------|
| (2,1)    | [0,0.423176)        | –             | –             | –             |
| (2,1)    | [0.423176,0.483393) | 128; 10.658   | –             | –             |
| (2,1)    | [0.483393,0.846352) | 128; 10.658   | 128; 8.458    | 128; 9.817    |
| (2,1)    | [0.846352,0.962257) | 64; 1.458     | 128; 8.458    | 128; 9.817    |
| (2,1)    | [0.962257,1.6927)   | 64; 1.458     | 64; 1.134     | 64; 1.514     |
| (2,1)    | [1.6927,1.90651)    | 32; 0.290     | 64; 1.134     | 64; 1.514     |
| (2,1)    | [1.90651,2.79584)   | 32; 0.290     | 32; 0.269     | 32; 0.399     |

|    | $(T, p)$ | Tolerance interval   | W: $N$ ; sec. | F: $N$ ; sec. | A: $N$ ; sec. |    |
|----|----------|----------------------|---------------|---------------|---------------|----|
| 1  |          |                      |               |               |               | 1  |
| 2  |          |                      |               |               |               | 2  |
| 3  | (2,1)    | [2.79584, 3.21615)   | 16; 0.092     | 32; 0.269     | 32; 0.399     | 3  |
| 4  | (2,1)    | [3.21615, 5.59167)   | 16; 0.092     | 16; 0.119     | 16; 0.176     | 4  |
| 5  | (2,1)    | [5.59167, 6.13235)   | 8; 0.034      | 16; 0.119     | 16; 0.176     | 5  |
| 6  | (2,1)    | [6.13235, 10.0042)   | 8; 0.034      | 8; 0.061      | 8; 0.090      | 6  |
| 7  | (2,1)    | [10.0042, 10.6876)   | 4; 0.014      | 8; 0.061      | 8; 0.090      | 7  |
| 8  | (2,1)    | [10.6876, $\infty$ ) | 4; 0.014      | 4; 0.029      | 4; 0.042      | 8  |
| 9  | (2,4)    | [0, 0.45093)         | —             | —             | —             | 9  |
| 10 | (2,4)    | [0.45093, 0.506837)  | 128; 10.689   | —             | —             | 10 |
| 11 | (2,4)    | [0.506837, 0.901859) | 128; 10.689   | 128; 8.453    | 128; 9.797    | 11 |
| 12 | (2,4)    | [0.901859, 1.00914)  | 64; 1.452     | 128; 8.453    | 128; 9.797    | 12 |
| 13 | (2,4)    | [1.00914, 1.80372)   | 64; 1.452     | 64; 1.134     | 64; 1.509     | 13 |
| 14 | (2,4)    | [1.80372, 2.00023)   | 32; 0.290     | 64; 1.134     | 64; 1.509     | 14 |
| 15 | (2,4)    | [2.00023, 2.92224)   | 32; 0.290     | 32; 0.268     | 32; 0.397     | 15 |
| 16 | (2,4)    | [2.92224, 3.31506)   | 16; 0.092     | 32; 0.268     | 32; 0.397     | 16 |
| 17 | (2,4)    | [3.31506, 5.84447)   | 16; 0.092     | 16; 0.118     | 16; 0.177     | 17 |
| 18 | (2,4)    | [5.84447, 6.33036)   | 8; 0.034      | 16; 0.118     | 16; 0.177     | 18 |
| 19 | (2,4)    | [6.33036, 10.3185)   | 8; 0.034      | 8; 0.061      | 8; 0.091      | 19 |
| 20 | (2,4)    | [10.3185, 10.9232)   | 4; 0.014      | 8; 0.061      | 8; 0.091      | 20 |
| 21 | (2,4)    | [10.9232, $\infty$ ) | 4; 0.014      | 4; 0.029      | 4; 0.043      | 21 |
| 22 | (3,1)    | [0, 0.63562)         | —             | —             | —             | 22 |
| 23 | (3,1)    | [0.63562, 0.690808)  | 128; 15.635   | —             | —             | 23 |
| 24 | (3,1)    | [0.690808, 1.27124)  | 128; 15.635   | 128; 12.420   | 128; 14.176   | 24 |
| 25 | (3,1)    | [1.27124, 1.37488)   | 64; 2.070     | 128; 12.420   | 128; 14.176   | 25 |
| 26 | (3,1)    | [1.37488, 2.54248)   | 64; 2.070     | 64; 1.651     | 64; 2.148     | 26 |
| 27 | (3,1)    | [2.54248, 2.72357)   | 32; 0.401     | 64; 1.651     | 64; 2.148     | 27 |
| 28 | (3,1)    | [2.72357, 4.11404)   | 32; 0.401     | 32; 0.388     | 32; 0.568     | 28 |
| 29 | (3,1)    | [4.11404, 4.53322)   | 16; 0.126     | 32; 0.388     | 32; 0.568     | 29 |
| 30 | (3,1)    | [4.53322, 8.22807)   | 16; 0.126     | 16; 0.171     | 16; 0.252     | 30 |
| 31 | (3,1)    | [8.22807, 8.6154)    | 8; 0.046      | 16; 0.171     | 16; 0.252     | 31 |
| 32 | (3,1)    | [8.6154, 14.5143)    | 8; 0.046      | 8; 0.088      | 8; 0.130      | 32 |
|    | (3,1)    | [14.5143, 14.9762)   | 4; 0.019      | 8; 0.088      | 8; 0.130      |    |
|    | (3,1)    | [14.9762, $\infty$ ) | 4; 0.019      | 4; 0.041      | 4; 0.061      |    |

| $(T, p)$ | Tolerance interval  | W: $N$ ; sec. | F: $N$ ; sec. | A: $N$ ; sec. |
|----------|---------------------|---------------|---------------|---------------|
| (3,4)    | [0,0.678554)        | –             | –             | –             |
| (3,4)    | [0.678554,0.723386) | 128; 15.815   | –             | –             |
| (3,4)    | [0.723386,1.35711)  | 128; 15.815   | 128; 12.435   | 128; 14.257   |
| (3,4)    | [1.35711,1.44001)   | 64; 2.085     | 128; 12.435   | 128; 14.257   |
| (3,4)    | [1.44001,2.71422)   | 64; 2.085     | 64; 1.643     | 64; 2.169     |
| (3,4)    | [2.71422,2.85349)   | 32; 0.402     | 64; 1.643     | 64; 2.169     |
| (3,4)    | [2.85349,4.31287)   | 32; 0.402     | 32; 0.390     | 32; 0.569     |
| (3,4)    | [4.31287,4.66727)   | 16; 0.127     | 32; 0.390     | 32; 0.569     |
| (3,4)    | [4.66727,8.62574)   | 16; 0.127     | 16; 0.170     | 16; 0.253     |
| (3,4)    | [8.62574,8.88407)   | 8; 0.046      | 16; 0.170     | 16; 0.253     |
| (3,4)    | [8.88407,15.0203)   | 8; 0.046      | 8; 0.088      | 8; 0.131      |
| (3,4)    | [15.0203,15.306)    | 4; 0.019      | 8; 0.088      | 8; 0.131      |
| (3,4)    | [15.306, $\infty$ ) | 4; 0.019      | 4; 0.042      | 4; 0.061      |

Each time is the median complete prefix over three isolated repetitions. A dash is a capped failure. Adjacent bands with identical returned rungs are merged. Printed cutoffs are rounded; exact rational endpoints in the deposited frontier determine membership, including equality. The action and integration budgets are uniquely fixed by  $N$  in the common ladder. Minima, maxima and counts are deposited, not replaced by these medians.

TABLE K.III: Repeated common-accuracy comparison: target five, price one

| $d$ | $T$ | W pass | $N$ | sec.   | F pass | $N$ | sec.   |
|-----|-----|--------|-----|--------|--------|-----|--------|
| 2   | 2   | 3/3    | 16  | 0.092  | 3/3    | 16  | 0.119  |
| 2   | 3   | 3/3    | 16  | 0.126  | 3/3    | 16  | 0.171  |
| 3   | 2   | 3/3    | 16  | 2.424  | 3/3    | 16  | 2.074  |
| 3   | 3   | 3/3    | 16  | 3.508  | 3/3    | 16  | 3.032  |
| 4   | 2   | 3/3    | 16  | 58.569 | 3/3    | 16  | 60.809 |
| 4   | 3   | 3/3    | 16  | 82.156 | 3/3    | 16  | 90.099 |

All dimensions face the same coarse target five; caps and allocations remain explicit. Two-state observations are reused from the main catalogue and not counted twice. This finite comparison does not identify an asymptotic scaling law.

The original block A contains 60 services and 300 rungs. Repeated mathematical checkpoints are byte-identical. The pilot-curvature bisection generator realizes nonuniform axes in 0 of 252 date models; its name alone does not establish effective adaptation. At common successful declared target-by-repetition comparisons, witness is faster than uniform FVI in 13 of 39 and slower in 26. Against pilot-curvature bisection FVI, the corresponding counts are 30 faster and 9 slower out of 39. These counts summarize selected targets; Table K.II and the exact deposited partition, rather than a binary attainment headline, are the primary work comparison.

The new first-crossing direct-cost intervals identify lower witness cost in 0 cases, higher cost in 93, and leave 3 unresolved. For the separately evaluated favorable R48 work pair, the corresponding counts are 0, 0 and 8. A cheaper first crossing and a sharper bound therefore receive an actual economic comparison rather than an inferred ranking. The detailed intervals and initial actions are retained, including every adverse outcome.

Across the 24 declared controller-price decisions, the upper-net-cost selector is certified strictly cheaper than the all-state-bound minimizer in 24. Its cost-regret upper bounds relative to the best evaluated bit range from 0.00238686 to 0.00332242 in the normalized economic units. Where the comparison does not identify a strict improvement it remains unresolved; midpoint choices are descriptive, not true optima. This is a finite expected-net-cost conclusion under the uniform initial law, not a calibrated welfare result or a universal preference for coarse observation.

The direct and sensor study contains 200 sampled contrasts, totaling 52,428,800 interval paths, with a separately recorded joint evaluation time of 923.64 seconds. Identity rows use no simulated paths. This evaluation work is not allocated to one favored generator or hidden inside another method's construction clock.

### K.3. *Actual first-crossing policies and replacement fees*

For the four two-state economic cells and each target 4,2,1, the protocol selects repetition zero's first-crossing witness and uniform FVI policies when both attain.

It also selects the particular favorable R48 target-two pair:  $T = 3, p = 4$ , witness at  $N = 32$  and FVI at  $N = 64$ . Those old frozen policies are evaluated as an additional pair; their historical clocks are not reconstructed as part of the new service. Each selected pair is compared under the uniform initial law and point laws at  $(1/8, 1/8)$ ,  $(1/2, 1/2)$  and  $(7/8, 7/8)$ , with full information or eight-bit observations. Finite observations use robust cell-wide capacity repair and downward action quantum  $1/4096$ .

Every sampled contrast uses 262,144 common interval paths, 40-bit equal initial and innovation bins, and outward mean and variance bounds. All new policy-pair and sensor contrasts share one family of at most 300 with error  $1/100$  and log upper bound 13. The independent-bin model supplies the statistical assertion; the fixed PCG64 streams supply reproducibility. Old R48 intervals retain their separate original confidence event. Simultaneous interpretation of both old and new families is at least 98 percent by a union bound, not automatically 99 percent.

TABLE K.IV: Actual costs of the returned policies and fee sensitivity

| Pair | $\epsilon$ | Bits | $n$ | W<br>lower | W<br>higher | Unres. | Sufficient<br>fee range |
|------|------------|------|-----|------------|-------------|--------|-------------------------|
| R49  | 4          | E    | 16  | 0          | 14          | 2      | [0.00156,0.1042]        |
| R49  | 4          | 8    | 16  | 0          | 15          | 1      | [0.001565,0.1047]       |
| R49  | 2          | E    | 16  | 0          | 16          | 0      | [0.007903,0.05741]      |
| R49  | 2          | 8    | 16  | 0          | 16          | 0      | [0.007853,0.05892]      |
| R49  | 1          | E    | 16  | 0          | 16          | 0      | [0.003374,0.02665]      |
| R49  | 1          | 8    | 16  | 0          | 16          | 0      | [0.003379,0.02688]      |
| R48  | 2          | E    | 4   | 0          | 0           | 4      | [0.001746,0.001875]     |
| R48  | 2          | 8    | 4   | 0          | 0           | 4      | [0.001746,0.001886]     |

E denotes full information. Counts cover every declared initial law and economic cell in the group; they do not rank all initial states. The fee is sufficient to certify that neither directional switch pays. It is not the smaller threshold that merely removes identified profitability. All new intervals share one 99 percent family conditional on the independent-bin model.

The complete numerical supplement reports the directional replacement gains for every pair and the fee sufficient to certify that neither switch pays. It also reports the lower threshold that merely removes an identified profitable switch. The old  $1/64$  decision is preserved, but no conclusion depends on treating that fee as calibrated. The deposited exact fee frontiers permit any nonnegative fee to be assessed on the same confidence event. Lower regret certificates are never described as observed policy-cost improvements.

#### K.4. *Purchasing information on the evaluated grid*

For each first-target-one controller, the protocol evaluates every integer precision  $b = 4, \dots, 16$  under the uniform initial law. Each policy is compared with its own 16-bit implementation; the reference self-comparison is exactly zero. Prices per coordinate-bit are  $1/16384, 1/4096, 1/1024$ . The deterministic all-state-account minimizer and the upper-net-cost selector of Theorem 38 use precisely that same grid. Neither is compared with an unobserved continuous optimum.

TABLE K.V: Information choice on the same evaluated bit grid

| $T;p$ | Method | Bound-minimizing bits | Net-selected bits | Net-regret bound range |
|-------|--------|-----------------------|-------------------|------------------------|
| 2;1   | W      | 16,14,12              | 6,6,4             | [0.002563,0.002596]    |
| 2;1   | F      | 16,14,12              | 7,4,4             | [0.002387,0.002426]    |
| 2;4   | W      | 16,14,12              | 6,6,4             | [0.002691,0.00272]     |
| 2;4   | F      | 16,14,12              | 4,4,4             | [0.002564,0.002564]    |
| 3;1   | W      | 16,14,12              | 7,4,4             | [0.003092,0.003107]    |
| 3;1   | F      | 16,14,12              | 6,4,4             | [0.002992,0.003036]    |
| 3;4   | W      | 16,15,13              | 4,4,4             | [0.003322,0.003322]    |
| 3;4   | F      | 16,14,12              | 4,4,4             | [0.003237,0.003237]    |

Bit triples correspond, in order, to prices  $1/16384, 1/4096, 1/1024$  per coordinate-bit. The net selector minimizes the upper actual-net-cost interval. Its regret bound is against the best bit in the finite catalogue, not an estimated unconstrained optimum. Every price-specific value and all nonnegative-price envelopes are deposited.

For each economic cell, the record additionally gives all nonnegative information prices by exact lower and upper affine envelopes. A retrospective price query within that family is allowed by simultaneous coverage and is not a new experiment. Initial-law dependence remains substantive: a uniform-state expected-cost decision need not be optimal at a particular initial state, and a conservatively optimal all-state bound may rationally purchase more information.

#### K.5. *The retained evidence and its interpretation*

The new service does not rewrite the original nonlinear training studies. The R42 tighter-target neural attainments, six of twelve scalar and five of six coupled services, remain capped-service outcomes; the R44 residual refinements of seven frozen failures remain diagnostics, not retraining successes. The 24 direct neural-minus-ridge intervals remain unresolved. The R45 constructive comparison still has nine of twelve neural and twelve of twelve spline attainments at its tightest target. Fixed64 and adaptive precision still certify all 36 historical precision services, with adaptive slower in 26 and about 0.98 percent slower in aggregate.

The R48 target-one count, twelve of twelve for witness versus six of twelve for each FVI method, also remains unchanged. At a relaxed tolerance 1.014 the old FVI near misses all attain; the release supplies a complete sensitivity table rather than reclassifying those original failures. The new common  $N = 128$  state rung uses separately declared action and integration budgets. It is not the unexecuted diagonal (128,128,128) experiment. The original 48 constrained-policy intervals, including six favoring FVI and none favoring witness, remain available with newly exposed fee frontiers. Their adverse point-state pattern motivates inspection of actual policies; it is not removed by a favorable all-state certificate.

The complete development edition, unchanged historical companions and machine-readable preservation map retain all of these results and their proofs. This revision adds a controlled resource allocation and actual implementation decision, not a selectively favorable replacement record.

## APPENDIX L: COST-DIRECTED CONSTRUCTION AND IMPLEMENTATION EVIDENCE

We now execute the distinction between constructing a safe policy and improving its actual economic performance. The new policies use the same nonlinear investment primitives as Section K; neither the price parameter nor the terminal penalty is changed to favor the proposed action rule. All earlier continuation weights, owners, policies, clocks and adverse contrasts retain their historical identities. The recovered R50 main experiment and common-accuracy amendment had separate local pre-execution freezes. R51 is an explicitly known-design reproduction, with its own Git source identity and runner clocks; the same draws do not provide additional independent observations.

### L.1. *Prospective resource selection and a residual-driven comparator*

For the primary block, select powers of two  $N \in \{4, 8, 16, 32, 64\}$  and  $K, M \in \{1, 2, 4, 8, 16\}$  before constructing a policy. Among allocations satisfying the primitive witness budget  $a/N + b/K + q/M \leq \varepsilon - 10^{-8}$ , minimize the exact integer query proxy  $T(N+1)^d(K+1)M$ , with lexicographic ties. The reserve is checked against the realized certificate; it is not a universal fixed-precision theorem. This rule supplies an executed version of the separated allocation, rather than a retrospective proxy calculation. Conventional FVI receives the same allocation but its own actual-modulus certificate. The witness planning formula alone does not guarantee FVI attainment.

The catalogue contains seven economic cells:  $d = 2, 3, 4$ ,  $T = 2$ ,  $p = 1$ ,  $\varepsilon = 5$ , and  $d = 2$ ,  $T = 2, 3$ ,  $p = 1, 4$ ,  $\varepsilon = 2$ . Witness and uniform FVI are each run on all cells. Two additional services choose the smallest isotropic power-of-two allocation  $N = K = M$  satisfying the same primitive target-five budget in  $d = 2$ ,  $T = 2$ ,  $p = 1$ . A further service in that cell uses residual-driven FVI. Each of these seventeen specifications is executed three times in separate single-CPU processes, with identical warm-up and rotated service order. Repetitions estimate local timing dispersion, not a distribution of economic tasks or neural initializations.

The residual-driven comparator queries its own future Bellman objective at interval endpoints and midpoints. It inserts coordinate knots where the largest interpolation surplus across the other coordinates is greatest, then equidistributes square-root surplus mass along each axis, with dyadic rounding and explicit monotone-knot checks. This last step follows the local quadratic interpolation-error scaling, rather than the curvature-mass rule of the earlier graded block. All pilot queries, residual comparisons, compilation and output are charged. Recorded geometry, not the method name, determines whether adaptation occurred. This is a coordinate-tensor comparator, not a claim to have executed the whole adaptive sparse-grid literature.

The original R50 design and source were frozen locally before their measured execution. Its early bisection-only development test and subsequent equidistribution correction are retained with their original records. R51 freezes the recovered algorithm before a known-design reproduction; it is not presented as a new blind experiment. The present clocks belong only to this execution, and the earlier R49 evidence is unchanged.

TABLE L.I: Primary prospective allocations and realized certificates

| $d;T;p$ | Method | $N,K,M$ | $\epsilon$ | Bound  | Pass | Sec.   | Sec. range      |
|---------|--------|---------|------------|--------|------|--------|-----------------|
| 2;2;1   | W      | 32,2,2  | 2          | 1.9713 | 3/3  | 0.0847 | [0.0841,0.0852] |
| 2;2;1   | W I    | 8,8,8   | 5          | 4.6296 | 3/3  | 0.0195 | [0.0193,0.0200] |
| 2;2;1   | W      | 16,1,1  | 5          | 3.9427 | 3/3  | 0.0240 | [0.0238,0.0268] |
| 2;2;4   | W      | 32,8,1  | 2          | 1.9317 | 3/3  | 0.1086 | [0.1072,0.1105] |
| 2;3;1   | W      | 64,4,2  | 2          | 1.7567 | 3/3  | 0.5059 | [0.5001,0.5075] |
| 2;3;4   | W      | 64,4,2  | 2          | 1.9149 | 3/3  | 0.5058 | [0.4962,0.5095] |
| 3;2;1   | W      | 16,1,1  | 5          | 4.0581 | 3/3  | 0.4424 | [0.4335,0.4610] |
| 4;2;1   | W      | 8,8,2   | 5          | 4.8628 | 3/3  | 1.8441 | [1.8348,1.9009] |
| 2;2;1   | A      | 16,1,1  | 5          | 4.2586 | 3/3  | 1.0709 | [1.0636,1.0941] |
| 2;2;1   | F      | 32,2,2  | 2          | 2.1589 | 0/3  | 0.0644 | [0.0631,0.0646] |
| 2;2;1   | F I    | 8,8,8   | 5          | 5.3703 | 0/3  | 0.0232 | [0.0227,0.0233] |
| 2;2;1   | F      | 16,1,1  | 5          | 4.2199 | 3/3  | 0.0243 | [0.0242,0.0246] |
| 2;2;4   | F      | 32,8,1  | 2          | 2.0923 | 0/3  | 0.0900 | [0.0869,0.0918] |

| $d;T;p$ | Method | $N,K,M$ | $\varepsilon$ | Bound  | Pass | Sec.   | Sec. range      |
|---------|--------|---------|---------------|--------|------|--------|-----------------|
| 2;3;1   | F      | 64,4,2  | 2             | 1.7761 | 3/3  | 0.3650 | [0.3495,0.3661] |
| 2;3;4   | F      | 64,4,2  | 2             | 1.9038 | 3/3  | 0.3606 | [0.3590,0.3723] |
| 3;2;1   | F      | 16,1,1  | 5             | 4.3316 | 3/3  | 0.3890 | [0.3831,0.4258] |
| 4;2;1   | F      | 8,8,2   | 5             | 5.653  | 0/3  | 2.4360 | [2.4275,2.4645] |

W: native witness with equivalent ReLU realization; F: uniform FVI; A: residual-driven FVI; I: isotropic allocation. Each specification has three identical mathematical checkpoints and three distinct clocks. The primitive planning guarantee is witness-specific; FVI uses its actual critic modulus and certificate. Decimal rounding never determines attainment.

The primary block certifies 24 of 24 witness services, 12 of 24 uniform FVI services and 3 of three residual-driven services. These counts include timing repetitions, not independent policy draws. The residual comparator has nonuniform axes in all 3 date models of its repeated checkpoint. Its bound is 4.2586, compared with 4.2199 for uniform FVI at the same quota. Its pilot work is retained.

In the isotropic comparison cell, the planned witness uses 1,156 target innovation evaluations, versus 11,664 for the isotropic service. The latter has fewer state nodes; query savings are therefore not a claim of less work in every category. The table reports measured complete clocks rather than inferring a speed ratio from this count.

## L.2. A separate common-accuracy completion

Inspection of the primary block showed that sharing a witness-sufficient allocation does not make the FVI certificate pass. We therefore froze a separate, explicit service: in  $d = 2, 3, 4$  at  $T = 2, p = 1, \varepsilon = 5$ , begin with the same planner; if a method's actual certificate fails, double only  $N$ , reconstruct from primitives and retain the failed attempt. Stop at first success or at  $N = 64$ . Both methods face this rule. There are three new isolated repetitions per method and dimension. The source freeze and results are separate from the primary catalogue, and old failures remain failures.

TABLE L.II: Separate common-target completion: horizon two,  
price one, target five

| $d$ | Method | $N$ range | Tries | Bound  | Pass | Sec.    | Sec. range        | MiB   |
|-----|--------|-----------|-------|--------|------|---------|-------------------|-------|
| 2   | W      | 16–16     | 1     | 3.9427 | 3/3  | 0.0242  | [0.0223,0.0281]   | 41.4  |
| 2   | F      | 16–16     | 1     | 4.2199 | 3/3  | 0.0244  | [0.0236,0.0275]   | 41.7  |
| 3   | W      | 16–16     | 1     | 4.0581 | 3/3  | 0.4383  | [0.4155,0.4386]   | 44.3  |
| 3   | F      | 16–16     | 1     | 4.3316 | 3/3  | 0.3913  | [0.3840,0.3935]   | 42.3  |
| 4   | W      | 8–8       | 1     | 4.8628 | 3/3  | 1.7418  | [1.7139,1.8030]   | 50.5  |
| 4   | F      | 8–16      | 2     | 3.1181 | 3/3  | 29.0946 | [28.9063,29.3447] | 169.8 |

All failed state refinements are included in the complete prefix. MiB is the maximum process peak over three repetitions, not just model storage. The amendment followed the primary outcomes and was frozen before its own execution. Neither block overwrites the other. Host exclusivity and CPU frequency were not controlled.

The returned policies receive new direct actual-cost comparisons below. These observations provide a common target in every reported dimension. They remain a finite low-dimensional tensor experiment. The construction and corner factors  $(N + 1)^d$  and  $2^d$  are not removed by the compiler, the terminal repair or this allocation rule.

### L.3. Direct policy costs and the source of removable loss

The primary cost experiment fixes twenty-five groups. Sixteen use the original first-target-two R49 witness and FVI policies in all four price–horizon cells, evaluated under the uniform law and each of the three declared point laws. Seven use the new planned services under uniform initial states; one compares residual-driven and uniform FVI; one uses the isotropic allocations. A separate three-group family evaluates the common-accuracy amendment’s returned policies. Every group contains both original and final-repaired policies, with the same eight-bit observation and  $1/4096$  action contract. Both generators receive the same improvement.

Each group uses 131,072 continuous-law interval paths and analytic final-innovation integration. Four absolute costs and all six pair contrasts are recorded.

The primary 250 estimands and supplemental 30 estimands are covered by the conservative endpoint union-bound budget of 512 and log upper bound 13. The supplemental groups have their own conditional independent-bin model within the declared simultaneous family; the policies may depend on earlier construction information. Fixed streams establish reproducibility only. The older R49 family retains its own separate error account.

TABLE L.III: Actual policy differences and removable final-action loss

| Case     | $J_L - J_R$           | $J_{L+} - J_{R+}$      | $J_L - J_{L+}$         |
|----------|-----------------------|------------------------|------------------------|
| C2;2;1;U | [0.023033,0.028988]   | [0.012051,0.017775]    | [0.011448,0.014567]    |
| C3;2;1;U | [0.01796,0.023776]    | [0.010129,0.015794]    | [0.0077535,0.010757]   |
| C4;2;1;U | [0.033636,0.039473]   | [0.015912,0.021433]    | [0.016432,0.019542]    |
| I2;2;1;U | [0.042713,0.048868]   | [0.018209,0.023856]    | [0.023541,0.026849]    |
| O2;2;1;M | [0.0064146,0.011415]  | [-0.0023843,0.0023843] | [0.007622,0.010239]    |
| O2;2;1;L | [0.054972,0.060123]   | [0.044007,0.048792]    | [0.0099411,0.012722]   |
| O2;2;1;H | [0.0042173,0.0090667] | [0.0022567,0.007049]   | [0.0014934,0.003946]   |
| O2;2;1;U | [0.013161,0.018416]   | [0.0064562,0.011605]   | [0.0056474,0.0083165]  |
| O2;2;4;M | [0.035287,0.040496]   | [0.030381,0.035526]    | [0.0036933,0.0063836]  |
| O2;2;4;L | [0.020345,0.025506]   | [0.017976,0.023095]    | [0.0010828,0.0036974]  |
| O2;2;4;H | [0.039236,0.044497]   | [0.034549,0.039692]    | [0.0034416,0.006155]   |
| O2;2;4;U | [0.011181,0.016658]   | [0.0058295,0.011256]   | [0.004188,0.0069254]   |
| O2;3;1;M | [0.015697,0.021868]   | [0.012485,0.018641]    | [0.0016956,0.0047777]  |
| O2;3;1;L | [0.026367,0.032516]   | [0.022928,0.029055]    | [0.0020031,0.005116]   |
| O2;3;1;H | [0.011221,0.017268]   | [0.0095515,0.015586]   | [0.00030976,0.0033545] |
| O2;3;1;U | [0.0095127,0.015773]  | [0.0065302,0.012761]   | [0.0014967,0.0045926]  |
| O2;3;4;M | [0.010433,0.016948]   | [0.0089337,0.015429]   | [9.4671e-05,0.0033779] |
| O2;3;4;L | [0.0095446,0.01608]   | [0.0074164,0.013916]   | [0.0005048,0.0037993]  |
| O2;3;4;H | [0.0073415,0.013898]  | [0.0046083,0.011119]   | [0.0011154,0.0044219]  |
| O2;3;4;U | [0.0056055,0.012273]  | [0.0039105,0.010565]   | [0.00021625,0.0035063] |
| Q2;2;1;U | [0.01186,0.017187]    | [0.0056983,0.010907]   | [0.0055893,0.0083122]  |
| P2;2;1;U | [0.022852,0.028801]   | [0.011835,0.01755]     | [0.011497,0.014615]    |

| Case     | $J_L - J_R$            | $J_{L^+} - J_{R^+}$    | $J_L - J_{L^+}$        |
|----------|------------------------|------------------------|------------------------|
| Q2;2;4;U | [0.0086864,0.014074]   | [0.004022,0.0093656]   | [0.003535,0.0062382]   |
| Q2;3;1;U | [0.01136,0.017753]     | [0.0081953,0.014553]   | [0.0017403,0.0048791]  |
| Q2;3;4;U | [0.010559,0.017349]    | [0.0074835,0.014248]   | [0.0020091,0.0053568]  |
| P3;2;1;U | [0.017874,0.023687]    | [0.0099988,0.015657]   | [0.0078066,0.010814]   |
| P4;2;1;U | [0.033196,0.039029]    | [0.015676,0.021193]    | [0.016374,0.019482]    |
| A2;2;1;U | [-0.0024517,0.0023493] | [-0.0024125,0.0023985] | [0.00062241,0.0030972] |

A case is block/dimension; horizon; price; initial law. O: original R49 target-two policies; P: primary target-five plan; Q: primary target-two plan; C: common-accuracy amendment; I: isotropic; A: surplus FVI versus uniform FVI. Except A, left is witness and right is uniform FVI. U is uniform; L,M,H are point states 1/8,1/2,7/8. Superscript + denotes the identical terminal safety rule. Five-significant-digit display is not an exact endpoint; the rational records govern all conclusions.

All 27 declared witness comparison groups identify a strictly positive reduction of the witness incumbent's actual expected cost. The simultaneous gain lower endpoints range from 9.4671e-05 to 0.023541; upper endpoints range from 0.0033545 to 0.026849. This is a law-specific statement for the stated policies, not strict improvement at every initial state or 27 independent training successes.

Before repair, witness cost is identified as higher in 27 groups. After both methods receive the same repair, it remains higher in 26, is lower in 0, and is unresolved in 1. The residual-driven FVI comparison is unresolved both before and after repair. Thus the terminal construction repairs a genuine economic loss without manufacturing a reversal of the strong conventional comparison.

The active evaluator records 3,670,016 interval paths, 7,340,032 cell-gate queries and 5,447,560 accepted changes across the two incumbents. No acquisition ambiguity or broad-hull fallback is recorded. Its joint direct-evaluation time before durable record output is 127.95 seconds. That total is evaluation work, not a speed comparison between independently executed policies.

For each incumbent the direct reduction is the centered final contrast. The remaining witness-FVI difference after both repairs is an actual comparison of the two

changed policies. The identity

$$J_W - J_F = (J_W - J_{W+}) + (J_{W+} - J_{F+}) - (J_F - J_{F+})$$

separates removable final-action losses from the difference that remains. It does not attribute every remaining discrepancy to a single representation mechanism. The deposited date-level action means, terminal state means, accepted and blocked proposals, and capacity/quantum repair counts permit this distinction to be checked. A finite-sample positive gain can justify a strictly positive replacement charge up to its simultaneous lower endpoint; a deterministic nonincrease alone does not justify an arbitrary implementation fee.

#### L.4. *Actual-cost frontiers and complete resource boundaries*

The supplement reports every group's four absolute cost intervals and every direct contrast. For each new construction, its resource coordinate includes the full own-future service and, separately, the joint evaluation charge. The reader can use either a shared fixed evaluation cost for the catalogue decision or the complete joint cost required to certify a particular comparison. We do not claim to have measured independent marginal costs for four separately simulated policies when the implementation shares their earlier trajectories. The pair clock is an explicitly charged object, not an unpriced convenience.

TABLE L.IV: Construction work and actual cost of new implementations

| Case     | Method | Build sec. | Original cost     | Repaired cost     | Pair sec. |
|----------|--------|------------|-------------------|-------------------|-----------|
| C2;2;1;U | W      | 0.0242     | [0.6242,0.6378]   | [0.61117,0.62482] | 3.799     |
| C2;2;1;U | F      | 0.0244     | [0.59821,0.61177] | [0.59628,0.60988] | 3.799     |
| C3;2;1;U | W      | 0.4383     | [0.58992,0.60068] | [0.58062,0.59146] | 6.102     |
| C3;2;1;U | F      | 0.3913     | [0.56904,0.57982] | [0.56768,0.57849] | 6.102     |
| C4;2;1;U | W      | 1.7418     | [0.62188,0.63199] | [0.60399,0.61391] | 9.790     |
| C4;2;1;U | F      | 29.0946    | [0.5855,0.59525]  | [0.5854,0.59515]  | 9.790     |
| I2;2;1;U | W      | 0.0195     | [0.63959,0.65386] | [0.61456,0.6285]  | 3.744     |

| Case     | Method | Build sec. | Original cost     | Repaired cost     | Pair sec. |
|----------|--------|------------|-------------------|-------------------|-----------|
| I2;2;1;U | F      | 0.0232     | [0.59407,0.60781] | [0.59363,0.60737] | 3.744     |
| Q2;2;1;U | W      | 0.0847     | [0.60972,0.62332] | [0.60276,0.61638] | 3.870     |
| Q2;2;1;U | F      | 0.0644     | [0.59518,0.60882] | [0.59444,0.6081]  | 3.870     |
| P2;2;1;U | W      | 0.0240     | [0.6222,0.63577]  | [0.60912,0.62274] | 3.901     |
| P2;2;1;U | F      | 0.0243     | [0.59639,0.60993] | [0.59445,0.60803] | 3.901     |
| Q2;2;4;U | W      | 0.1086     | [0.69243,0.70702] | [0.68754,0.70213] | 3.919     |
| Q2;2;4;U | F      | 0.0900     | [0.68106,0.69562] | [0.68086,0.69542] | 3.919     |
| Q2;3;1;U | W      | 0.5059     | [0.72237,0.73696] | [0.71906,0.73365] | 4.406     |
| Q2;3;1;U | F      | 0.3650     | [0.70782,0.7224]  | [0.70769,0.72228] | 4.406     |
| Q2;3;4;U | W      | 0.5058     | [0.86908,0.88468] | [0.86539,0.881]   | 4.513     |
| Q2;3;4;U | F      | 0.3606     | [0.85511,0.87074] | [0.85451,0.87015] | 4.513     |
| P3;2;1;U | W      | 0.4424     | [0.58981,0.60055] | [0.58046,0.59128] | 6.294     |
| P3;2;1;U | F      | 0.3890     | [0.56902,0.57978] | [0.56764,0.57844] | 6.294     |
| P4;2;1;U | W      | 1.8441     | [0.61951,0.62957] | [0.60168,0.61155] | 7.713     |
| P4;2;1;U | F      | 2.4360     | [0.58357,0.59328] | [0.58332,0.59303] | 7.713     |
| A2;2;1;U | A      | 1.0709     | [0.59772,0.61129] | [0.59585,0.60945] | 3.797     |
| A2;2;1;U | F      | 0.0243     | [0.59777,0.61134] | [0.59585,0.60946] | 3.797     |

Pair sec. is a shared cost of the whole four-policy comparison, repeated in the display for clarity, not independently attributable to each row. Construction medians use the same cell and fresh records; no historical clocks are spliced. Absolute-cost intervals are wider than centered pair intervals. Charges enter Proposition 39 with their explicitly chosen shared or marginal accounting convention.

All stored coefficients, original owners, action scalars, model evaluation categories, transform relaxations, interpolation/corner operations, terminal derivative queries, whole-cell tests, exact fallbacks, numerical interval widths, serialized bytes and process peak resident memory remain in the records. The counters are primitive categories, not a complete FLOP or bit-operation count. The latter would require an additional cost model. Direct comparison time is not added to clocks from another host or to a historical failure in order to manufacture a new complete service.

The retained sensor catalogue has a structural interpretation through equation (105). Its prices remain normalized primitives, and its confidence-envelope

regret is relative to the best observed implementation in the finite catalogue. This revision neither estimates a sensor technology nor substitutes a sample-mean winner for the true expected-net-cost minimizer.

#### L.5. *Historical findings that the new evidence does not replace*

The earlier 93 of 96 strictly higher witness-cost contrasts remain visible in Table K.IV. Original curvature-FVI never changed its grid in the realized block. The later genuinely graded method also remains an adverse conventional-comparator experiment: it was less precise in all twelve matched final services and slower in all twelve. These two findings concern the two specified heuristics, not all adaptive dynamic programming. The present surplus comparator adds an independently recorded realized partition, while uniform FVI remains the strong baseline.

Likewise, a successful final-action repair does not establish better earlier continuation learning, universal neural superiority, a guarantee for unrestricted non-convex training, or a dimension-free policy construction. What is established is a cost-directed feasible improvement with an explicit safety condition, its executed original-economy specialization, and source-bound resource and actual-cost comparisons. These results extend the NBO policy-evaluation and improvement program without replacing its economic subject or erasing its previous evidence.

### APPENDIX M: EXHAUSTIVE ROBUSTNESS IN THE ORIGINAL INVESTMENT ECONOMY

The action-null result permits a diagnostic in which the true policy and its economic cost are held fixed while a critic's irrelevant component changes. This avoids confusing a new representation with a new economic policy. Before execution, we fixed all eight R49 target-two checkpoint policies retained in the R51 package: witness and uniform FVI for every two-state price-horizon cell. We enumerate every one of the  $256^2$  closed eight-bit observation cells for each policy. There is no selection of cells by favorable cost or by the new results.

For each cell we retain the original robust capacity repair, the action lattice 1/4096, and the convex final-date proposal. The centered outward interval for its actual

conditional cost difference is the same object used in the inherited terminal gate. We compare the global-width and contrast-aware gates after replacing  $g$  by  $g - n_M$ , for  $M \in \{0, 1, 16, 4096\}$ . Proposition 33 establishes the zero action-error allowance algebraically; it is not estimated from this grid. Closed-cell enclosures cover boundary states, while execution retains the original observation tie convention. Symbolic cancellation precedes interval evaluation.

TABLE M.I

EXHAUSTIVE ACTION-NULL ROBUSTNESS ON THE ORIGINAL INVESTMENT ECONOMY

| Incumbent | $T$ | $p$ | $M = 0$ | $M = 1$ | $M = 16$ | $M = 4096$ |
|-----------|-----|-----|---------|---------|----------|------------|
| Witness   | 2   | 1   | 48872   | 0       | 0        | 0          |
| Witness   | 2   | 4   | 60574   | 0       | 0        | 0          |
| Witness   | 3   | 1   | 48634   | 0       | 0        | 0          |
| Witness   | 3   | 4   | 60114   | 0       | 0        | 0          |
| FVI       | 2   | 1   | 42198   | 0       | 0        | 0          |
| FVI       | 2   | 4   | 54249   | 0       | 0        | 0          |
| FVI       | 3   | 1   | 37574   | 0       | 0        | 0          |
| FVI       | 3   | 4   | 54249   | 0       | 0        | 0          |

Each row enumerates all 65,536 closed eight-bit cells. Entries count strict changes accepted by the global-width gate. The contrast-aware gate accepts exactly the  $M = 0$  column at every amplitude, with identical returned actions. Earlier policy dates are unchanged. Counts concern cells, not additional expected-cost observations.

The records retain the incumbent and proposed action indices, the returned contrast-aware index, the exact binary64 endpoint encodings, and every global-width decision for every cell. An independent record audit checks cell completeness, cell-wide capacity, all gates, counts and accepted upper bounds. The contrast-aware actions are also compared with a fresh call to the unmodified inherited terminal implementation on every cell. Thus the reported invariance concerns the actual stored policy, not equality of two rounded average costs. The exact finite-state suite separately exercises nonzero action-error bounds, rejection, and all passes of the finite-sweep theorem.

The experiment measures resistance to irrelevant continuation error. It is not a new estimate of  $J^{\text{witness}} - J^{\text{FVI}}$ . The contrast-aware policies are identical to the unperturbed terminal-repaired incumbents, so repeating their cost simulation would not create additional independent economic observations. The prior actual-cost intervals, their confidence events, and the 93 witness-higher, zero witness-lower and three unresolved R49 comparisons remain unchanged. Safe improvement of either incumbent remains different from superiority over the competing policy.

Every service includes input loading, proposals, true-cost interval comparisons, independent inherited-policy replay, amplitude decisions and durable cell output. Machine-independent query counters, serialization time and peak memory are retained separately. The supplementary work table is the cost of this exhaustive diagnostic, not the construction cost of an alternative learned model. A full  $T$ -pass investment-economy execution is not inferred from a final-date ablation. The general finite-sweep result has explicit verified-error hypotheses; the continuous economic experiment executes only the declared specialization.

## APPENDIX N: ECONOMIC SCOPE AND CONCLUSION

The economic contribution is an explicit connection between a policy's construction resources and its implementable cost. A primitive error budget can determine whether to refine the state cover, action search or integration. A complete finite tolerance frontier states when one generator returns a certified policy earlier. Independent comparisons of the returned policies then determine what can be said about controller replacement and purchased information. These are different steps; none can be replaced by the favorable outcome of another.

The constructive backend and exact compiler make positive deterministic statements under explicit moduli and effective-computability conditions. Their guarantees are not success probabilities for unrelated nonconvex training algorithms. The matched experiments and direct-cost intervals make finite, source-bound claims about the declared economies, resource ladders and initial laws. Common-accuracy dimension cells do not establish an asymptotic scaling law, and endogenous precision within a finite sensor catalogue is not an optimum over all information

technologies. The original broad theory, applications and adverse experimental record remain part of the complete NBO development.

The cost-directed result adds a further positive statement: a feasible acquired policy can be improved without sacrificing its incumbent cost guarantee, provided its centered advantage is verified. In the executed final-stage specialization every studied witness incumbent has a strictly positive identified reduction under the declared initial law. Most witness–FVI differences remain unfavorable after both methods are repaired. The improvement theorem and this comparative observation are consistent: safe improvement relative to an incumbent is not dominance over another method.

Neural Bellman Operators therefore provide a way to organize continuation construction, feasible implementation, certification and economic resource choice around the policy actually returned. Whether a particular representation or generator is preferable is decided by its full certified frontier and actual economic consequences, rather than by the name of its function class.

#### APPENDIX O: SUBMISSION AND PRESERVATION MAP

This article and its technical supplement are the active R52 exposition of the same paper. Every R49 main-article and supplementary label is retained in its corresponding current document. The complete development editions retain the R48 main and supplementary bodies and add the current theory and evidence, with a reading notice distinguishing historical wording from the active empirical account. The original R49 sources remain unchanged under `preserved/R49`; R48 sources and earlier theory/application companions are likewise retained. Preservation is not a new mathematical validation of every historical application.

The local revision contains ordinary sources, both immutable R49 evidence archives, the R48 publication input, newly executed records, generated tables, a point-by-point response and compiled documents. The direct numerical correction retains its original observations and replays the same streams with explicitly outward cross-policy subtraction. The corrected intervals, not their raw predecessors, govern the active direct-cost tables. Source and evidence identities are checked

before a clean offline Git-archive rebuild. The local delivery record distinguishes a completed build from a remote publication: no remote branch or commit is claimed without a confirmed push. This limitation is administrative, not a substitute for the substantive revised manuscript.

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